Optimal Confidence Procedures for Partially Identified Parameters*

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Abstract

I construct optimal confidence procedures in a flexible class of partially identified models. Under my decision-theoretic framework, reporting a confidence set incurs a loss equal to its excess length: the length of the part that does not overlap with the identified set. A confidence procedure is optimal if it minimizes worst-case expected excess length. I show that the problem of constructing such a procedure reduces to solving a single optimal testing problem and inverting the resulting test. This theoretical insight allows me to leverage recent computational advances in optimal testing to numerically approximate the optimal procedure. In a series of calibrated simulations, I quantify the suboptimality of existing inference techniques across a variety of empirical settings. Unlike the optimal procedure, methods with poor performance often direct power towards alternatives that are inconsistent with the model. I apply the optimal procedure to a recent event study design and find that it yields confidence sets that are 15 to 22 percent shorter than existing approaches.

Keywords: Optimal Inference; Partial Identification

JEL classification codes: C12, C21, C44

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1 Introduction

Partially identified models are becoming increasingly prevalent in empirical economics, with applications spanning a wide range of topics. Examples include labor supply responses (Kline and Tartari, 2016), experimenter demand effects (de Quidt et al., 2018), and export market entry (Morales et al., 2019). The partial identification paradigm enables researchers to dispense with implausible assumptions that might otherwise be necessary to achieve point identification. It also introduces considerable challenges in forming confidence sets (Canay and Shaikh, 2017). An enormous body of work has developed confidence procedures for partially identified parameters with good theoretical coverage properties. However, the optimality properties of such procedures are not well-understood. How can researchers construct optimal confidence procedures under partial identification, and how far from optimal are existing methods?

In this paper, I address these questions by leveraging a particular structure that arises in a broad class of partially identified models. In Section 2.2, I highlight several empirical examples with this structure, including sensitivity analysis in difference-in-difference designs (Rambachan and Roth, 2023) and settings with either intersection bounds or union bounds (Chernozhukov et al., 2013; Bei, 2025). In all of these examples, the distribution of the data is informative about a scalar target parameter through a vector of moments. An identification problem arises because the moments are contaminated by additively separable nuisance parameters. Depending on the empirical setting, the nuisance parameters may represent violations of the parallel trends assumption, or slack in a collection of bounds. Although the nuisance parameters are unknown, they must lie in a known set. I leave the specification of this set entirely up to the researcher's discretion, thereby accommodating a wide range of applications whose common structure has not yet been recognized.

To develop a theory of optimal inference for the class of models I consider, I take a decision theoretic-perspective on confidence procedures (Wald, 1950). From this point of view, reporting a confidence set incurs a loss that reflects the researcher's preferences over performance. I propose using excess length loss, which naturally extends length loss to the partially identified case. The excess length of a confidence set is the length of the part that does not overlap with the identified set. The idea is that researchers do not consider it costly to report values of the target parameter that lie in the identified set. While my main specification defines an optimal confidence procedure as one that has minimax expected excess length, I also accommodate length loss, minimax regret, and Bayes risk. To abstract from asymptotic considerations, I suppose that the researcher observes normally distributed moment estimates centered around the true moments with known covariance. I place no restrictions on the class of confidence procedures under consideration, aside from the usual frequentist coverage requirement for the target parameter.

My main theoretical result establishes that an optimal confidence procedure can be constructed by inverting a single optimal test. Test inversion is the predominant method of forming confidence sets for partially identified parameters (Canay and Shaikh, 2017). Under this approach, a $100(1-\alpha)\%$ confidence set comprises all values of the target parameter that are *not* rejected by a level-α test. Although many such tests are available in the literature, it is difficult for researchers to know which one to use: a uniformly most powerful test typically does not exist, and existing tests seldom dominate their competitors in terms of power. I address this challenge by relating the power properties of the test to the performance of the resulting

confidence procedure via the Ghosh–Pratt identity (Ghosh, 1961; Pratt, 1961). Under point identification, this identity links power to expected length; I show that under partial identification, an analogous identity links power to expected excess length. The form of the optimal testing problem I derive provides researchers explicit guidance on how best to direct power, given their preferences over confidence procedure performance. In particular, in Section 4.3 I show that it is never optimal to direct power towards alternatives that are inconsistent with the model.

To approximate the optimal test, I build upon numerical methods developed in the literature on weighted average power optimal testing. In recent work on this subject, Aradillas Fernández et al. (2025) propose a computationally attractive algorithm and provide theoretical guarantees on its performance. However, the authors require researchers to specify a set of weights that dictate how they are willing to trade off power across different alternatives. In my setting, the primitive of interest is the test inversion confidence procedure, rather than the test itself, so researchers' preferences over power are endogenously determined. I apply duality theory to summarize these implicit preferences by a least favorable prior over moments that index confidence procedure performance. I then extend the algorithm of Aradillas Fernández et al. (2025) to simultaneously approximate the least favorable prior while solving for an approximately optimal test according to that prior. Finally, I establish a theoretical performance guarantee for the proposed algorithm, which motivates concrete recommendations regarding its implementation.

Equipped with these tools, I quantify the suboptimality of existing inference techniques in a series of calibrated simulations, along with two empirical applications. In a simulation study based on that of Rambachan and Roth (2023), I show that existing efficiency bounds are highly conservative in their setting: they suggest that the authors' recommended procedure could be improved upon by as much as 33%, while the margin is in fact closer to 10%. In an application to de Quidt et al. (2018), I find that the misspecification-robust confidence interval of Stoye (2020) turns out to be nearly minimax optimal under correct specification. Finally, I conduct a sensitivity analysis on the event study of Iovino et al. (2026), incorporating a credible restriction on the direction of parallel trends violations. The optimal confidence sets are 15 to 22 percent shorter than those of existing approaches, and the optimal procedure is 22 to 26 percent more efficient in calibrated simulations.

This paper is related to work by Armstrong and Kolesár (2018, 2021), who study optimal inference in a framework that applies to a special case of my proposed class of models. The authors' main result concerns settings in which the set of possible nuisance parameters is both convex and centrosymmetric (that is, closed under negation). In this case, they show that a computationally tractable fixed length confidence interval due to Donoho (1994) is nearly minimax expected length optimal. My approach differs from theirs in two respects. First, I place no restrictions on the set of possible nuisance parameters beyond a basic measurability requirement. This generalization is empirically relevant: most of the empirical examples I discuss in Section 2.2 do not satisfy their convexity and centrosymmetry requirement. Second, rather than providing a lower bound on the efficiency of an existing procedure, I characterize the optimal procedure and approximate it numerically. In settings where the optimality theory of Armstrong and Kolesár (2018, 2021) applies, I find that their procedure often performs dramatically better than their efficiency bounds would suggest, making it a particularly attractive option for practitioners; see Section A.1. Otherwise, to the

best of my knowledge, there is no existing optimality theory available for researchers to rely on. This paper develops such a theory, delivering both a numerical approximation to the optimal confidence procedure and a benchmark against which to assess the performance of existing methods.

The literature on weighted average power optimal testing represents another approach to optimal inference under partial identification. This literature takes as given a researcher-specified set of weights over the alternative, and seeks to maximize the corresponding weighted average of power subject to a size constraint (Andrews and Ploberger, 1994; Müller, 2011; Moreira and Moreira, 2019; Montiel Olea, 2020). Work on composite null hypotheses is particularly salient to the partially identified case; for instance, Elliott et al. (2015) numerically approximate an optimal test in the presence of nuisance parameters and apply it to the Imbens–Manski problem, which I study in Section 7.1. More recently, Aradillas Fernández et al. (2025) propose a similar numerical approach with theoretical performance guarantees, and Ketz et al. (2025) diagnose whether an existing test is weighted average power optimal for some choice of weights. In certain special cases, a test may have provably optimal local asymptotic power for *any* choice of weights (Kaido and Santos, 2014; Rambachan and Roth, 2023). However, in most cases the form of a weighted average power optimal test depends on the choice of weights, which can appear somewhat ad hoc in practice (Elliott et al., 2015, §5.4). The theory I develop provides a principled method of determining the weights if the researcher's ultimate goal is to form a confidence set via test inversion; see Section 4.3. On the other hand, in Section 5 I demonstrate that the computational insights of Aradillas Fernández et al. (2025) transfer to my setting. In this way, the current work complements and extends existing work on weighted average power optimal testing from both a theoretical and computational perspective.

The remainder of the paper is organized as follows. Section 2 sets up the econometric framework and maps it onto several applications. Section 3 introduces the statistical decision problem that defines an optimal confidence procedure. Section 4 establishes that researchers may construct an optimal confidence procedure by inverting a single optimal test. Section 5 describes the numerical approach I take to approximate the optimal procedure and compute efficiency bounds, while Section 6 conducts a simulation study. Finally, Section 7 compares optimal confidence sets to existing alternatives in two empirical applications, and Section 8 concludes. Additional details and proofs are available in Appendix Sections A to D.

2 Framework and Examples

I study optimal inference in an econometric framework with enough structure to keep the problem tractable, but enough flexibility to accommodate a wide variety of empirical applications. The framework consists of both a model and an experiment. First, I introduce the model and demonstrate its applicability in a series of examples. Next, I introduce the experiment and discuss its relationship to the situation researchers will encounter in finite samples.

2.1 Model

The researcher's model relates observed moments $\mu \in \mathbb{R}^{k_\mu}$ to unknown structural parameters $\tau \in \mathbb{R}^{k_\tau}$ and nuisance parameters $\delta \in \mathbb{R}^{k_\mu}$. The researcher would like to learn about a scalar summary $\theta \in \mathbb{R}$ of the structural parameters. I restrict attention to models in which the structural parameters enter linearly and the nuisance parameters are additively separable, so

$$\begin{aligned} \mu &= K\tau + \delta, \\ \theta &= l'\tau \end{aligned} \quad \text{for some} \quad (\tau, \delta) \in \mathbb{R}^{k_\tau} \times \Delta \tag{1}$$

where the coefficients $K \in \mathbb{R}^{k_\mu \times k_\tau}$ and $l \in \mathbb{R}^{k_\tau}$ are known. The set of possible nuisance parameters $\Delta \subseteq \mathbb{R}^{k_\mu}$ must not depend on τ, but is otherwise up to the researcher's discretion. In the classical setting of linear regression with a fixed design matrix, τ is a vector of regression coefficients, K is full rank with $k_\mu \geq k_\tau$, and there are no nuisance parameters, so $\Delta = \{0\}$. Hence, the target parameter is point identified and given by the familiar formula $\theta = l'(K'K)^{-1}K'\mu$. In the more challenging settings that motivate this paper, Δ is not a singleton, so the target parameter is only partially identified: that is, many values of θ may be consistent with the same value of μ. I proceed by mapping (1) onto several concrete examples.

2.2 Examples

Example 2.1 (Relaxing Parallel Trends)**.** Rambachan and Roth (2023) study identification and inference under relaxations of the parallel trends assumption in difference-in-differences (DiD) and event study designs. For the sake of concreteness, consider a canonical non-staggered DiD design following Rambachan and Roth (2023, Example 1). In this case, the researcher has a balanced panel $\{Y_{it}, D_i\}_{i=1,\, t=-\underline{T}}^{n,\ \overline{T}}$ in which units with $D_i = 1$ are treated from period $t = 1$ onwards, and units with $D_i = 0$ are never treated. Setting period 0 as the reference period, the DiD estimand in period t is given by

$$\mu_t \equiv (\mathbb{E}[Y_{it} \mid D_i = 1] - \mathbb{E}[Y_{it} \mid D_i = 0]) - (\mathbb{E}[Y_{i0} \mid D_i = 1] - \mathbb{E}[Y_{i0} \mid D_i = 0]).$$

Suppose the researcher adopts the potential outcomes model $Y_{it} = (1 - D_i)Y_{it}(0) + D_i Y_{it}(1)$ and rules out treatment anticipation so $Y_{it}(0) = Y_{it}(1)$ for all $t < 1$. Then the DiD estimand may be expressed as

$$\mu_t = \underbrace{\mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid D_i = 1]}_{\equiv \tau_t} + \underbrace{\mathbb{E}[Y_{it}(0) - Y_{i0}(0) \mid D_i = 1] - \mathbb{E}[Y_{it}(0) - Y_{i0}(0) \mid D_i = 0]}_{\equiv \delta_t}.$$

Let $\mu^{\text{post}} \equiv (\mu_{\overline{T}}, \ldots, \mu_1)'$, $\mu^{\text{pre}} \equiv (\mu_{-1}, \ldots, \mu_{-\underline{T}})'$, and define τ^{post}, τ^{pre} and δ^{post}, δ^{pre} analogously. The no-anticipation assumption ensures that $\tau^{\text{pre}} = 0$, so (1) holds with

$$\mu = K\tau + \delta \quad \text{for} \quad \mu \equiv \begin{bmatrix} \mu^{\text{post}} \\ \mu^{\text{pre}} \end{bmatrix}, \quad K \equiv \begin{bmatrix} \mathbf{I}_{\overline{T} \times \overline{T}} \\ \mathbf{0}_{\underline{T} \times \overline{T}} \end{bmatrix}, \quad \tau \equiv \tau^{\text{post}}, \quad \text{and} \quad \delta \equiv \begin{bmatrix} \delta^{\text{post}} \\ \delta^{\text{pre}} \end{bmatrix}.$$

The structural parameters τ represent post-period average treatment effects on the treated, while the nuisance parameters δ represent average differences in trends (relative to period 0) in untreated potential outcomes between the treated and untreated units. The coefficients $l \in \mathbb{R}^{k_\tau}$ are typically chosen so that $\theta = l'\tau$ represents a weighted average of τ.

Under parallel trends, $\delta \in \Delta = \{0\}$, so $\theta = l'\tau = l'\mu^{\text{post}}$ is point-identified and the researcher may apply classical inference techniques. If the researcher does not find it credible to assume that parallel trends holds exactly, Rambachan and Roth (2023) propose expanding the set of possible differences in trends Δ, building on work by Manski and Pepper (2018). The authors emphasize that the researcher's choice of Δ should be motivated by domain knowledge specific to their empirical setting. While Rambachan and Roth (2023) provide confidence procedures for θ, they focus primarily on ensuring uniform coverage in a computationally tractable manner, with results on (near) optimality limited to certain special cases; see Section 6.2 for further discussion. □

Example 2.2 (Counterfactuals in Judge Designs)**.** Kolesár et al. (2026) investigate identification of average outcomes under counterfactual policies in judge designs. To fix ideas, consider a setting in which two randomly assigned judges $Z \in \{0, 1\}$ make pretrial release decisions $D \in \{0, 1\}$ in order to prevent pretrial misconduct $Y \in \{0, 1\}$. Kolesár et al. (2026) study quota policies under which judges are required to release at least $100q\%$ of defendants for some researcher-specified $q \in [0, 1]$. Let $D(z, a)$ denote the release decision judge z would have made under policy $a \in \{0, 1\}$, where $a = 0$ represents the status quo and $a = 1$ represents the quota policy. Similarly, write $Y(d)$ for the defendant's misconduct under release decision d, embedding an exclusion restriction, and suppose that the instrument Z is exogenous, meaning

$$\{(Y(d), D(z, a))\}_{d,z,a \in \{0,1\}} \perp\!\!\!\perp Z.$$

Finally, instead of the usual Imbens and Angrist (1994) monotonicity assumption, Kolesár et al. (2026) consider variations of a stronger "policy invariance" assumption under which judges make release decisions according to the following latent index model,

$$D(z, a) = \mathbb{1}\{p_z + \max\{0, q - p_z\}a \geq U\}.$$

The policy-invariant uniform random variable U captures judges common ranking of defendants. Judge z has release rate $p_z = \Pr(D = 1 \mid Z = z)$ under the status quo, which must increase to hit the quota if $p_z < q$ but is otherwise unchanged. The target parameter $\theta \equiv \mathbb{E}[Y(D(Z, 1))]$ is the misconduct rate under the quota policy.

In judge designs, it is not known a priori whether one judge is more lenient than another, substantially complicating identification and inference. To see why, suppose for the sake of simplicity that $Y(0) = 0$ (a defendant cannot commit misconduct while detained), and case shares $w_z \equiv \Pr(Z = z)$ are known and policy-invariant.[1] Writing $y_z \equiv \Pr(Y = 1 \mid Z = z)$ for status quo misconduct rates and collecting the moments $\mu \equiv (p_0 - q, y_0, p_1 - q, y_1)'$, I show in Section A.3 that the identified set for θ is given by

$$\Theta(\mu) = \begin{cases} \{w_0 y_0 + w_1 y_1\} & \text{if } q \leq p_0 \leq p_1, \\ w_1 y_1 + w_0[\max\{y_0, y_1 - (p_1 - q)\}, \min\{y_0 + (q - p_0), y_1\}] & \text{if } p_0 < q \leq p_1, \\ [y_1, y_1 + (q - p_1)] & \text{if } p_0 \leq p_1 < q. \end{cases}$$

If $p_1 \leq p_0$, an analogous result holds with the indices swapped. The form of the identified set depends strongly on the unknown ordering of p_0, p_1, and q, ranging from point identification under a non-binding quota to Manski bounds under universal release. Since this dependence renders many existing inference techniques inapplicable, Kolesár et al. (2026) form confidence sets for θ via projection inference, which ensures uniform coverage but may be quite conservative (Romano and Shaikh, 2008). In Section A.3, I establish that (1) holds with $K \equiv (0, 1, 0, 1)'$, $l \equiv 1$, and Δ defined as the union of six polyhedra: one for each possible ordering. Hence, the unknown ordering manifests as a non-convexity in the set of possible nuisance parameters. □

Example 2.3 (Intersection Bounds)**.** Chernozhukov et al. (2013) study inference on θ given a collection of lower and upper bounding moments μ.[2] Let $\mathcal{J}_\ell$ and $\mathcal{J}_u \subseteq \{1, \ldots, k_\mu\}$ index the lower and upper bounds, respectively. Since all of the bounds are valid, the identified set for θ takes the form of intersection bounds

$$\Theta(\mu) = \Big[\max_{j \in \mathcal{J}_\ell} \mu_j,\ \min_{j \in \mathcal{J}_u} \mu_j\Big].$$

In this case, (1) holds with

$$K \equiv \mathbf{1} \in \mathbb{R}^{k_\mu}, \quad l \equiv 1, \quad \text{and} \quad \Delta = \Big\{\delta : \max_{j \in \mathcal{J}_\ell} \delta_j \leq 0 \leq \min_{j \in \mathcal{J}_u} \delta_j\Big\}.$$

The nuisance parameter $\delta_j = \mu_j - \theta$ represents the slackness of the jth bound. Intersection bounds appear in a wide variety of empirical settings, ranging from the effect of arrest on domestic violence recidivism (Siddique, 2013) to the trade-off between distance and price in hospital referrals (Ho and Pakes, 2014); see Chernozhukov et al. (2013) for additional applications. □

[1] Alternatively, the researcher may exogenously determine (rather than estimate) what the case shares would be under the quota policy. Kolesár et al. (2026, Appendix B.2) implicitly assume that case shares are known when conducting inference.

[2] Chernozhukov et al. (2013) allow the collection of bounds to be uncountably infinite, which necessitates a different technical approach from the one I develop for the finite case.

Example 2.4 (Union Bounds)**.** Bei (2025) studies inference on θ when its identified set takes the form of union bounds, meaning

$$\Theta(\mu) = \left[\min_{j \in \mathcal{J}_\ell} \mu_j,\ \max_{j \in \mathcal{J}_u} \mu_j\right]$$

where $\mathcal{J}_\ell$ and $\mathcal{J}_u$ are defined as in Example 2.3.[3] The researcher knows that at least one of the candidate lower bounds is valid, and similarly for the upper bounds. In this case, (1) holds with

$$K \equiv \mathbf{1} \in \mathbb{R}^{k_\mu}, \quad l \equiv 1, \quad \text{and} \quad \Delta = \left\{\delta : \min_{j \in \mathcal{J}_\ell} \delta_j \leq 0 \leq \max_{j \in \mathcal{J}_u} \delta_j\right\}.$$

As in Example 2.3, the nuisance parameter $\delta_j = \mu_j - \theta$ represents the slackness of the jth bound. Union bounds often arise when the researcher wishes to relax a canonical assumption that ensures point identification, but that may be implausible or inconsistent with the data. Intuitively, the idea is to calibrate the degree of relaxation based on the largest magnitude of several observed specification errors. For instance, Masten and Poirier (2021) take this approach to relax exclusion and exogeneity in instrumental variable models. Several of the specifications of Δ in Rambachan and Roth (2023) result in union bounds as well, such as when post-period parellel trends violations are bounded in magnitude by the largest observed pre-period violation. Finally, Kolesár and Rothe (2018) apply the same reasoning to relax functional form extrapolation in regression discontinuity designs; see Bei (2025) for additional applications. □

2.3 Experiment

I consider a statistical experiment in which the researcher observes a noisy estimate X of the moments μ. In particular, I suppose that X is normally distributed around μ with known covariance matrix Σ,

$$X \sim \mathcal{N}(\mu, \Sigma) \equiv P_\mu. \tag{2}$$

This is referred to as a Gaussian shift experiment, which acts as the starting point for an extensive literature on optimal inference. The experiment in (2) is motivated by the central limit theorem, and captures the heart of the inference problem while abstracting from asymptotic considerations. In Section C, I formalize how (2) arises under local asymptotics, and define Σ as the asymptotic covariance matrix of an efficient moment estimator. I establish that under suitable regularity conditions, optimality in (2) translates to local asymptotic optimality of a feasible confidence procedure. In Section D, I provide additional conditions under which the feasible procedure also satisfies a uniform asymptotic coverage requirement.

My proposed econometric framework dates back to the approximately linear model of Sacks and Ylvisaker (1978), who characterize minimax mean squared error estimation of θ when Σ is diagonal and Δ is rectangular. This model was subsequently generalized by Donoho (1994) and Armstrong and Kolesár (2018, 2021), who establish near minimax expected length optimality of a computationally tractable confidence procedure for θ when Δ is convex and centrosymmetric ($\delta \in \Delta$ implies $-\delta \in \Delta$). A key advantage of my approach is that it places only a bare measurability requirement on Δ. In all of the examples from Section 2.2, Δ is either non-convex or non-centrosymmetric. In Example 2.1, Rambachan and Roth (2023) consider Δ given by finite

[3] See Section A.4 for the somewhat more general setup of Bei (2025), and how it maps into (1).

unions of polyhedra: only one of the nine specifications they propose is both convex and centrosymmetric. In Example 2.2, Δ is non-convex due to the unknown ordering of propensity scores. In the intersection bounds of Example 2.3, Δ is convex but non-centrosymmetric, since the sign of each slack parameter δ_j is known. Finally, in the union bounds of Example 2.4, Δ is centrosymmetric when $\mathcal{J}_\ell = \mathcal{J}_u$, but non-convex as a union of cones.

2.4 Assumptions

Before placing assumptions on the primitives defined in Section 2, it will be helpful to introduce a few pieces of terminology that frequently appear when discussing identification and inference. First, the *null parameter space* under $H_0 : \theta = t$ refers to the set of moments consistent with the null hypothesis and (1),

$$\mathcal{M}(t) \equiv \big\{\mu \in \mathbb{R}^{k_\mu} : \mu = K\tau + \delta \text{ and } t = l'\tau \text{ for some } (\tau, \delta) \in \mathbb{R}^{k_\tau} \times \Delta\big\}.$$

A given vector of moments is consistent with (1) if and only if it lies in $\mathcal{M}(t)$ for some $t \in \mathbb{R}$. The set of moments consistent with (1), which I refer to as the *parameter space*, is therefore given by

$$\begin{aligned}\mathcal{M} \equiv \bigcup_{t\in\mathbb{R}} \mathcal{M}(t) &= \big\{\mu \in \mathbb{R}^{k_\mu} : \mu = K\tau + \delta \text{ for some } (\tau, \delta) \in \mathbb{R}^{k_\tau} \times \Delta\big\} \\ &= \text{range}(K) + \Delta,\end{aligned}$$

where $\text{range}(K) = \{K\tau : \tau \in \mathbb{R}^{k_\tau}\}$. Whenever I add two sets, as above, this denotes their Minkowski sum: that is, $A + B = \{a + b : (a, b) \in A \times B\}$. Finally, for $\mu \in \mathcal{M}$, the *identified set* for θ under P_μ refers to the values of θ that are consistent with the distribution of observables, $\Theta(\mu) \equiv \big\{t \in \mathbb{R} : \mu \in \mathcal{M}(t)\big\}$.

The first assumption holds by convention in order to simplify the expression for $\mathcal{M}(t)$.

Assumption 2.1 (Normalization)**.** $l \in \mathbb{R}^{k_\tau}$ is a nonzero vector normalized so that $\|l\|_2 = 1$.

In Section A.6, I verify that any optimal confidence procedure will remain so under a change of units, meaning the normalization in Assumption 2.1 is without loss of generality since l is known. Under Assumption 2.1, $t = l'\tau$ if and only if $\tau = tl + m$ for some $m \in \ker(l')$, where $\ker(l') = \{m \in \mathbb{R}^{k_\tau} : l'm = 0\}$. The null parameter space may therefore be expressed as

$$\begin{aligned}\mathcal{M}(t) &= K(tl + \ker(l')) + \Delta \\ &= tKl + K\ker(l') + \Delta \\ &= tKl + \mathcal{M}(0)\end{aligned} \tag{3}$$

for $\mathcal{M}(0) = K\ker(l') + \Delta$. That is, increasing t by one unit simply translates the null parameter space by the vector Kl. This translation equivariance property will turn out to be essential for the tractability of the optimal inference problem developed in Section 3. It also implies that the identified set for θ may be written as

$$\Theta(\mu) = \big\{t \in \mathbb{R} : \mu - tKl \in \mathcal{M}(0)\big\},$$

so the identification content of the researcher's model derives from the geometry of the set $\mathcal{M}(0)$.

In certain models, some directions in moment space may be uninformative about the target parameter. These directions form a linear subspace, which I refer to as the *lineality space* of $\mathcal{M}(0)$,

$$V \equiv \{v \in \mathbb{R}^{k_\mu} : tv + \mathcal{M}(0) = \mathcal{M}(0) \text{ for all } t \in \mathbb{R}\}. \tag{4}$$

It is apparent from the expression for $\Theta(\mu)$ that if $Kl \in V$, then $\Theta(\mu) = \mathbb{R}$ for all $\mu \in \mathcal{M}$. In Section A.6, I show that the reverse direction also holds, so it is necessary and sufficient to have $Kl \notin V$ in order for the model to be informative about the target parameter. The second assumption simply states this basic identification requirement.

Assumption 2.2 (Informativeness)**.** $Kl \notin V$.

Since K, l, and Δ are all known, the researcher may derive V and check whether Assumption 2.2 holds in any particular application; if not, the problem of identification and inference is trivial.

Examples 2.1 to 2.4 showcase the diversity of nuisance parameter sets that may arise in practice. However, from a technical perspective, allowing Δ to be an arbitrary set would cause measurability problems. The third assumption places a mild regularity condition on Δ in order to rule out such pathologies.

Assumption 2.3 (Nuisance Parameter Set)**.** $\Delta \subseteq \mathbb{R}^{k_\mu}$ is non-empty and σ-compact.

A σ-compact set is one that can be written as a countable union of compact sets. Assumption 2.3 is therefore a minimal technical requirement that can always be met if the researcher were to replace Δ with its closure, since closed sets in $\mathbb{R}^{k_\mu}$ are σ-compact. By Lemma B.1, Assumption 2.3 guarantees that the identified set is measurable.

The fourth and final assumption ensures that the sampling uncertainty in X is not implicitly lower-dimensional than μ, meaning no linear combination of the moments is known to the researcher.

Assumption 2.4 (Positive Definiteness)**.** The covariance matrix Σ is positive definite, $\Sigma \succ 0$.

Lemma B.2 considers the case in which Σ is only positive semi-definite, and shows that Assumption 2.4 still holds under a suitable reparameterization of the model so long as $\text{range}(K) \subseteq \text{range}(\Sigma)$. This technique closely resembles how perfect multicolinearity is handled in linear regression models, as pointed out by Cox and Shi (2023) in the context of testing moment inequalities. To see how Assumptions 2.1 to 2.4 can be verified in practice, it will be useful to revisit the empirical setting of Example 2.1.

Example 2.1 (Relaxing Parallel Trends, Continued)**.** Consider a four-period DiD design with two pre-periods and two post-periods. Recall that the DiD estimand μ_0 in the final pre-period is normalized to zero, and suppose that the target parameter is the average treatment effect on the treated τ_2 in the final post-period. Then (1) holds with

$$\begin{bmatrix} \mu_2 \\ \mu_1 \\ \mu_{-1} \end{bmatrix} = K \begin{bmatrix} \tau_2 \\ \tau_1 \end{bmatrix} + \begin{bmatrix} \delta_2 \\ \delta_1 \\ \delta_{-1} \end{bmatrix} \quad \text{for} \quad K \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{and} \quad \theta = l' \begin{bmatrix} \tau_2 \\ \tau_1 \end{bmatrix} \quad \text{for} \quad l \equiv \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

so the model satisfies Assumption 2.1. In one of several possible specifications of the nuisance parameter set Δ, Rambachan and Roth (2023) propose bounding the magnitude of the post-period parallel trends violations by the largest observed magnitude in the pre-period scaled by a factor $\bar{M} > 0$,

$$\Delta^{\text{RM}}(\bar{M}) \equiv \{(\delta_2, \delta_1, \delta_{-1})' : |\delta_1| \leq \bar{M}|\delta_{-1}|,\ |\delta_2 - \delta_1| \leq \bar{M}|\delta_{-1}|\}.$$

Since worst-case violations of magnitude $\bar{M}|\delta_{-1}| = \bar{M}|\mu_{-1}|$ can add up over the two post-periods, the identified set for θ is given by $\Theta(\mu) = \left[\mu_2 \pm 2\bar{M}|\mu_{-1}|\right]$. Alternatively, we may write

$$\Theta(\mu) = \left[\min\{\mu_2 - 2\bar{M}\mu_{-1}, \mu_2 + 2\bar{M}\mu_{-1}\}, \max\{\mu_2 + 2\bar{M}\mu_{-1}, \mu_2 - 2\bar{M}\mu_{-1}\}\right],$$

so the identified set is an instance of union bounds, as in Example 2.4.

By definition, $0 \in \Theta(\mu)$ if and only if $\mu \in \mathcal{M}(0)$, so the null parameter space under $H_0 : \theta = 0$ is given by

$$\mathcal{M}(0) = \{(\mu_2, \mu_1, \mu_{-1})' : |\mu_2| \leq 2\bar{M}|\mu_{-1}|\}.$$

The above expression makes clear that the event study coefficient in the first post-period μ_1 is uninformative about the target parameter. In Section A.6, I verify that this is in fact the only uninformative direction in moment space, so $V = \{(0, s, 0)' : s \in \mathbb{R}\}$. Consequently, $Kl = (1, 0, 0)' \notin V$, so Assumption 2.2 holds. Moreover, $\Delta^{\text{RM}}(\bar{M})$ is non-empty and closed, hence σ-compact, so Assumption 2.3 also holds. Finally, to verify Assumption 2.4, recall from Section 2.3 that Σ represents the asymptotic covariance matrix of an efficient moment estimator; see Section C for details. In the absence of additional parametric assumptions, the analog estimator is efficient, and Assumption 2.4 maintains that its asymptotic covariance matrix is positive definite. □

3 Statistical Decision Problem

I formalize what it means for a confidence procedure to be optimal by adopting the decision-theoretic framework of Wald (1950). Before observing the statistic X, the researcher would like to choose among confidence procedures satisfying a frequentist coverage requirement. A confidence procedure specifies what confidence set to report based on the statistic X and an independent uniform random variable U, which allows for randomization. Upon reporting a confidence set, the researcher incurs a loss given by its excess length: the length of the part that lies outside of the identified set. A minimax optimal confidence procedure minimizes the risk, or expected loss, faced by the researcher under the least favorable DGP.

3.1 Action Space and Confidence Procedure

The researcher takes an action by reporting a confidence set, so the action space $\mathcal{A}$ is the collection of all confidence sets under consideration. To allow for randomization, define an auxiliary uniform random variable $U \sim \text{Unif}[0, 1]$ independent of X and denote their joint law by $P_{\mu,U}$. A (randomized) confidence procedure is

then a measurable map from the sample space to the action space,

$$S : \mathbb{R}^{k_\mu} \times [0,1] \to \mathcal{A},$$

letting $\mathcal{S}$ denote the set of all confidence procedures. To make this precise, I must specify the action space and say what it means for a collection of sets to be measurable. Formally, I take the action space to be the measure algebra: the collection $\mathcal{B}$ of Borel sets of $\mathbb{R}$ modulo null sets,

$$\mathcal{A} \equiv \mathcal{B}/\{A \in \mathcal{B} : \lambda(A) = 0\},$$

where $\lambda(\cdot)$ denotes Lebesgue measure, or length.[4] Joshi (1969) first recognized the necessity of viewing confidence sets as equivalent up to null sets in order for an admissible confidence procedure to exist. I adopt this formalism to arrive at a general action space that is nevertheless topologically well-behaved. In particular, the researcher need not report an interval unless it is optimal to do so.

3.2 Class of Confidence Procedures with Correct Coverage

I constrain the class of confidence procedures under consideration to satisfy a conventional frequentist $1-\alpha$ coverage requirement for some fixed $\alpha \in (0,1)$, as in Evans et al. (2005). Aside from this requirement, I leave the class entirely unrestricted. I emphasize that researchers may use decision theory to guide their choice of a confidence procedure within this class, without resorting to ad hoc restrictions for the sake of tractability.[5]

Defining coverage in the usual way is complicated by the fact that the indicator

$$(t,x,u) \mapsto \mathbb{1}\{t \in S(x,u)\}$$

is not well-defined on $\mathcal{A}$. This is because the sets $S(x,u)$ and $S(x,u) \setminus \{t\}$ represent the same element of $\mathcal{A}$, since $\lambda(\{t\}) = 0$. In Section A.7, I provide a rigorous definition of the above indicator and denote it by $I_S(t,x,u)$. Analogously, in lieu of writing $\Pr_{P_{\mu,U}}(t \in S(X,U)) = \mathbb{E}_{P_{\mu,U}}[\mathbb{1}\{t \in S(X,U)\}]$, let the *coverage probability function* of a confidence procedure $S \in \mathcal{S}$ under $P_{\mu,U}$ be given by

$$C(t,\mu,S) \equiv \mathbb{E}_{P_{\mu,U}}[I_S(t,X,U)].$$

For each $\mu \in \mathcal{M}$ and $S \in \mathcal{S}$, this function is determined up to almost everywhere equivalence in t by Lemma B.1.

[4] Unlike $\mathcal{B}$, the measure algebra is endowed with a natural topology under which it is a Polish space (Le Maître, 2026, Section 2.1), so I equip $\mathcal{A}$ with the corresponding Borel σ-algebra. By Pfanzagl (1994, Theorem 1.10.33), the fact that $\mathcal{A}$ is a Polish space licenses the use of an auxiliary randomization device in the first place, rather than adopting the more cumbersome definition of confidence procedures as Markov kernels.

[5] In particular, I do not seek to provide a decision-theoretic justification for the usual frequentist notion of coverage. See Manski (2021) for a discussion of the (de)merits of frequentist inference from a decision-theoretic perspective.

Following Imbens and Manski (2004), the $1-\alpha$ coverage requirement for a partially identified parameter θ is typically represented in the canonical form

$$\inf_{\mu\in\mathcal{M}}\inf_{t\in\Theta(\mu)} C(t,\mu,S)\geq 1-\alpha.$$

This makes clear that under any DGP consistent with the model ($\mu\in\mathcal{M}$), the confidence procedure S must cover each value of θ in the identified set ($t\in\Theta(\mu)$) with probability at least $1-\alpha$. On the other hand, swapping the order of the infima emphasizes the connection to test inversion,

$$\inf_{t\in\mathbb{R}}\inf_{\mu\in\mathcal{M}(t)} C(t,\mu,S)\geq 1-\alpha.$$

For each $t\in\mathbb{R}$, rearranging terms reveals that $\sup_{\mu\in\mathcal{M}(t)}(1-C(t,\mu,S))\leq\alpha$. In other words, the probability of rejecting $H_0:\theta=t$ (that is, not covering t) must be no greater than α under any DGP consistent with the null. By the definition of $C(t,\mu,S)$, this can only be verified for almost every $t\in\mathbb{R}$, so I replace the outer infimum over $t\in\mathbb{R}$ with an essential infimum as in Evans et al. (2005). The class of confidence procedures with correct coverage is therefore given by

$$\mathcal{S}_\alpha\equiv\{S\in\mathcal{S}:\operatorname*{ess\,inf}_{t\in\mathbb{R}}\inf_{\mu\in\mathcal{M}(t)} C(t,\mu,S)\geq 1-\alpha\}.$$

Note that both the essential infimum and the indicator I_S are merely technical devices needed to ensure that $\mathcal{S}_\alpha$ is a well-defined, without affecting the interpretation of the coverage requirement in practice.[6]

3.3 Loss and Risk Functions

There is a long tradition in statistics of seeking the shortest possible confidence set subject to a frequentist coverage requirement (Neyman, 1937; Wilks, 1938). However, classical results in this literature apply almost exclusively to point identified models.[7] Consequently, there is no consensus as to whether length is a satisfactory notion of loss under partial identification, and if not, what should be used instead. I propose using excess length: the length of a confidence set excluding the part that overlaps with the identified set. Unlike length loss, I find that excess length loss extends the renowned Ghosh-Pratt identity to partially identified models, and is therefore compatible with the Neyman–Pearson approach of inverting a family of tests that maximize power subject to a size constraint.

Let the excess length loss function $L:\mathcal{M}\times\mathcal{A}\to\overline{\mathbb{R}}_+$ be given by

$$L(\mu,A)\equiv\lambda(A\setminus\Theta(\mu)),$$

so the risk of a confidence procedure $S\in\mathcal{S}$ under $P_{\mu,U}$ is given by its expected excess length,

$$R(\mu,S)\equiv\mathbb{E}_{P_{\mu,U}}[L(\mu,S(X,U))].$$

[6] Formally, Part (v) of Lemma B.1 shows that $S\in\mathcal{S}_\alpha$ if and only if it admits a measurable representative $\overline{S}$ satisfying the usual Imbens and Manski (2004) coverage requirement. In practice, all of the representatives I construct possess this property.

[7] A notable exception is the strand of the literature initiated by Donoho (1994); see Section 4.4 for further discussion.

The above definition of risk has already appeared as a measure of performance in the literature: for instance, Rambachan and Roth (2023) use it to benchmark their proposed confidence procedures relative to an expression for oracle risk. Other authors have considered alternative loss functions such as length $\lambda(A)$ (Armstrong and Kolesár, 2021) and net length $\lambda(A) - \lambda(\Theta(\mu))$ (Stoye, 2020). To see why excess length is a particularly natural notion of loss from a frequentist point of view, notice that risk may be expressed in terms of the coverage probability function,

$$\begin{aligned} R(\mu, S) &= \mathbb{E}_{P_{\mu,U}}[\lambda(S(X,U) \setminus \Theta(\mu))] \\ &= \mathbb{E}_{P_{\mu,U}}[\lambda(S(X,U) \cap (\mathbb{R} \setminus \Theta(\mu)))] \\ &= \mathbb{E}_{P_{\mu,U}}\left[\int_{\mathbb{R}\setminus\Theta(\mu)} I_S(t,X,U)\,\mathrm{d}t\right] \\ &= \int_{\mathbb{R}\setminus\Theta(\mu)} C(t,\mu,S)\,\mathrm{d}t, \end{aligned} \tag{5}$$

where the last equality follows from Tonelli's theorem. In words, the expected excess length of a confidence procedure is equal to the integrated probability of false coverage. This is precisely the Ghosh-Pratt identity (Ghosh, 1961; Pratt, 1961), which I extend to allow for partial identification.

Unlike excess length loss, under length or net length loss the domain of integration in the analog of (5) would be the entire real line, meaning the integral would also include true coverage probabilities. In terms of test inversion, this is discordant with the Neyman–Pearson approach of maximizing power subject to a size constraint, since it implies that the researcher would also like to maximize size per se. As shown in Section A.1, this means that when the identified set is an interval that is known to be long relative to sampling error, the researcher will find it optimal to poke holes in their reported confidence set to decrease its expected length. Excess length loss does not produce such strange behavior, since the researcher does not consider it costly to report values of θ that are consistent with their model.

3.4 Aggregating Risk

In order to determine whether one confidence procedure is preferable to another, the researcher must specify how to aggregate risk across the parameter space. I allow for three widely used decision criteria: minimax risk, minimax regret, and Bayes risk. To fix ideas, I focus on minimax risk as my main specification, and defer discussion of minimax regret and Bayes risk to Section A.8. A confidence procedure $S^\star \in \mathcal{S}_\alpha$ is minimax expected excess length optimal if it solves

$$\inf_{S\in\mathcal{S}_\alpha} \sup_{\mu\in\mathcal{M}} R(\mu, S). \tag{6}$$

The particular functional form of the researcher's model has no bearing on the above definition of optimality, which applies to confidence procedures for any scalar parameter, partially identified or otherwise, under the experiment in (2). However, decision problem (6) is difficult to operationalize in its current form: the choice variable is a function that outputs a set based on (X, U), subject only to basic measurability and

coverage requirements. In the following section, I take advantage of the structure in (1) to make decision problem (6) tractable.

4 Reduction to an Optimal Testing Problem

The predominant method of forming confidence sets that satisfy the Imbens and Manski (2004) coverage requirement is via test inversion (Canay and Shaikh, 2017). Under this approach, the researcher conducts a level-α test of $H_0 : \theta = t$ for each $t \in \mathbb{R}$. They then form a $1 - \alpha$ confidence set for θ by collecting all of the candidate values t that cannot be rejected. In general, the performance of such a confidence procedure depends on the joint behavior of the entire family of tests across $t \in \mathbb{R}$, so constructing an optimal procedure remains intractable in the absence of additional structure.

However, by leveraging the translation symmetries in (3) and (4), along with the Ghosh–Pratt identity in (5), Theorem 4.1 establishes that decision problem (6) admits a solution obtained by inverting a single optimal test. While the test statistic depends on the value of t, the acceptance region of the test does not, meaning the optimal testing problem must only be solved once. By relating the power properties of the test to the risk properties of the corresponding confidence procedure, I connect two burgeoning but as yet separate literatures on optimal testing and optimal confidence procedures under partial identification.

4.1 Setting Up the Optimal Testing Problem

Consider the problem of constructing a level-α test of $H_0 : \theta = t$ for some posited value $t \in \mathbb{R}$ of the target parameter. Under the framework from Section 2, this is equivalent to testing whether the moments μ are consistent with $\theta = t$ under the researcher's model, $H_0 : \mu \in \mathcal{M}(t)$. The structure in (1) then enables two reductions of the testing problem. First, by the translation equivariance property in (3), we know that

$$\mu \in \mathcal{M}(t) \iff \mu - tKl \in \mathcal{M}(0),$$

meaning it suffices to test whether the recentered moments $\mu - tKl$ are consistent with $\theta = 0$. Second, the definition of V in (4) suggests that we may reduce the dimensionality of the problem by projecting out uninformative directions. In particular, let Q denote a matrix whose rows form a basis of $V^{\perp}$, the orthogonal complement of V. Then by Part (i) of Lemma B.3,

$$\mu - tKl \in \mathcal{M}(0) \iff Q(\mu - tKl) \in Q\mathcal{M}(0).$$

So, instead of testing $H_0 : \mu \in \mathcal{M}(t)$ based on X, we may define the *reduced moments*

$$\nu(t) \equiv Q(\mu - tKl)$$

and test $H_0 : \nu(t) \in Q\mathcal{M}(0)$ based on the analogous *reduced statistic*

$$Y \equiv Q(X - tKl) \sim \mathcal{N}(\nu(t), Q\Sigma Q') \equiv P_{\nu(t)}.$$

The dimension of both quantities is given by the number of informative directions in moment space, $k_\nu \equiv \dim(V^\perp) = k_\mu - \dim(V)$.

I characterize randomized tests in the reduced problem by their critical functions φ, where $\varphi(y)$ represents the rejection probability upon observing $Y = y$. Accordingly, let

$$\Phi \equiv \left\{\varphi \in L^\infty(\mathbb{R}^{k_\nu}) : 0 \leq \varphi \leq 1\right\} \quad \text{and} \quad \Phi_\alpha \equiv \left\{\varphi \in \Phi : \sup_{\nu \in Q\mathcal{M}(0)} \mathbb{E}_{P_\nu}[\varphi(Y)] \leq \alpha\right\}$$

denote the set of tests and level-α tests, respectively.

Consider simplifying decision problem (6) by restricting attention to test inversion confidence procedures based on $\varphi \in \Phi_\alpha$. For a given $\varphi \in \Phi_\alpha$, the corresponding procedure S_φ takes the form

$$S_\varphi(x, u) \equiv \{t \in \mathbb{R} : u \leq 1 - \varphi(Q(x - tKl))\}.$$

Hence, the simplified decision problem may be reframed as an optimal testing problem by taking the outer infimum over tests, $\inf_{\varphi \in \Phi_\alpha} \sup_{\mu \in \mathcal{M}} R(\mu, S_\varphi)$. To better understand the researcher's implied preferences over tests, it will be instructive to unpack the definition of risk by applying the Ghosh–Pratt identity in (5),

$$\begin{aligned} R(\mu, S_\varphi) &= \int_{\mathbb{R} \setminus \Theta(\mu)} C(t, \mu, S_\varphi)\, \mathrm{d}t \\ &= \int \mathbb{1}\{\mu \notin \mathcal{M}(t)\} \mathbb{E}_{P_\mu}[1 - \varphi(Q(X - tKl))]\, \mathrm{d}t \\ &= \int \mathbb{1}\{\nu(t) \notin Q\mathcal{M}(0)\} \mathbb{E}_{P_{\nu(t)}}[1 - \varphi(Y)]\, \mathrm{d}t. \end{aligned}$$

The second equality expresses coverage probabilities as acceptance probabilities of the test, while the third equality adopts notation from the reduced testing problem. In prose, the risk of a test is equal to the integrated type II error probability: the probability of accepting the null hypothesis, $\mathbb{E}_{P_{\nu(t)}}[1 - \varphi(Y)]$, when it is false, $\nu(t) \notin Q\mathcal{M}(0)$.

To further simplify the expression for risk, notice that only the direction QKl in reduced moment space depends on the value of t. By Part (iii) of Lemma B.3, we may choose Q so that $QKl = e_1$, the first basis vector in $\mathbb{R}^{k_\nu}$. Under this convention, write $Q = \left[\begin{smallmatrix} Q_1 \\ Q_{-1} \end{smallmatrix}\right]$, where Q_1 denotes the first row and Q_{-1} denotes the remaining rows, so that $\nu = \left[\begin{smallmatrix} \nu_1 \\ \nu_{-1} \end{smallmatrix}\right] = \left[\begin{smallmatrix} Q_1\mu - t \\ Q_{-1}\mu \end{smallmatrix}\right]$, suppressing dependence on t from the notation. The change of variables $\nu_1 = Q_1\mu - t$ reveals that $\nu_{-1} = Q_{-1}\mu$ determines the risk of a test,

$$R(\mu, S_\varphi) = \int \mathbb{1}\{\nu \notin Q\mathcal{M}(0)\} \mathbb{E}_{P_\nu}[1 - \varphi(Y)]\, \mathrm{d}\nu_1 \equiv R(\nu_{-1}, \varphi).$$

The simplified decision problem may therefore be expressed as the following optimal testing problem,

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi). \tag{7}$$

To fix ideas, it will be helpful to derive the primitives of (7) in the context of Example 2.1.

Example 2.1 (Relaxing Parallel Trends, Continued)**.** Recall the four-period DiD design from Example 2.1 in which $\theta = \tau_2$ and

$$\begin{aligned}
\Delta^{\mathrm{RM}}(\bar{M}) &= \{(\delta_2, \delta_1, \delta_{-1})' : |\delta_1| \leq \bar{M}|\delta_{-1}|,\ |\delta_2 - \delta_1| \leq \bar{M}|\delta_{-1}|\}, \\
\mathcal{M}(0) &= \{(\mu_2, \mu_1, \mu_{-1})' : |\mu_2| \leq 2\bar{M}|\mu_{-1}|\}, \\
V &= \{(0, s, 0)' : s \in \mathbb{R}\}.
\end{aligned}$$

The matrix Q must be constructed so that its rows form a basis of $V^\perp$ with $QKl = (1,0)'$. Since $Kl = (1,0,0)'$, simply set the rows of Q to be the first and third basis vectors,

$$Q \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The reduced moments are then given by $\nu(t) = (\mu_2 - t, \mu_{-1})'$, and

$$Q\mathcal{M}(0) = \{(\nu_1, \nu_{-1})' \in \mathbb{R}^2 : |\nu_1| \leq 2\bar{M}|\nu_{-1}|\}.$$

The model does not place any a priori restrictions on the moments, so $\mathcal{M} = \mathbb{R}^3$ and $Q_{-1}\mathcal{M} = \mathbb{R}$. Risk depends on $\nu_{-1} = \mu_{-1}$, the pre-period parallel trends violation. □

4.2 Inverting an Optimal Test yields an Optimal Confidence Procedure

Theorem 4.1 establishes that the proposed simplification from Section 4.1 is in fact without loss. In other words, inverting a solution to the optimal testing problem (7) solves decision problem (6).

Theorem 4.1. *Under Assumptions 2.1 to 2.4, there exists a solution to the optimal testing problem (7),*

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi).$$

Moreover, if $\varphi^\star$ solves (7), then the test inversion confidence procedure $S_{\varphi^\star}$ has minimax expected excess length, meaning it solves decision problem (6),

$$\inf_{S \in \mathcal{S}_\alpha} \sup_{\mu \in \mathcal{M}} R(\mu, S).$$

Finally, the two problems share the same value, that is, $\sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi^\star) = \sup_{\mu \in \mathcal{M}} R(\mu, S_{\varphi^\star})$.

Theorem 4.1 justifies the conventional wisdom that inverting a test with good power properties yields a confidence procedure with good length properties. In particular, Theorem 4.1 states that inverting a test with minimax integrated type II error probability yields a confidence procedure with minimax expected excess length. In Section A.8, I show that an analogous result holds under length loss, but the integral that defines risk in the optimal testing problem also includes type I error probabilities. This supports the claim from Section 3.3 that excess length loss is more closely compatible with the Neyman–Pearson approach of maximizing power (i.e. minimizing type II error probabilities) subject to a size constraint. In addition, Theorem 4.1 easily generalizes to accommodate minimax regret, as well as certain forms of Bayes risk; see Section A.8 for details.

Theorem 4.1 contributes to a long line of work in statistical decision theory on generalizations of the Hunt–Stein theorem (Bondar and Milnes, 1981). The classical Hunt–Stein theorem provides conditions under which there exists a test with maximin power that is invariant to a group of transformations (Hunt and Stein, 1946). Analogously, Theorem 4.1 establishes that there exists a confidence procedure with minimax expected excess length that is equivariant to a group of translations: for $(s, v) \in \mathbb{R} \times V$, notice that $S_\varphi(x + sKl + v, u) = s + S_\varphi(x, u)$. Inverting tests of the form $\varphi \in \Phi_\alpha$ therefore provides an explicit method of constructing translation equivariant confidence procedures. Existing work has restricted attention to procedures with this property for the sake of simplicity (Li and Müller, 2021). Theorem 4.1 shows that doing so is without loss in terms of minimax expected excess length.

Reducing decision problem (6) to an optimal testing problem has several computational advantages. First, as described in Section 5, it allows me to leverage recent advances in computing nearly optimal tests of composite null hypotheses (Aradillas Fernández et al., 2025). Second, reducing the dimension of the problem by $\dim(V)$ helps mitigate the curse of dimensionality, which hinders numerical methods that rely on discretizing the null parameter space. Third, inverting a nearly optimal test amounts to evaluating the test at the reduced statistic $Y = Q(X - tKl)$ across candidate values t. By contrast, in much of the literature on test inversion under partial identification, critical values must be recomputed for each t, creating a computational bottleneck (Canay et al., 2023). Although approximating the optimal critical function $\varphi^\star$ may be computationally demanding, it only needs to be done once.

4.3 Relationship to Weighted Average Power Optimal Testing

Uniformly most powerful tests of $H_0 : \theta = t$ typically do not exist in partially identified models. This is because many vectors of moments may be consistent with the same value t of the target parameter, so both the null $H_0 : \mu \in \mathcal{M}(t)$ and the alternative hypothesis $H_1 : \mu \in \mathcal{M} \setminus \mathcal{M}(t)$ are composite. For instance, in moment inequality models, Romano et al. (2014) show that there does not exist a uniformly most powerful test of $H_0 : \mu \leq 0$ under the experiment in (2). Moreover, the classical approach of deriving uniformly most powerful tests subject to side conditions is not fruitful: in this case, the only unbiased test is trivial (Lehmann, 1952), and tests that are similar on the boundary have poor power (Andrews, 2012). So, when selecting a method of inference, the researcher must trade off power against different alternatives.

The weighted average power criterion makes this trade-off explicit. To construct such a criterion, the researcher must specify a set of non-negative weights over the alternative, directing power towards certain alternatives by giving them more weight. A weighted average power optimal test then maximizes this criterion subject to a size constraint. To see how this approach relates to the minimax optimal testing problem in Theorem 4.1, let Π denote the set of probability distributions over $Q_{-1}\mathcal{M}$, and denote Bayes risk under a prior $\pi \in \Pi$ by

$$\begin{aligned} R(\pi, \varphi) &\equiv \int R(\nu_{-1}, \varphi)\, \pi(\mathrm{d}\nu_{-1}) \\ &= \iint \mathbb{E}_{P_\nu}[1 - \varphi(Y)] \underbrace{\mathbb{1}\{\nu \notin Q\mathcal{M}(0)\}\mathrm{d}\nu_1\, \pi(\mathrm{d}\nu_{-1})}_{\equiv \omega_\pi(\mathrm{d}\nu)}. \end{aligned}$$

Then Bayes risk is a non-negatively weighted integral of one minus power, and a minimax optimal test $\varphi^\star$ minimizes Bayes risk under a *least favorable prior*, if one exists.[8]

Lemma 4.1 (Bayes-Minimax)**.** *The optimal testing problem (7) satisfies*

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi) = \inf_{\varphi \in \Phi_\alpha} \sup_{\pi \in \Pi} R(\pi, \varphi) = \sup_{\pi \in \Pi} \inf_{\varphi \in \Phi_\alpha} R(\pi, \varphi).$$

Suppose $\pi^\star$ is least favorable, meaning it attains the right-hand side supremum. Then if $\varphi^\star$ solves (7), it also solves $\inf_{\varphi \in \Phi_\alpha} R(\pi^\star, \varphi)$.

Although the formulation in Lemma 4.1 closely resembles a weighted average power optimal testing problem, the details differ. The problem is phrased in terms of minimizing one minus power, rather than maximizing power. The distinction between the two is not just cosmetic. The weighted integral of power $\int \mathbb{E}_{P_\nu}[\varphi(Y)]\omega_{\pi^\star}(\mathrm{d}\nu)$ is typically infinite, since the weights themselves are improper. Ultimately, the researcher cares about expected excess length, which implies that the weights must be flat along the first coordinate, hence improper. In addition, the weights are endogenously determined, rather than exogenously set by the researcher as is commonly assumed (Elliott et al., 2015). The least favorable prior $\pi^\star$ derives from the minimax decision criterion; a different decision criterion, such as minimax regret, would induce a different prior. In this way, the researcher's preferences over confidence procedures determine how they should optimally allocate power.

Lemma 4.1 has an insightful implication for the form of the alternative hypothesis, and how it relates to model misspecification. Since the least favorable prior $\pi^\star$ is supported on $Q_{-1}\mathcal{M}$ and $Q\mathcal{M} = \mathbb{R} \times Q_{-1}\mathcal{M}$, the weights $\omega_{\pi^\star}$ are supported on the alternative $Q\mathcal{M} \setminus Q\mathcal{M}(0)$. In other words, a minimax optimal test never directs power towards alternatives that are inconsitent with the model, $\nu \notin Q\mathcal{M}$; the same is true under minimax regret or Bayes risk, which may alter the prior but not its support. Tests based on conventional statistics need not share this property, since the alterative is usually taken to be unrestricted (Canay and Shaikh, 2017, §3). Consequently, existing test inversion confidence procedures may inadvertently act as specification tests: if the confidence set is empty, the model is rejected by the data. There has been some

[8] As is typical of minimax decision problems, a least favorable prior may not exist, in which case the claim still holds in a limiting sense.

debate in the literature as to whether such behavior is desirable or not. Several authors have advocated against mixing parameter inference together with specification testing, either for conceptual reasons (Hahn and Ridder, 2015), or due to misspecification concerns (Stoye, 2020; Andrews and Kwon, 2024). Lemma 4.1 supports the same conclusion by a different line of reasoning that has not yet been articulated in the literature: under correct specification, an optimal confidence procedure does not implement a specification test. Intuitively, this is because specification testing wastes power that could be used to improve performance instead.

4.4 Relationship to Nearly Optimal Fixed Length Confidence Procedures

An influential line of work dating back to Donoho (1994) studies the performance of affine fixed-length confidence intervals (FLCIs). An affine FLCI takes the form

$$S(x,u) = \left[c'x + d - \chi, c'x + d + \chi\right]$$

for some constants $c \in \mathbb{R}^{k_\mu}$, $d \in \mathbb{R}$, and $\chi \in \overline{\mathbb{R}}_+$. Thus, affine FLCIs resemble conventional two-sided confidence intervals, with $c'X + d$ acting as an affine estimator of θ. For any choice of coefficients c and d, the radius $\chi(c,d)$ can be chosen no larger than necessary in order to satisfy the coverage requirement. Since length is then deterministic and equal to $2\chi(c,d)$, the problem of constructing a length-optimal affine FLCI boils down to choosing c and d so that $\chi(c,d)$ is as small as possible. This amounts to trading off the estimator's variance against its worst-case bias, which can be achieved via convex programming (Low, 1995). In addition to being computationally tractable, optimal affine FLCIs are known to be nearly optimal relative to broader classes of confidence procedures in certain special cases. In seminal work on this subject, Donoho (1994) showed that an FLCI centered around any estimator cannot substantially improve upon the length of an optimal affine FLCI when Δ is convex. In subsequent work, Armstrong and Kolesár (2018, 2021) show that optimal affine FLCIs in fact have nearly minimax expected length among all $1-\alpha$ confidence procedures when Δ is convex and centrosymmetric.

Theorem 4.1 both sharpens and extends existing optimality theory for affine FLCIs, at the cost of additional computational complexity. When Δ is convex and centrosymmetric, the efficiency bound developed by Armstrong and Kolesár (2018) works by comparing the optimal affine FLCI length to the smallest possible expected length under P_μ achievable by any $1-\alpha$ confidence procedure, where μ satisfies $Q_{-1}\mu = \nu_{-1} = 0$. This quantity clearly provides a lower bound for the value of minimax expected length, but it is not known how tight this lower bound is in practice. That is, the optimal affine FLCI may in fact be much closer to minimax expected length optimal than is guaranteed by existing results, where applicable. By modifying Theorem 4.1 to use length loss as in Section A.8, I am able to quantify this slack numerically.

Moreover, as demonstrated by Examples 2.1 to 2.4, in many empirical settings Δ is either non-convex or non-centrosymmetric, so efficiency bounds are either unavailable or uninformative. Indeed, whenever the identified set may be arbitrarily long, any FLCI with correct coverage must have infinite length. In such cases, any non-trivial confidence procedure must adapt to the unknown length of the identified set; Theorem 4.1 shows how to do so optimally. The main drawback of minimax optimal test inversion relative to constructing an optimal affine FLCI is computational. While the program in (7) is convex, it imposes no parametric

structure on the critical function, and is therefore infinite dimensional. In the following section, I outline my proposed approach to numerically approximate a minimax optimal test.

5 Numerical Approach

Computation is one of the primary impediments to the use of statistical decision theory in economic applications (Manski, 2021). While Theorem 4.1 greatly simplifies the problem of constructing an optimal confidence procedure, solving even a single minimax optimal testing problem remains a formidable task, especially in high-dimensional settings. Fortunately, in recent years rapid progress has been made in the development of computational techniques to approximate minimax decision rules (e.g. Guggenberger and Huang, 2026; Montiel Olea and Zubova, 2026). In particular, Aradillas Fernández et al. (2025) propose a numerical method designed to approximate weighted average power optimal tests, and provide theoretical guarantees for its performance. Building on the reformulation from Section 4.3, I adapt this method to my setting and demonstrate its desirable properties from both a theoretical and practical perspective.

5.1 The Dual Problem

The idea of applying duality theory to solve minimax optimal testing problems dates back to Krafft and Witting (1967). To motivate this approach, fix a prior $\pi \in \Pi$ and recall from Section 4.3 the problem of minimizing Bayes risk among level-α tests, $\inf_{\varphi \in \Phi_\alpha} R(\pi, \varphi)$. Given a test $\varphi \in \Phi$, it may be difficult to check whether $\varphi \in \Phi_\alpha$. Indeed, establishing size control in a Gaussian shift experiment is the central challenge faced by much of the testing literature (Cox and Shi, 2023; Andrews et al., 2023). To avoid searching directly over the space Φ_α, I write down the Lagrangian and take the supremum over Lagrange multipliers,

$$\inf_{\varphi \in \Phi} \sup_{\Gamma \in \mathcal{G}} R(\pi, \varphi) + \int \big(\mathbb{E}_{P_\nu}[\varphi(Y)] - \alpha\big)\, \Gamma(\mathrm{d}\nu).$$

Since the size constraint is over $Q\mathcal{M}(0)$, the Lagrange multipliers Γ are finite non-negative Borel measures on $Q\mathcal{M}(0)$, the set of which I denote by $\mathcal{G}$. By construction, the inner supremum is equal to $R(\pi, \varphi)$ whenever $\varphi \in \Phi_\alpha$, and equal to ∞ otherwise. What makes this useful is that we may swap the infimum and supremum without changing the value of the problem, just as in Lemma 4.1. Now, consider the infimum of the Lagrangian over tests $\varphi \in \Phi$, taking both π and Γ as given,

$$\inf_{\varphi \in \Phi} R(\pi, \varphi) + \int \big(\mathbb{E}_{P_\nu}[\varphi(Y)] - \alpha\big)\, \Gamma(\mathrm{d}\nu).$$

The primary advantage of the dual formulation is that the above problem admits a closed-form solution, by the same reasoning as in the proof of the Neyman–Pearson lemma (Lehmann and Romano, 2022, §3.2).

To derive the solution, let p_ν denote the density of $P_\nu = \mathcal{N}(\nu, Q\Sigma Q')$, and for a measure w on $\mathbb{R}^{k_\nu}$ write $p_w(y) \equiv \int p_\nu(y)\, w(\mathrm{d}\nu)$ for the corresponding mixture of densities. Expressing Bayes risk as in Section 4.3 and

exchanging the order of integration, the Lagrangian may be written as

$$\begin{aligned} &\iint (1-\varphi(y))p_\nu(y)\,\mathrm{d}y\,\omega_\pi(\mathrm{d}\nu) + \int \Big(\int \varphi(y)p_\nu(y)\,\mathrm{d}y - \alpha\Big)\,\Gamma(\mathrm{d}\nu) \\ = &\int \big[(1-\varphi(y))p_{\omega_\pi}(y) + \varphi(y)p_\Gamma(y)\big]\,\mathrm{d}y - \alpha\Gamma(Q\mathcal{M}(0)). \end{aligned}$$

Hence, a test of the Neyman–Pearson form $\varphi_{\pi,\Gamma} \equiv \mathbb{1}\{p_{\omega_\pi} > p_\Gamma\}$ minimizes the Lagrangian, since it minimizes the above integrand pointwise. The intuition is that the Lagrange multipliers $\Gamma(\mathrm{d}\nu)$ represent shadow prices of the constraints $\mathbb{E}_{P_\nu}[\varphi(Y)] \leq \alpha$ in units of excess length. The Neyman–Pearson test rejects at $Y = y$ when the marginal benefit $p_{\omega_\pi}(y)\mathrm{d}y$ of the reduction in Bayes risk exceeds the marginal cost $p_\Gamma(y)\mathrm{d}y$ of the increase in type I error probabilities. The following lemma makes use of this fact in conjunction with Lemma 4.1 to establish strong duality and recover a minimax optimal test from the dual problem.

Lemma 5.1 (Duality)**.** *The optimal testing problem* (7) *shares its value with the dual problem*

$$\sup_{\pi\in\Pi}\sup_{\Gamma\in\mathcal{G}} R(\pi,\varphi_{\pi,\Gamma}) + \int \big(\mathbb{E}_{P_\nu}[\varphi_{\pi,\Gamma}(Y)] - \alpha\big)\,\Gamma(\mathrm{d}\nu), \tag{8}$$

where $\varphi_{\pi,\Gamma} \equiv \mathbb{1}\{p_{\omega_\pi} > p_\Gamma\}$*. Moreover, if this value is positive and* $(\pi^\star, \Gamma^\star)$ *solves* (8)*, then* $\varphi_{\pi^\star,\Gamma^\star}$ *is the unique solution to* (7)*.*

Lemma 5.1 is the analog of Krafft and Witting (1967, Theorem 4) under the minimax integrated type II error probability criterion of (7), as opposed to the classical maximin power criterion. Letting $\gamma^\star \equiv \Gamma^\star(Q\mathcal{M}(0))$, the probability distribution $\Gamma^\star/\gamma^\star$ is referred to as the *least favorable distribution* over the null. The problem of numerically approximating $\gamma^\star$ and $\Gamma^\star/\gamma^\star$ has been carefully studied under the weighted average power criterion (Elliott et al., 2015; Aradillas Fernández et al., 2025). Lemma 5.1 makes clear that solving (7) requires simultaneously approximating the least favorable prior $\pi^\star$.

Lemma 5.1 does not establish that a solution to the dual problem (8) exists. Just as in Lemma 4.1, the issue is that the least favorable configuration of the parameters could correspond to a limiting case if either $Q_{-1}\mathcal{M}$ or $Q\mathcal{M}(0)$ are unbounded. To approach the supremum in (8), then, the mass of π or Γ may escape to infinity, so the supremum is not attained. Since it is possible for many tests to achieve optimal performance in the limiting case, the solution to (7) need not be unique. In such a situation, minimax regret or Bayes risk might produce a more definitive answer. Otherwise, Lemma 5.1 shows that whenever the optimal testing problem (7) is non-trivial, its solution is unique and given by the Neyman–Pearson test associated with a solution to the dual problem (8).

5.2 Discretizing the Dual

The dual problem (8) maximizes over measures on $Q_{-1}\mathcal{M}$ and $Q\mathcal{M}(0)$, which are infinite-dimensional objects. For the sake of computational tractability, in practice I restrict their supports to fixed finite grids $\{\nu_{-1}^{(1)},\ldots,\nu_{-1}^{(L)}\} \subseteq Q_{-1}\mathcal{M}$ and $\{\nu^{(1)},\ldots,\nu^{(M)}\} \subseteq Q\mathcal{M}(0)$, following Elliott et al. (2015). Writing $\tilde{\pi} \in \mathbb{R}^L$ and

$\tilde{\Gamma} \in \mathbb{R}^M$ for the corresponding mass vectors, the discretized analog of (8) is

$$\sup_{\tilde{\pi},\tilde{\Gamma}} \sum_{l=1}^{L} \tilde{\pi}_l R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}) + \sum_{m=1}^{M} \tilde{\Gamma}_m \big(\mathbb{E}_{P_{\nu(m)}}[\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y)] - \alpha\big) \quad \text{s.t.} \quad (\tilde{\pi},\tilde{\Gamma}) \geq 0, \quad \mathbf{1}'\tilde{\pi} = 1. \tag{9}$$

Just as (8) is dual to the optimal testing problem (7), (9) is dual to a discretized testing problem in which risk is maximized over the risk grid and size is controlled over the null grid,

$$\inf_{\varphi \in \Phi} \max_{l \leq L} R(\nu_{-1}^{(l)}, \varphi) \quad \text{s.t.} \quad \max_{m \leq M} \mathbb{E}_{P_{\nu(m)}}[\varphi(Y)] \leq \alpha. \tag{10}$$

Unlike numerical approaches that discretize the primal problem directly (Chiburis, 2008), the critical functions in (10) remain infinite-dimensional. The following corollary establishes that (9) admits a solution, and transfers Lemma 5.1 to the discrete case.

Lemma 5.2 (Discretization). *There exists a solution $(\tilde{\pi}^\star, \tilde{\Gamma}^\star)$ to the discretized dual problem (9), which shares its value with (10). Moreover, if this value is positive, then $\varphi_{\tilde{\pi}^\star,\tilde{\Gamma}^\star}$ is the unique solution to (10).*

Lemma 5.2 leverages the fact that the optimal test must be of Neyman–Pearson form to solve the infinite-dimensional primal problem (10) by way of its $(L+M)$-dimensional dual. In addition, Lemma 5.2 demonstrates that the delicate issue of non-existence of a dual solution disappears once the dual is discretized, since the mass of $\tilde{\pi}$ and $\tilde{\Gamma}$ cannot escape to infinity by construction. This suggests that non-existence of a solution to (8) will manifest as mass concentrating at extremal grid points.

Unfortunately, formal results bounding the suboptimality and size distortions induced by discretization are scarce. In theory, size can be controlled between points in the null grid, since $\nu \mapsto \mathbb{E}_{P_\nu}[\varphi(Y)]$ is globally Lipschitz in ν with Lipschitz constant independent of φ (Chiburis, 2008). In certain settings, size can also be controlled outside of the null grid by "switching" to a test that is known to control size in a limiting case, taking care to account for any size distortions that may be induced by the switch itself (Elliott et al., 2015). Both of these strategies quickly become impractical as the dimension of the problem increases. For this reason, I pursue a numerical approach that has performance guarantees for the discretized problem and scales well with the number of grid points, following Aradillas Fernández et al. (2025). I then assess the magnitude of discretization error by conducting a sensitivity analysis on the grid specifications.

The discretized dual problem (9) is a concave program amenable to gradient ascent methods. The following proposition records this property.

Proposition 5.1 (Concave Program). *The objective of the discretized dual problem (9) is jointly concave in $(\tilde{\pi},\tilde{\Gamma})$. For some finite $\overline{\gamma} > 0$, it is without loss to impose the constraint $\mathbf{1}'\tilde{\Gamma} \leq \overline{\gamma}$, so the feasible set*

$$\{(\tilde{\pi},\tilde{\Gamma}) \in \mathbb{R}^{L+M} : (\tilde{\pi},\tilde{\Gamma}) \geq 0, \mathbf{1}'\tilde{\pi} = 1, \mathbf{1}'\tilde{\Gamma} \leq \overline{\gamma}\}$$

is convex and compact. Finally, the gradient of the objective at $(\tilde{\pi},\tilde{\Gamma}) > 0$ is given by

$$\big(R(\nu_{-1}^{(1)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}), \ldots, R(\nu_{-1}^{(L)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}), \mathbb{E}_{P_{\nu(1)}}[\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y)] - \alpha, \ldots, \mathbb{E}_{P_{\nu(M)}}[\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y)] - \alpha\big).$$

The fact that the objective is *jointly* concave in its arguments means that it is possible to simultaneously optimize over $\tilde{\pi}$ and $\tilde{\Gamma}$. In related work on the weighted average power criterion, Ketz et al. (2025) recommend alternating between optimizing $\tilde{\Gamma}$ with the weights fixed and updating the weights with $\tilde{\Gamma}$ fixed, which can be far more computationally demanding. Compactness of the feasible set is important from a theoretical perspective, as pointed out by Aradillas Fernández et al. (2025, §2.3); see the proof of Lemma 5.2 for an explicit construction of $\overline{\gamma}$. Finally, the expression for the gradient follows from the envelope theorem, since the Neyman–Pearson test $\varphi_{\tilde{\pi},\tilde{\Gamma}}$ minimizes the Lagrangian at $(\tilde{\pi},\tilde{\Gamma})$ and the objective of (9) is the Lagrangian evaluated at $\varphi_{\tilde{\pi},\tilde{\Gamma}}$.

To iteratively increase the objective of (9) starting from a feasible point, we must contend with the fact that moving in the direction of the gradient would violate the feasibility constraints. In addition, even for moderate grid sizes, it will be computationally prohibitive to repeatedly approximate the gradient in Proposition 5.1 to a high degree of precision. Aradillas Fernández et al. (2025) circumvent both of these difficulties by applying the method of *stochastic mirror ascent*.

5.3 Stochastic Mirror Ascent

Stochastic mirror ascent takes gradient steps based on noisy but unbiased gradient estimates, and measures how far one iterate is from the next according to a carefully chosen divergence. In particular, let $(\tilde{\pi}_t,\tilde{\Gamma}_t) > 0$ denote the current iterate, $(\hat{R}_t,\hat{\alpha}_t-\alpha)$ the noisy gradient estimate, and $D(\,\cdot\,,\,\cdot\,)$ the divergence. Then for step sizes $\eta_L,\eta_M > 0$, the next iterate maximizes a local linearization of the objective in (9) subject to feasibility constraints and a penalty for straying too far from the current iterate according to D,

$$(\tilde{\pi}_{t+1},\tilde{\Gamma}_{t+1}) = \arg\max_{\tilde{\pi},\tilde{\Gamma}} \hat{R}_t'\tilde{\pi} + (\hat{\alpha}_t-\alpha)'\tilde{\Gamma} - \frac{1}{\eta_L}D(\tilde{\pi},\tilde{\pi}_t) - \frac{1}{\eta_M}D(\tilde{\Gamma},\tilde{\Gamma}_t)$$
$$\text{s.t.} \quad (\tilde{\pi},\tilde{\Gamma}) \geq 0, \quad \mathbf{1}'\tilde{\pi} = 1, \quad \mathbf{1}'\tilde{\Gamma} \leq \overline{\gamma}.$$

Projected stochastic gradient ascent is a special case in which the divergence is given by squared Euclidean distance. Aradillas Fernández et al. (2025) use the generalized Kullback–Leibler divergence instead, which improves the rate at which their performance guarantees scale with the number of grid points.[9] Under this choice of divergence, the gradient updates correspond to exponential tilting,

$$\tilde{\pi}_{t+1} \propto \tilde{\pi}_t \odot \exp(\eta_L\hat{R}_t), \quad \tilde{\Gamma}_{t+1} \propto \tilde{\Gamma}_t \odot \exp(\eta_M(\hat{\alpha}_t-\alpha)),$$

where $\odot$ denotes the elementwise product, $\tilde{\pi}_{t+1}$ is normalized so that $\mathbf{1}'\tilde{\pi}_{t+1} = 1$, and $\tilde{\Gamma}_{t+1}$ is normalized so that $\mathbf{1}'\tilde{\Gamma}_{t+1} \leq \overline{\gamma}$, if necessary.

Unbiased gradient estimates can be cheaply computed via Monte Carlo simulations. Forming an unbiased estimate of the type I error rate $\mathbb{E}_{P_{\nu(m)}}[\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y)]$ is as simple as drawing $Y \sim P_{\nu(m)}$ and computing $\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y) = \mathbb{1}\{p_{\omega_{\tilde{\pi}}}(Y) > p_{\tilde{\Gamma}}(Y)\}$. Forming an unbiased estimate of the risk $R(\nu_{-1}^{(l)},\varphi_{\tilde{\pi},\tilde{\Gamma}})$ is less straightforward. In

[9] The generalized Kullback–Leibler divergence is the Bregman divergence associated with negative entropy. See the proof of Lemma 5.3 for additional details.

Section A.9, I derive and discuss an unbiased estimator $\hat{R}(Y_{-1}; \nu_{-1}^{(l)}, \tilde{\pi}, \tilde{\Gamma})$ based on $Y_{-1} \sim \mathcal{N}(\nu_{-1}^{(l)}, Q_{-1}\Sigma Q_{-1}')$. Computing $\hat{R}$ is inexpensive whenever the identified set is a bounded interval, and in particular does not involve inverting the test $\varphi_{\tilde{\pi},\tilde{\Gamma}}$. Both estimators can be made more precise by averaging over repeated draws $n = 1, \ldots, N$. The proposed algorithm, which extends that of Aradillas Fernández et al. (2025), is as follows.

Algorithm 1 Stochastic mirror ascent for (9)

Require: Iterations T, draws per iteration N, step sizes $\eta_L, \eta_M > 0$, bound $\overline{\gamma}$.

1: Initialize $\tilde{\pi}_{1,l} \leftarrow 1/L$ for $l \leq L$ and $\tilde{\Gamma}_{1,m} \leftarrow \min\{\exp(-1), \overline{\gamma}/M\}$ for $m \leq M$.
2: **for** $t = 1, \ldots, T$ **do**
3: Draw $Y^{(1)}, \ldots, Y^{(N)} \sim \mathcal{N}(0, Q\Sigma Q')$.
4: $\hat{R}_t \leftarrow \mathrm{vec}_{l \leq L} \left\{ \frac{1}{N} \sum_{n=1}^{N} \hat{R}(\nu_{-1}^{(l)} + Y_{-1}^{(n)}; \nu_{-1}^{(l)}, \tilde{\pi}_t, \tilde{\Gamma}_t) \right\}$.
5: $\hat{\alpha}_t \leftarrow \mathrm{vec}_{m \leq M} \left\{ \frac{1}{N} \sum_{n=1}^{N} \varphi_{\tilde{\pi}_t, \tilde{\Gamma}_t}(\nu^{(m)} + Y^{(n)}) \right\}$.
6: $\tilde{\pi}_{t+1} \leftarrow \tilde{\pi}_t \odot \exp(\eta_L \hat{R}_t) / \mathbf{1}'\big(\tilde{\pi}_t \odot \exp(\eta_L \hat{R}_t)\big)$
7: $\tilde{\Gamma}_{t+1} \leftarrow \tilde{\Gamma}_t \odot \exp(\eta_M(\hat{\alpha}_t - \alpha)) \min\{1, \overline{\gamma}/\mathbf{1}'\big(\tilde{\Gamma}_t \odot \exp(\eta_M(\hat{\alpha}_t - \alpha))\big)\}$
8: **end for**
9: **return** $\hat{\pi}_T \leftarrow \frac{1}{T}\sum_{t=1}^{T} \tilde{\pi}_t$ and $\hat{\Gamma}_T \leftarrow \frac{1}{T}\sum_{t=1}^{T} \tilde{\Gamma}_t$.

Notice that the approximate solution $(\hat{\pi}_T, \hat{\Gamma}_T)$ produced by Algorithm 1 is the *average* over all iterates, rather than the final iterate. This is what enables the algorithm to perform well even if $N = 1$, so long as the number of iterations T is large enough and the step sizes η_L, η_M are not too large. Indeed, Aradillas Fernández et al. (2025, Theorem 2) prove that in weighted average power optimal testing problems, a properly tuned stochastic mirror ascent algorithm produces an approximately least favorable distribution $\hat{\Gamma}_T / \mathbf{1}'\hat{\Gamma}_T$ with high probability after finitely many iterations. This result transfers to my setting under Bayes risk, in which case Algorithm 1 can be modified to treat $\tilde{\pi}$ as known. However, the theory developed by Aradillas Fernández et al. (2025) does not accommodate simultaneously approximating the least favorable prior.[10] The following lemma establishes the rate at which Algorithm 1 converges in expectation, under a uniform bound on a measure of variability of the risk estimates.

Lemma 5.3 (Convergence)**.** *Let $\overline{\gamma}$ be as in Proposition 5.1, and suppose that there exists $\overline{r} > 0$ such that for every $(\tilde{\pi}, \tilde{\Gamma}) > 0$ in the feasible set of Proposition 5.1, the unbiased risk estimator in line 4 of Algorithm 1 computed at $(\tilde{\pi}, \tilde{\Gamma})$ satisfies*

$$\sqrt{\mathbb{E}\Big[\tfrac{1}{4}\big(\max_{l \leq L} \hat{R}_l - \min_{l \leq L} \hat{R}_l\big)^2\Big]} \leq \overline{r},$$

where the expectation is taken over Monte Carlo simulations. Write f for the objective of (9) and $f^\star$ for its optimal value. Then if $\alpha \leq \frac{1}{2}$, the output of Algorithm 1 with step sizes

$$\eta_L = \frac{1}{\overline{r}}\sqrt{\frac{2\log(L)}{T}} \quad \textit{and} \quad \eta_M = \frac{1}{1-\alpha}\sqrt{\frac{2\max\{1,\, 1 + \log(\overline{\gamma}),\, \log(M)\}}{T}}$$

satisfies

$$f^\star - \mathbb{E}\big[f(\hat{\pi}_T, \hat{\Gamma}_T)\big] \leq \overline{r}\sqrt{\frac{2\log(L)}{T}} + (1-\alpha)\,\overline{\gamma}\sqrt{\frac{2\max\{1,\, 1 + \log(\overline{\gamma}),\, \log(M)\}}{T}}.$$

[10] Aradillas Fernández et al. (2025) require a uniform bound on the L^∞-norm of the gradient estimates, which is not available for the risk estimates.

Lemma 5.3 bounds the expected suboptimality of the approximate solution after T iterations, and provides theoretical guidance on the choice of step sizes. Algorithm 1 scales well with the number of grid points in the sense that the number of iterations T need only grow faster than $\log(L)$ and $\log(M)$ for the bound to tighten.[11] This is of practical importance since in order to control discretization error, the number of grid points must increase exponentially in the dimension of the reduced moments. The number of draws N only appears indirectly through the variability of the risk estimates, and does not affect the algorithm's rate of convergence. In particular, the theory does not preclude setting $N = 1$ if it is computationally beneficial to do so.

The suboptimality bound in Lemma 5.3 depends on two constants: a bound $\overline{r}$ on the root mean squared half-range of the risk estimates, and a bound $\overline{\gamma}$ on the mass of the Lagrange multipliers at the optimum, $\tilde{\gamma}^\star = \mathbf{1}'\tilde{\Gamma}^\star$. This suggests that Algorithm 1 may exhibit worse performance if the risk estimates are more variable either within or across grid points, or if the optimum is more sensitive to the coverage constraint. The latter interpretation follows from the envelope theorem, since $\tilde{\gamma}^\star$ is a subgradient of the optimum $f^\star$ with respect to the coverage level $1 - \alpha$. In practice, Algorithm 1 performs far better than the guarantee of Lemma 5.3, since $\overline{r}$ and $\overline{\gamma}$ are worst-case bounds that almost never bind for iterates $(\tilde{\pi}_t, \tilde{\Gamma}_t)$ that the algorithm actually reaches. I therefore calibrate η_L and η_M by replacing $\overline{r}$ and $\overline{\gamma}$ in the expressions from Lemma 5.3 with their empirical counterparts from a pilot run, which are much less conservative.

Motivated by Lemma 5.3, I define an approximately minimax optimal test as $\varphi_{\hat{\pi}_T, \hat{\Gamma}_T}$, the primal solution corresponding to the approximate solution $(\hat{\pi}_T, \hat{\Gamma}_T)$ of the discretized dual problem (9). Bounding the resulting approximation error is challenging for two main reasons. First, there is the discretization error induced by replacing the optimal testing problem (7) with its discretized counterpart (10), as discussed in Section 5.2. Second, there is the issue of showing that an approximate solution of (9) translates to a test that is not only approximately optimal for (10), but also approximately feasible in a suitable sense. Aradillas Fernández et al. (2025, Theorem 3) establish such a result in their setting for $\frac{1}{T}\sum_{t=1}^{T} \varphi_{\tilde{\pi}_t, \tilde{\Gamma}_t}$, the average of the Neyman–Pearson tests across iterates, rather than the Neyman–Pearson test at the averaged iterate. I avoid this construction since it has the undesirable feature of being randomized, despite the fact that the true solution must be non-randomized by Lemma 5.2. In practice, I find that $\varphi_{\hat{\pi}_T, \hat{\Gamma}_T}$ tends to perform just as well as its randomized counterpart on average, while exhibiting far less variability. I leave a formal analysis of the differences between these two methods to future research.

6 Simulation Study

To benchmark the performance of existing methods against an approximately minimax optimal confidence procedure, I return to the difference-in-differences setting of Example 2.1. I base my simulation study on that of Rambachan and Roth (2023), who also measure performance in terms of expected excess length. The tests they recommend are derived from Andrews et al. (2023), and reflect the current state of the literature on inference under partial identification.

[11] Strictly speaking, the constants $\overline{r}$ and $\overline{\gamma}$ may also depend on the grid specification, so this statement is only rigorous for a given sequence of grids if the corresponding sequence of constants is bounded.

6.1 Simulation Design

Rambachan and Roth (2023) calibrate their simulations using the estimated covariance matrices from twelve event study papers surveyed by Roth (2022). For the sake of computational tractability, I apply my proposed approach to the six papers with at most four pre-periods in addition to the reference period. Following Rambachan and Roth (2023), I set all treatment effects to zero, $\tau = 0$, and specify that parallel trends is violated in the final pre-period, so that $\mu = \delta$ with μ_{-1} its only nonzero component. The target parameter is the average treatment effect on the treated in the first post-period, $\theta = \tau_1$. For each paper, the covariance matrix Σ of the Monte Carlo simulations $\hat{\mu} \sim \mathcal{N}(\mu, \Sigma)$ is given by the paper's estimated covariance matrix, normalized so that the standard error of $\hat{\mu}_1$ is equal to one.

To make the magnitude of the parallel trends violation $\mu_{-1} = \delta_{-1}$ comparable across studies, I measure it relative to the standard error of $\hat{\mu}_{-1}$ given the other pre-period estimates,

$$\tilde{\sigma}_{-1} \equiv \sqrt{\text{Var}(\hat{\mu}_{-1} \mid \hat{\mu}_{-2}, \ldots, \hat{\mu}_{-\underline{T}})}.$$

This choice of scaling is motivated by the common practice of testing for pre-trends in applied work. In particular, the most powerful level-α pre-trends test against μ^{pre} has power $\Phi(\mu_{-1}/\tilde{\sigma}_{-1} - z_{1-\alpha})$, where Φ and $z_{1-\alpha}$ denote the standard normal CDF and its $1-\alpha$ quantile. Hence, $\mu_{-1}/\tilde{\sigma}_{-1}$ measures how large the violation is by how difficult it is to detect; see Section A.10 for additional details.

I consider two classes of nuisance parameter sets, just as in Rambachan and Roth (2023, Figure 3). The first I have already introduced in the four-period setting of Example 2.1, while the second is constructed analogously from second differences, rather than first differences,

$$\begin{aligned}
\Delta^{\text{RM}}(\bar{M}) &\equiv \left\{\delta : |\delta_{t+1} - \delta_t| \leq \bar{M} \max_{s<0} |\delta_{s+1} - \delta_s| \text{ for } t \geq 0\right\}, \\
\Delta^{\text{SDRM}}(\bar{M}) &\equiv \left\{\delta : |(\delta_{t+1} - \delta_t) - (\delta_t - \delta_{t-1})| \leq \bar{M} \max_{s<0} |(\delta_{s+1} - \delta_s) - (\delta_s - \delta_{s-1})| \text{ for } t \geq 0\right\}.
\end{aligned}$$

As before, $\Delta^{\text{RM}}(\bar{M})$ bounds the magnitude of post-period violations by $\bar{M}$ times the largest observed magnitude of any pre-period violation. The set $\Delta^{\text{SDRM}}(\bar{M})$ behaves similarly while accommodating linear differences in trends. In particular, $\Delta^{\text{SDRM}}(0)$ corresponds to a model that allows for linear pre-trends and extrapolates them into the post-period.

I compare my proposed approach to two test inversion confidence procedures from Andrews et al. (2023) that Rambachan and Roth (2023) adapt to the setting of Example 2.1. I implement all three methods in the reduced testing problem of Section 4.1 so that they are on an equal footing. This separates the question of how to project out uninformative directions, which Andrews et al. (2023) handle via linear programming, from the question of how different methods perform once those directions have been removed. Since the authors' tests are designed for moment inequality models, they can be applied by representing $Q\mathcal{M}(0)$ as a finite union of polyhedra, testing each polyhedron separately, and rejecting the reduced null hypothesis whenever all of the tests reject. This technique is known as the intersection-union method, which controls size

but may be conservative (Berger, 1997). I derive the reduced null and decompose it into a union of polyhedra of the form $\{\nu : A\nu - b \leq 0\}$ in Section A.10.

The first test compares a canonical test statistic to a least favorable critical value. Specifically, it tests the moment inequalities $H_0 : A\nu - b \leq 0$ using the studentized maximum statistic (Canay and Shaikh, 2017, Equation (15)) with critical values computed as if all of the inequalities were binding. This approach has a long history in the moment inequality testing literature, and enjoys certain optimality properties (Canay and Shaikh, 2017, Example 4.1). The second test represents a more modern approach based on the conditional inference technique of Lee et al. (2016). In recent years, a burgeoning literature has applied this technique to several long-standing inference problems in econometrics (Cox and Shi, 2023; Andrews et al., 2024; Bei, 2025). These methods often have desirable computational features and are insensitive to inequalities that are far from binding, unlike least favorable tests. However, the literature has not yet reached a consensus regarding their performance. While Rambachan and Roth (2023) provide conditions under which the conditional test of Andrews et al. (2023) has optimal local asymptotic power, in certain settings conditional test inversion is known to produce confidence sets with infinite expected length (Kivaranovic and Leeb, 2021).

6.2 Simulation Results

For each paper, I compute expected excess length over $\mu_{-1}/\tilde{\sigma}_{-1} \in [0, 3]$ and report the median performance across all six papers in Figure 1, setting $\bar{M} = 1$ and $\alpha = 0.05$.

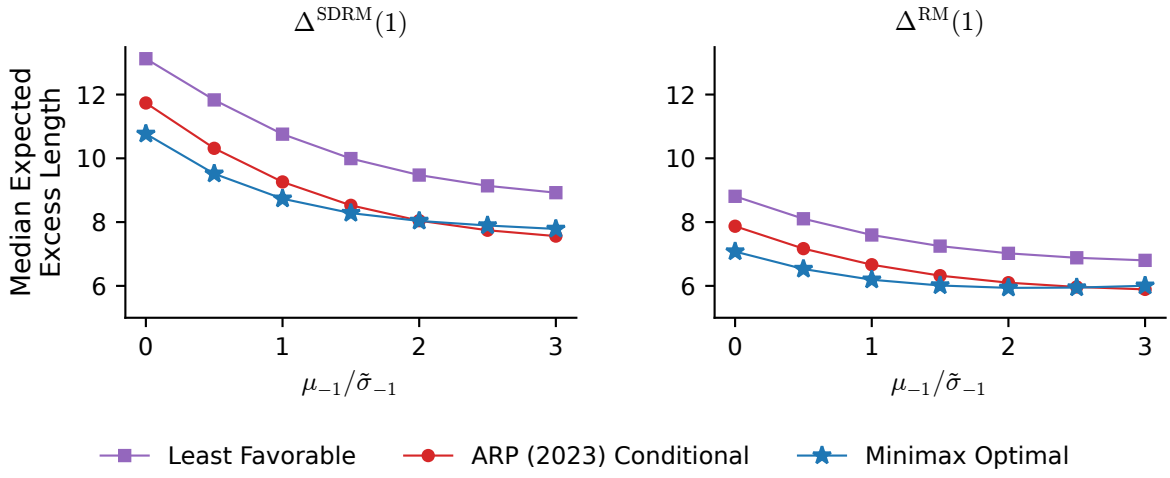


Figure 1: Simulation Results for $\Delta^{\text{SDRM}}(1)$ and $\Delta^{\text{RM}}(1)$

Notes: Median across papers of expected excess length for competing confidence procedures under $\Delta^{\text{SDRM}}(1)$ and $\Delta^{\text{RM}}(1)$ with $\alpha = 0.05$; see Table 2 in Section A.10 for a list of the papers, and Andrews et al. (2023, §3) for a description of the tests. The target parameter is τ_1, the ATT in the first post-period, and excess length is measured in standard errors of $\hat{\mu}_1$. Simulations are drawn under the assumption of zero treatment effects, with the only nonzero component of the event study coefficients $\mu = \delta$ given by μ_{-1}. The parallel trends violation μ_{-1} is measured relative to the conditional standard error $\tilde{\sigma}_{-1}$ of $\hat{\mu}_{-1}$ given the other pre-period estimates. For each paper, expected excess length is estimated from 10,000 Monte Calro simulations, yielding standard errors between 0.01 and 0.05.

Figure 1 shows that all three methods exhibit their worst performance under parallel trends. This phenomenon is not just a quirk of the simulation design: numerically, I find that in each paper, the risk of all three methods is uniquely maximized under parallel trends relative to *any* other configuration of μ^{pre}.[12] This gives the minimax optimal procedure a simple interpretation in terms of adaptivity: it maintains correct coverage under any parallel trends violation in the nuisance parameter set, while minimizing expected excess length under parallel trends.[13] Strikingly, both the minimax optimal and conditional approaches dominate the least favorable approach; this is also the case in every paper. The intersection-union method is largely to blame for this conservatism: the least favorable test of a given polyhedron tends to direct power towards alternatives that are in the null of a different polyhedron.

The median over papers of each procedure's maximum expected excess length can be read off of Figure 1 at $\mu_{-1}/\tilde{\sigma}_{-1} = 0$. By construction, the optimal procedure has the best performance there, at the expense of slightly worse performance than the conditional procedure for large violations. This is to be expected, since Rambachan and Roth (2023, §3.5.3) establish that the conditional test has optimal power as $\mu_{-1}/\tilde{\sigma}_{-1} \to \infty$ under $\Delta^{\text{SDRM}}(1)$. The intuition for this result is that as $\mu_{-1}/\tilde{\sigma}_{-1}$ increases, the length of the identified set grows large relative to sampling error, and its bounds are determined by a single pre-period violation that is well-separated from the rest.[14] Much like in Imbens and Manski (2004), the testing problem therefore reduces to two one-sided tests of a Gaussian mean, and the conditional test provides the usual one-sided critical value due to its insensitivity to slack moments. Hence, the optimality results in this paper are complementary to those in Rambachan and Roth (2023): I focus on the case in which it is unclear whether parallel trends holds, while they focus on the case in which parallel trends is clearly violated and the source of that violation is apparent.

6.3 Efficiency Ratios

I quantify the suboptimality of the least favorable and conditional approaches by approximating their efficiency ratios. The true efficiency ratio of a procedure is the ratio between the optimal value of decision problem (6) and the procedure's maximum expected excess length. Rambachan and Roth (2023, §B.1) report efficiency ratios based on a theoretical lower bound for the optimal value of decision problem (6), which I refer to as the theoretical efficiency ratio. The lower bound is based on Theorem 3.2 of Armstrong and Kolesár (2018), and it is not yet known whether any feasible procedure can come close to achieving it. Duality theory enables me to numerically approximate a much sharper lower bound. To see how, let f denote the objective of the discretized dual problem (9). Then $f(\hat{\pi}_T, \hat{\Gamma}_T)$ is a lower bound for $f(\pi^\star, \Gamma^\star)$, the value of (9), which in turn is a lower bound for the value of the non-discretized dual problem (8), since the discretization restricts the feasible set. By Lemma 5.1, this is equal to the the value of the optimal testing problem (7), which coincides with the value of decision problem (6) by Theorem 4.1. For each paper, I use a Monte Carlo estimate of the lower bound $f(\hat{\pi}_T, \hat{\Gamma}_T)$ to compute numerical efficiency ratios. I report the median across all six papers of both numerical and theoretical efficiency ratios in Table 1.

[12] Under $\Delta^{\text{SDRM}}(1)$, the uniqueness holds up to perfectly linear violations of parallel trends, which are immaterial.
[13] See Armstrong and Kolesár (2018) and references therein for a discussion of the literature on adaptivity.
[14] See Section A.10 for a derivation of the identified set.

Table 1: Median across Papers of Numerical and Theoretical Efficiency Ratios

	$\Delta^{\text{SDRM}}(1)$		$\Delta^{\text{RM}}(1)$	
Test	Numerical Efficiency Ratio	Theoretical Efficiency Ratio	Numerical Efficiency Ratio	Theoretical Efficiency Ratio
Least Favorable	0.81	0.59	0.80	0.60
ARP (2023) Conditional	0.92	0.67	0.89	0.67
Minimax Optimal	1.00	0.73	1.00	0.74

Notes: Median across papers of both numerical and theoretical efficiency ratios for competing confidence procedures under $\Delta^{\text{SDRM}}(1)$ and $\Delta^{\text{RM}}(1)$ with $\alpha = 0.05$; see Table 2 in Section A.10 for a list of the papers, and Andrews et al. (2023, §3) for a description of the tests. A numerical efficiency ratio is the ratio between $f(\hat{\pi}_T, \hat{\Gamma}_T)$ and a procedure's maximum expected excess length, where $(\hat{\pi}_T, \hat{\Gamma}_T)$ comes from Algorithm 1 and f denotes the objective of (9). Theoretical efficiency bounds are computed analogously, with the theoretical lower bound from Rambachan and Roth (2023, §B.1) in place of $f(\hat{\pi}_T, \hat{\Gamma}_T)$. For each paper, maximum expected excess length and $f(\hat{\pi}_T, \hat{\Gamma}_T)$ are estimated from 10,000 Monte Carlo simulations, with reported ratios accurate up to the second decimal place.

Table 1 shows that the theoretical efficiency ratios Rambachan and Roth (2023) use to benchmark their inference techniques are in fact highly conservative. Taking the median across papers, the optimal procedure improves upon the performance of the conditional approach by 8% under $\Delta^{\text{RM}}(1)$ and 11% under $\Delta^{\text{SDRM}}(1)$, rather than 33% as the authors report in their Figure 3.[15] Existing optimality theory therefore cannot reliably determine how much performance a given method leaves on the table in this setting. By contrast, the numerical efficiency bound I propose is sharp in the sense that the optimal procedure attains it up to two decimal places. However, unlike the conditional and least favorable approaches, the optimal procedure need not be exactly feasible for decision problem (6), since it is subject to approximation and discretization error. If the researcher does not find this acceptable, they may apply several techniques recommended by Elliott et al. (2015) to control size, such as inflating the mass of $\hat{\Gamma}_T$ and using importance sampling to evaluate size off of the null grid. I emphasize that concerns about size control do *not* affect the validity of $f(\hat{\pi}_T, \hat{\Gamma}_T)$ as a lower bound on performance. In other words, even if the researcher does not wish to apply the optimal procedure, they may nevertheless find it useful to estimate $f(\hat{\pi}_T, \hat{\Gamma}_T)$ in order to assess the suboptimality of their preferred procedure. The computational burden of doing so can be reduced by decreasing the number of grid points and iterations, at the cost of producing a less informative bound.

[15] I replicate the same 33% number in both theoretical efficiency ratio columns, despite only using six out of the twelve papers Rambachan and Roth (2023) consider.

7 Applications

7.1 The Imbens–Manski Problem in de Quidt et al. (2018)

In seminal work, Imbens and Manski (2004) posed the quintessential problem of inference under partial identification. Suppose the researcher is interested in a scalar parameter θ, but the data only provide a lower and an upper bound, $\theta \in [\mu_\ell, \mu_u]$. Given a joint asymptotically normal estimator of the bounds, how should the researcher go about forming a confidence set for θ? Despite being revisited several times over the past two decades (e.g. Stoye, 2009, 2020; Elliott et al., 2015; Müller and Norets, 2016), this question has not yet been definitively answered. The central challenge lies in determining how best to adapt to the unknown length of the identified set, $\mu_u - \mu_\ell$. This immediately disqualifies the fixed length confidence interval of Donoho (1994): any fixed length procedure must be infinitely long, since the identified set may be arbitrarily wide.

Stoye (2020) proposes a computationally simple confidence procedure and demonstrates that it performs better than existing methods in simulations. He finds its competitive performance particularly striking given that the procedure was motivated by concerns about misspecification, rather than performance. The procedure not only satisfies a conventional coverage requirement, but also covers a certain pseudo-true parameter under misspecification; see Stoye (2020) for details. Since imposing an additional constraint can only weakly decrease the performance of an optimal procedure, this suggests that existing methods must not have been nearly optimal to begin with. Comparing the misspecification-robust approach of Stoye (2020) to the optimal procedure under correct specification clarifies the mechanism underlying this phenomenon.

Stoye (2020) applies his proposed method to the work of de Quidt et al. (2018), who study experimenter demand: that is, the tendency of participants in an experiment to adjust their behavior based on what they believe the experimenter is looking for. Participants choose an action between zero and one (e.g. the proportion of a fixed sum to invest in a risky project), and are randomly informed that the experimenter expects them to choose a higher or lower action. Under the authors' behavioral model, the average action under the high (resp. low) signal is an upper (resp. lower) bound on the average "natural" action participants would have otherwise taken, which is the parameter of interest. Since the bounds are estimated from different populations, the estimated bounds may cross even if the true bounds do not.

Figure 2 displays confidence intervals for average natural actions across six different experimental tasks implemented by de Quidt et al. (2018). It compares the minimax optimal confidence set to those of Stoye (2020) and Romano et al. (2014), along with the least favorable approach from Section 6. The method of Romano et al. (2014) is popular in the moment inequality literature due to its ease of implementation and good performance relative to least favorable methods in practice (Canay et al., 2023). Following Stoye (2020), I use it to reflect the state of the established literature.

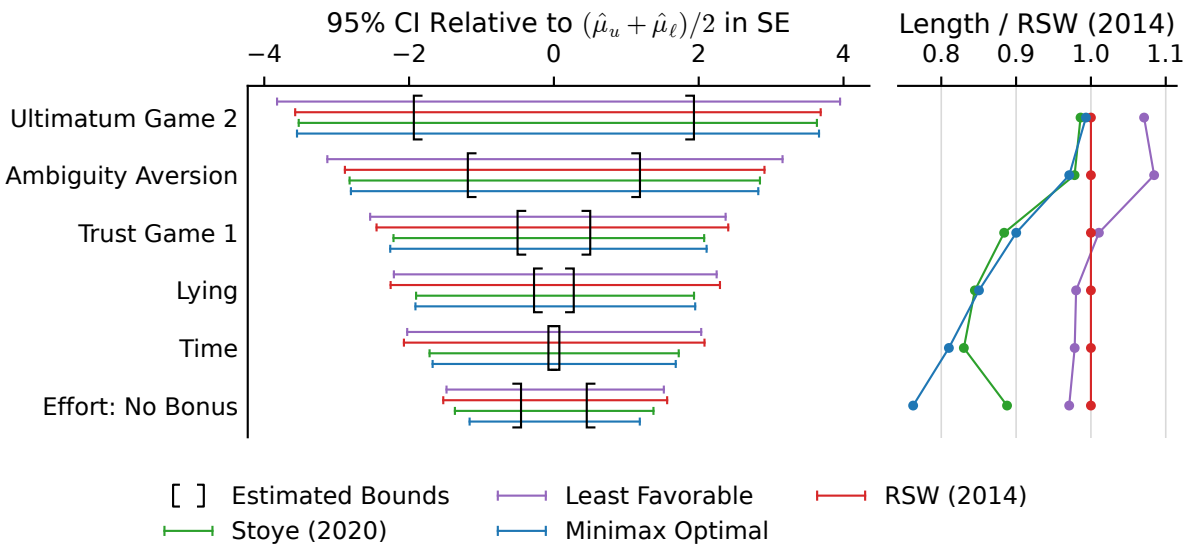


Figure 2: 95% Confidence Intervals for Average Natural Actions in de Quidt et al. (2018)

Notes: Competing 95% confidence intervals for participants' average natural action in six experimental tasks from de Quidt et al. (2018), ordered by the length of the estimated bounds. Intervals are reported net of the estimated midpoint $(\hat{\mu}_u - \hat{\mu}_\ell)/2$ and in units of its standard errors for the sake of comparability across tasks. The least favorable test uses the studentized maximum statistic with a least favorable critical value, as in Canay and Shaikh (2017, Example 4.1). See Romano et al. (2014) and Stoye (2020) for details on the other tests. For a given interval, the ratio between its length and that of Romano et al. (2014) is plotted on the right-hand side panel.

Figure 2 shows that researchers encounter a wide range of estimated identified set lengths in practice, from about -1 to 4 standard errors of the estimated midpoint $(\hat{\mu}_u + \hat{\mu}_\ell)/2$. This not only affects the length of a confidence interval in absolute terms, but also its length relative to competing methods.[16] The least favorable confidence interval is 2 to 3 percent shorter than that of Romano et al. (2014) when the estimated identified set is short, at the expense of being 7 to 8 percent longer when it is long. As pointed out by Stoye (2020), his misspecification-robust interval is shorter than that of Romano et al. (2014) for every experimental task, with a maximum length reduction of 17%. Strikingly, the length reductions of the optimal procedure only differ from that of Stoye (2020) by 1 or 2 percentage points, except for the "Effort: No Bonus" task. For this task, the estimated bounds cross, and the optimal and misspecification-robust procedures obtain length reductions of 24% and 11%, respectively. Figure 3 plots the corresponding acceptance regions to diagnose this discrepancy.

[16] Ideally, I would also like to report excess length, but the true identified set is unknown. In Figure 4, I assess expected excess length performance in a calibrated simulation. Throughout this section, I refer to confidence sets as confidence intervals, since it happens to be optimal to report an interval in this setting. As discussed in Section 3.1, this is an endogenous feature of the researcher's decision problem, rather than an a priori restriction.

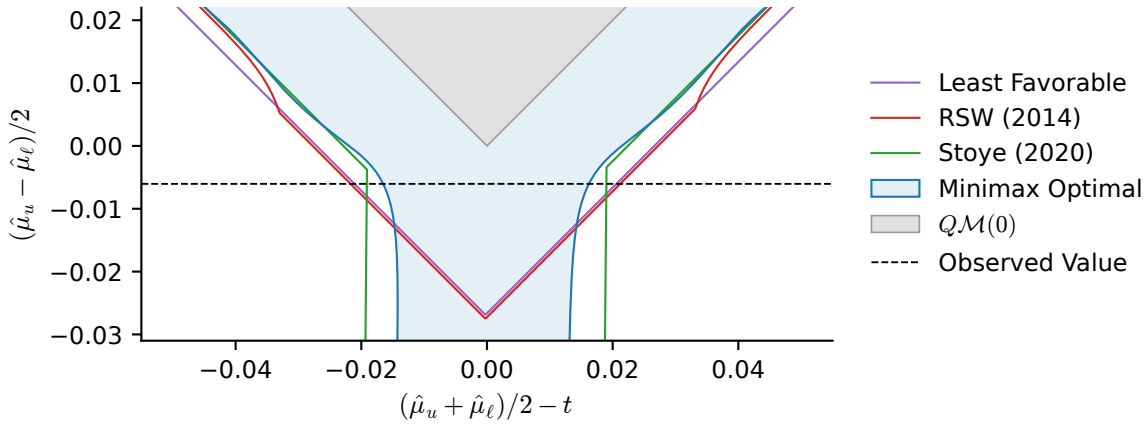


Figure 3: Acceptance Regions for the "Effort: No Bonus" Task in de Quidt et al. (2018)

Notes: Acceptance regions of competing test inversion confidence procedures for the average natural action in the "Effort: No Bonus" task in de Quidt et al. (2018), with $\alpha = 0.05$. All tests are as in Figure 2. The horizontal axis corresponds to the estimated midpoint of the identified set, net of a candidate value t of the average natural action θ. The vertical axis corresponds to the estimated half-length of the identified set. Both axes are in their original units, and the authors estimate $(\hat{\mu}_u + \hat{\mu}_\ell)/2 = 0.337$ and $(\hat{\mu}_u - \hat{\mu}_\ell)/2 = -0.006$, the latter being depicted with a dashed black line. Confidence intervals for each procedures can be obtained by tracing along the dashed black line and collecting candidate values t that are accepted. The null hypothesis $\mathcal{QM}(0)$ is plotted in gray, with parameters in the null given by the population analogs of the quantities on each axis.

Figure 3 reveals that the acceptance regions of the optimal and misspecification-robust procedures possess a similar structure, but the optimal acceptance region grows narrower as the estimated half-length of the identified set gets smaller. This is because the optimal procedure is not constrained to cover a pseudo-true parameter under misspecification, so it yields shorter intervals when the estimated half-length is negative in order to reduce expected excess length when $\mu_u = \mu_\ell$. Indeed, I find that the point identified case is least favorable for every experimental task. While the optimal confidence interval can be substantially shorter than that of Stoye (2020), it is also never empty: the length reduction relative to Stoye (2020) is at most 30%. This follows from the fact that the optimal test does not direct power towards alternatives with $\mu_u < \mu_\ell$, as discussed in Section 4.3. By contrast, Figure 3 shows that both the least favorable and Romano et al. (2014) tests are empty whenever $(\hat{\mu}_u - \hat{\mu}_\ell)/2 < -0.027$. This behavior is not only undesirable from the perspective of misspecification robustness, but also suboptimal under correct specification.

Assessing the length of competing confidence intervals in a particular application does not provide evidence of how they would perform in repeated samples. For this reason, I complement the length comparisons above with a calibrated simulation using the estimated covariance matrix from the "Effort: No Bonus" task. For each confidence procedure, Figure 4 plots expected excess length and coverage on the boundary of the null hypothesis against the half-length of the identified set.

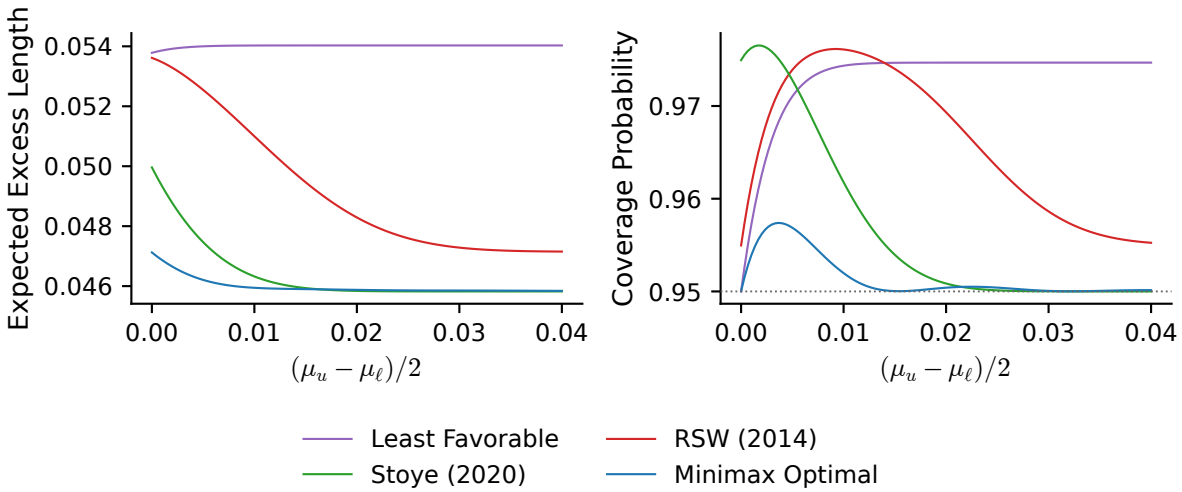


Figure 4: Risk and Coverage Calibrated to the "Effort: No Bonus" Task in de Quidt et al. (2018)

Notes: Expected excess length and coverage probabilities on the boundary of the null hypothesis for competing confidence procedures in a calibrated simulation based on the "Effort: No Bonus" task in de Quidt et al. (2018), with $\alpha = 0.05$. The covariance matrix Σ is calibrated to the estimated covariance matrix of the bound estimates $(\hat{\mu}_\ell, \hat{\mu}_u)$ for this task. Both expected excess length and coverage are plotted against the half-length of the identified set, which indexes risk. The standard error of the estimated half-length is about 0.01. Coverage is computed along the boundary of the null hypothesis where $(\mu_u - \mu_\ell)/2 = (\mu_u + \mu_\ell)/2 - t$, so $t = \mu_\ell$. In other words, the posited value of θ lies at the lower bound of the identified set.

Figure 4 shows that all four procedures are clearly ranked in terms of their performance. The procedure of Romano et al. (2014) dominates the least favorable procedure, achieving its largest gains when the identified set is long relative to sampling error. In terms of the numerical minimax efficiency ratios from Section 6.3, the gains are modest: the least favorable approach is 87% efficient, while that of Romano et al. (2014) is 88% efficient. On the other hand, the misspecification-robust procedure of Stoye (2020) is 94% efficient, and clearly dominates both of the aforementioned procedures. The optimal procedure removes the remaining inefficiency by using up slack in the coverage constraint. Although it is not similar on the boundary, its maximum overcoverage is less than 0.75 percentage points, compared to more than 2.5 percentage points for the other methods.

7.2 Event Study Sensitivity Analysis in Iovino et al. (2026)

Iovino et al. (2026) study the relationship between corporate income taxation and carbon dioxide (CO_2) emissions among firms in the United States. They document that the US corporate tax code implicitly subsidizes CO_2-intensive firms. Based on this evidence, they predict that a cut in the corporate income tax rate should disproportionately benefit clean firms. The authors test this prediction by exploiting the Tax Cuts and Jobs Act of 2017, which cut the federal corporate income tax rate from 35% to 21% beginning in 2018. They define a firm to be either CO_2-intensive or clean based on whether or not it is in the top quartile of the CO_2 intensity distribution in 2017. To quantify the reduction in CO_2 intensive firms' capital growth relative to clean firms, they implement an event study design with CO_2 intensive firms as the "treated" group and clean firms as the "control" group. Since the treatment affects both groups, the authors' target parameter $\theta = \tau_{2019}$ corresponds to the difference in group-specific average treatment effects on log capital in 2019; see Shahn and Hatfield (2025) for additional details.

Since clean firms are unlikely to constitute a perfect comparison group for CO_2-intensive ones, the authors assess the sensitivity of their conclusions to violations of the parallel trends assumption as recommended by Rambachan and Roth (2023). They consider sets of parallel trends violations given by $\Delta^{\text{RM}}(\bar{M})$ as defined in Section 6.1, and vary the scaling parameter $\bar{M}$; see their Figure A.4. I extend this analysis by incorporating a credible direction of the parallel trends violation into the specification of the nuisance parameter set. In particular, it seems plausible to assume that CO_2-intensive firms would have grown more slowly than clean ones even in the absence of the tax cut, due to the continuing decline of the coal industry, for instance. This is consistent with the observed pre-period differences in trends, and would suggest that the authors' reported estimates of the relative reduction in growth due to the tax cut are too large in magnitude, if anything. As pointed out by Rambachan and Roth (2023), including this assumption in the specification of the nuisance parameter set is simply a matter of intersecting $\Delta^{\text{RM}}(\bar{M})$ with the set of decreasing differential trends,

$$\Delta^{\text{RMD}}(\bar{M}) \equiv \Delta^{\text{RM}}(\bar{M}) \cap \left\{\delta : \delta_{t+1} - \delta_t \leq 0 \text{ for all } t\right\}.$$

Figure 5 depicts the resulting sensitivity analysis using the minimax optimal procedure, along with the least favorable and conditional approaches from Section 6.1.

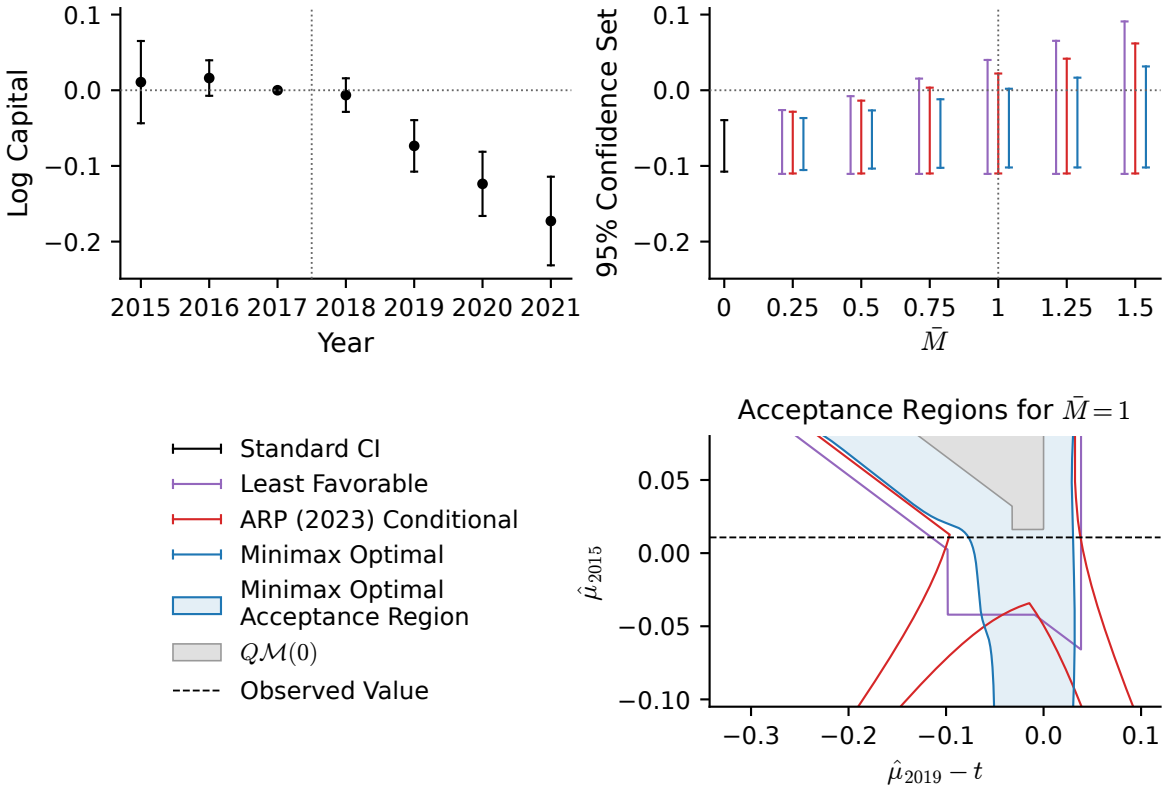


Figure 5: Event Study Senitivity Analysis in Iovino et al. (2026) under $\Delta^{\text{RMD}}(\bar{M})$

Notes: Sensitivity analysis using competing test inversion confidence sets for $\theta = \tau_{2019}$ under $\Delta^{\text{RMD}}(\bar{M})$, with $\bar{M}$ ranging from 0 to 1.5 and $\alpha = 0.05$. The upper left plot is the event study in Panel C of Iovino et al. (2026, Figure 1). The upper right plot depicts a sensitivity analysis on the same scale, comparing the optimal procedure to the least favorable and conditional procedures of Andrews et al. (2023), as described in Section 6.1. The bottom right plot displays slices of the acceptance regions with $\hat{\mu}_{2016}$ fixed at its observed value and $\bar{M} = 1$. Confidence intervals for each procedures can be obtained by tracing along the dashed black line and collecting candidate values t that are accepted. The null hypothesis $\mathcal{QM}(0)$ is plotted in gray, with parameters in the null given by the population analogs of the quantities on each axis.

The optimal procedure yields substantial length reductions relative to existing methods. As $\bar{M}$ ranges from 0.25 to 1.5, the optimal confidence intervals are 15 to 22 percent shorter than the conditional ones, and 18 to 33 percent shorter than the least favorable ones. Using the conditional confidence procedure, the authors' results are no longer significant at around $\bar{M} = 0.75$, while the optimal procedure has a breakdown value of around $\bar{M} = 1$.

To better understand the mechanism underlying these length reductions, it will be insightful to examine the form of the acceptance regions for each test under $\Delta^{\text{RMD}}(1)$. Since the tests depend on $\hat{\mu}_{2015}$, $\hat{\mu}_{2016}$, and $\hat{\mu}_{2019}$, I fix $\hat{\mu}_{2016}$ at its observed value and plot the resulting slice of the acceptance regions as the remaining coordinates vary in two dimensions. The length of each confidence interval can be obtained by tracing along the dashed black line corresponding to the observed value of $\hat{\mu}_{2015}$. Both the least favorable

and conditional approaches exhibit undesirable behavior as $\hat{\mu}_{2015}$ decreases. The least favorable confidence interval is long for $\hat{\mu}_{2015} > -0.042$ but empty for $\hat{\mu}_{2015} < -0.066$. Just as in Section 7.1, this suggests that the least favorable approach may be wasting power on alternatives that are inconsistent with the model. On the other hand, the conditional confidence interval grows longer and eventually splits into two disjoint intervals that grow increasingly far apart. This is a symptom of the intersection-union method, since each of the intervals corresponds to a different polyhedron in the null decomposition of Rambachan and Roth (2023).

To demonstrate that these favorable comparisons are not just idiosyncratic to a particular draw of the data, I complement the above analysis with a calibrated simulation using the estimated covariance matrix of the event study coefficients. For each confidence procedure, Figure 6 plots expected excess length and coverage on the boundary of the null hypothesis under $\Delta^{\text{RMD}}(1)$. I set $\mu_{2015} = 2\mu_{2016}$ so that pre-trends are linear, and index the horizontal axis by μ_{2016}.

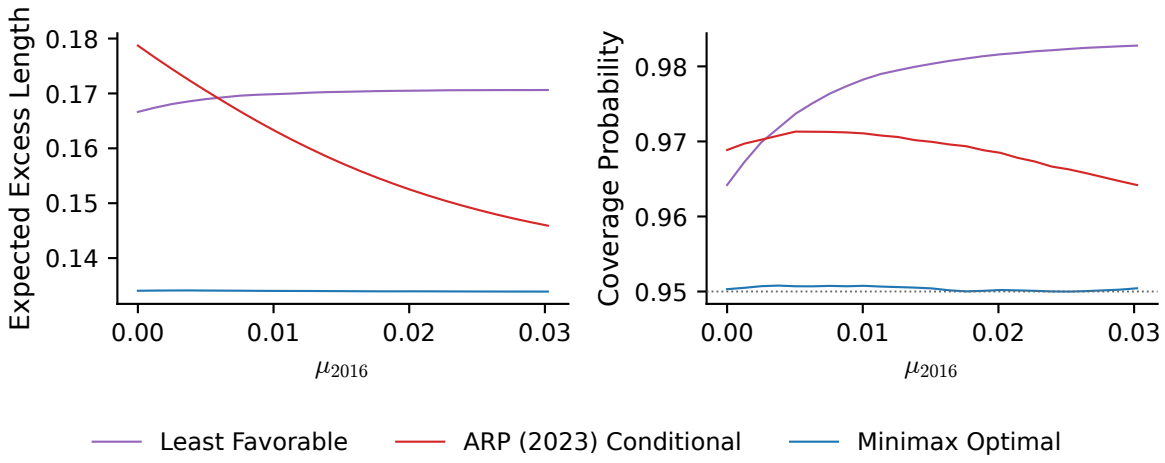


Figure 6: Risk and Coverage Calibrated to Iovino et al. (2026) under $\Delta^{\text{RMD}}(1)$ and Linear Pre-Trends

Notes: Expected excess length and coverage probabilities on the boundary of the null hypothesis for competing confidence procedures in a calibrated simulation based on Iovino et al. (2026) under $\Delta^{\text{RMD}}(1)$, with $\alpha = 0.05$. The covariance matrix Σ is calibrated to the estimated covariance matrix of the event study coefficients. Both expected excess length and coverage are plotted against μ_{2016}, with $\mu_{2015} = 2\mu_{2016}$ so that pre-trends are linear and decreasing, since $\mu_{2016} \geq 0$. Coverage is computed along the boundary of the null hypothesis where $t = \mu_{2019}$, so the posited value of θ lies at the lower bound of the identified set.

Figure 6 shows that both the least favorable and conditional procedures are dominated by the optimal procedure under linear pre-trends. Unlike in the examples from Section 6 and Section 7.1, I find that the least favorable prior is not a point mass: rather, it is spread out along the line $\mu_{2015} = 2\mu_{2016}$. This explains why the expected excess length of the optimal procedure is flat in Figure 6. The coverage constraint binds here as well, so the coverage of the optimal procedure is also flat and approximately equal to 95%. Unlike in Section 6.2, the conditional approach no longer dominates the least favorable one. In fact, in terms of the numerical minimax efficiency ratios from Section 6.3, the conditional procedure is 74% efficient, while the least favorable procedure is 78% efficient.

8 Conclusion

This paper develops a theoretical and computational framework for deriving and approximating optimal confidence procedures in a flexible class of partially identified models. I define optimality by embedding researchers' preferences over confidence procedure performance in a statistical decision problem. The class of competing procedures must satisfy a canonical coverage requirement, but is otherwise unrestricted. I establish that researchers may construct an optimal confidence procedure by inverting an optimal test. My numerical approach relies on duality theory not only to approximate the optimal procedure, but also to bound the suboptimality of alternative inference techniques. In a simulation study, I find that while the optimal procedure meaningfully improves upon the performance of existing methods, these methods are much closer to optimal than was previously known. In an application to de Quidt et al. (2018), I explain the puzzling empirical fact that the misspecification-robust approach of Stoye (2020) outperforms its non-robust counterparts. Finally, I apply my proposed approach to conduct a sensitivity analysis of an event study design in Iovino et al. (2026), and find that it substantially outperforms existing alternatives.

A Additional Details

A.1 Pathologies under Length Loss in Iovino et al. (2026)

This section is the subject of ongoing work. Please bear with me while I continue to provide updates via my website. The latest version of the paper is available at the link on the title page, under "Latest Version Here".

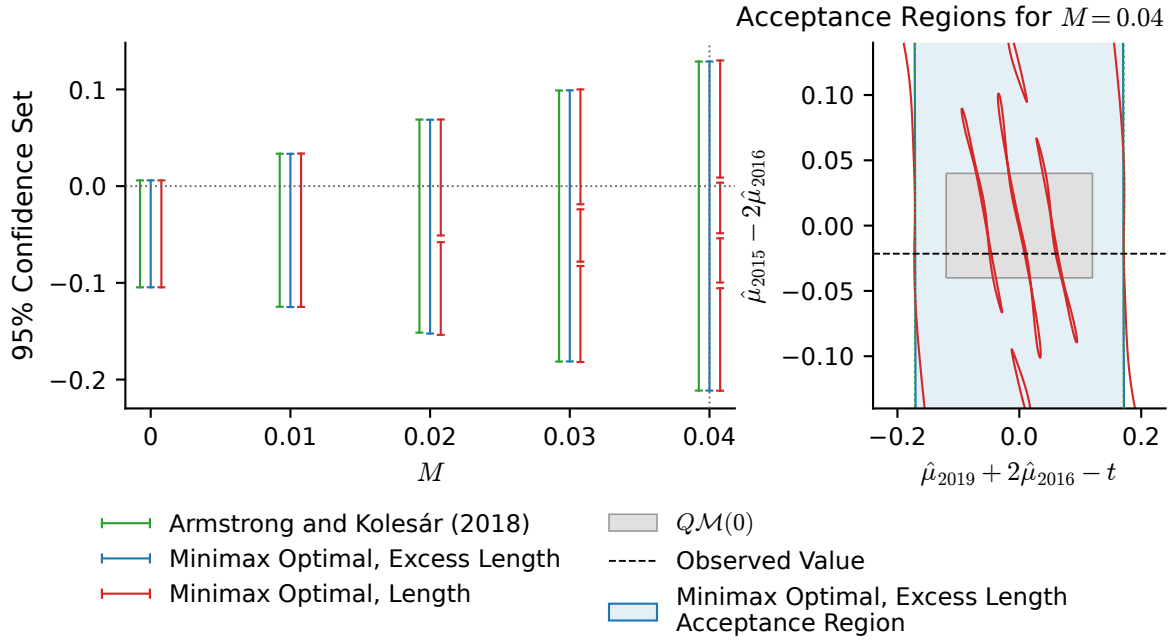


Figure 7: Iovino et al. (2026) $\Delta^{\text{SD}}(M)$

Notes:

	Expected Excess Length		Expected Length	
M	Numerical Efficiency Ratio	Theoretical Efficiency Ratio	Numerical Efficiency Ratio	Theoretical Efficiency Ratio
0.00	1.000	0.850	1.000	0.850
0.01	1.000	0.994	1.000	0.984
0.02	0.999	0.992	0.991	0.975
0.03	1.000	0.998	0.984	0.972
0.04	1.000	1.000	0.979	0.969

A.2 Omitted Variable Bias in Macchiavello and Morjaria (2015)

This section is the subject of ongoing work.

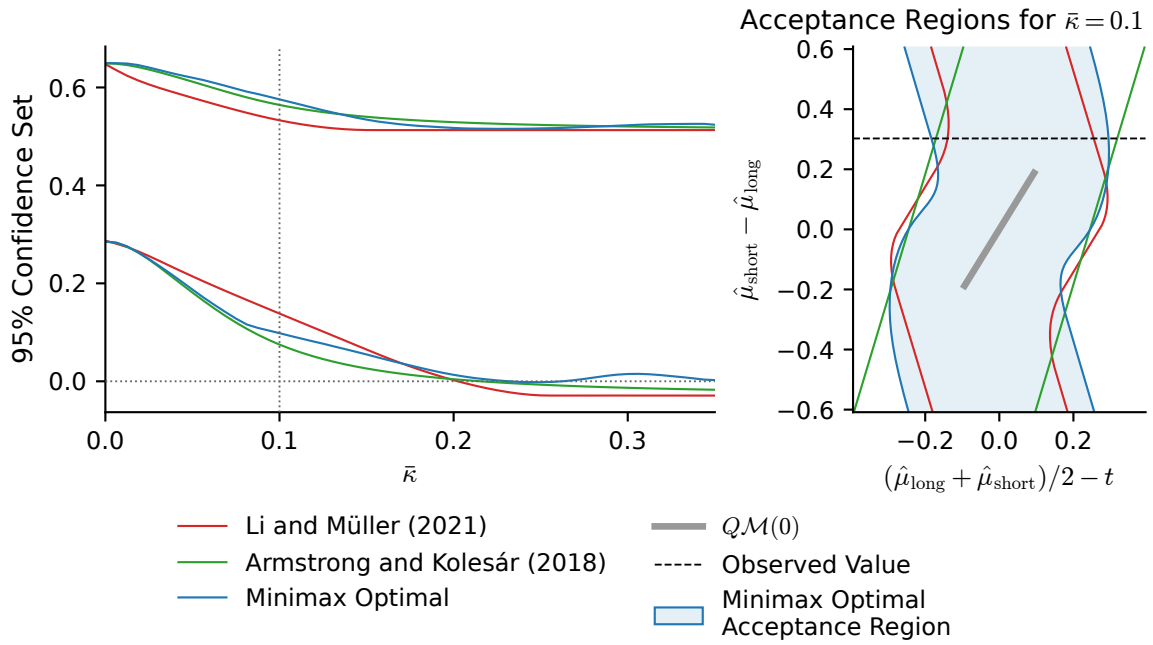


Figure 8: Macchiavello and Morjaria (2015)

Notes:

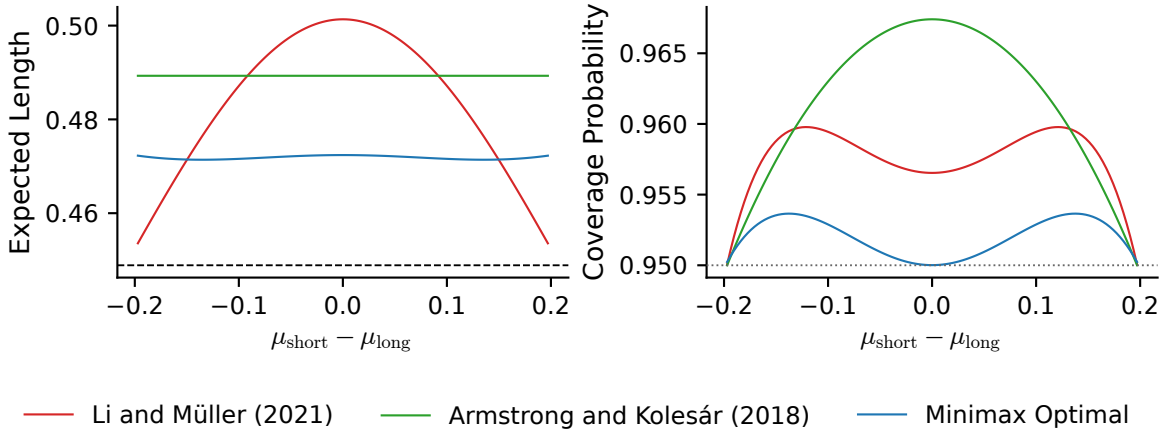


Figure 9: Macchiavello and Morjaria (2015)

Notes:

A.3 Quota Policy Identification

This section proves the claims from Example 2.2.

A.3.1 Identified Set

The identification problem boils down to what we can learn about the latent parameter

$$y_q \equiv \Pr(Y(1) = 1, U \leq q),$$

that is, a judge's misconduct rate if their release rate were q. By policy invariance, this quantity does not depend on the identity of the judge, nor whether the policy is in place or not. To see this, notice that the target parameter may be expressed as

$$\begin{aligned}\theta &= w_0(\mathbb{1}\{p_0 < q\}y_q + \mathbb{1}\{p_0 \geq q\}y_0) + w_1(\mathbb{1}\{p_1 < q\}y_q + \mathbb{1}\{p_1 \geq q\}y_1) \\ &= w_0\mathbb{1}\{p_0 \geq q\}y_0 + w_1\mathbb{1}\{p_1 \geq q\}y_1 + (w_0\mathbb{1}\{p_0 < q\} + w_1\mathbb{1}\{p_1 < q\})y_q.\end{aligned}$$

So, if $q \leq p_0 \leq p_1$, then θ is point identified and given by $w_0y_0 + w_1y_1$. If $p_0 < q \leq p_1$, then $\theta = w_1y_1 + w_0y_q$. By Kitagawa (2015), the following implications of the model are sharp,

$$0 \leq y_q - y_0 \leq q - p_0, \quad 0 \leq y_1 - y_q \leq p_1 - q.$$

In other words, the implied misconduct rate for $p_0 \to q$ compliers must be bounded between zero and one, and similarly for the $q \to p_1$ compliers. Hence,

$$\theta \in w_1y_1 + w_0[\max\{y_0, y_1 - (p_1 - q)\}, \min\{y_0 + (q - p_0), y_1\}].$$

Finally, if $p_0 \leq p_1 < q$, then $\theta = y_q$, since $w_0 + w_1 = 1$. The sharp implications are that

$$0 \leq y_1 - y_0 \leq p_1 - p_0, \quad 0 \leq y_q - y_1 \leq q - p_1,$$

so $\theta \in [y_1, y_1 + (q - p_1)]$. This establishes the form of the identified set given in Example 2.2. Of course, an analogous argument holds with the indices swapped if $p_1 \leq p_0$.

A.3.2 Nuisance Parameter Set

It remains to establish that the model fits into my framework. With $K \equiv (0, 1, 0, 1)'$ and $l \equiv 1$, the structural parameter is $\tau = \theta$, so (1) holds precisely when

$$\delta \equiv \mu - \theta K = (p_0 - q,\ y_0 - \theta,\ p_1 - q,\ y_1 - \theta)' \in \Delta.$$

Note that the ordering of p_0, p_1, and q is determined by the first and third components of δ. We may therefore split Δ into pieces based on this ordering. For a given ordering, we would like the corresponding piece of Δ to encode the model's sharp implications.

Suppose first that $q \leq p_0 \leq p_1$. Then

$$\begin{aligned} \theta = w_0 y_0 + w_1 y_1 &\iff 0 = w_0 \delta_2 + w_1 \delta_4, \\ 0 \leq y_1 - y_0 \leq p_1 - p_0 &\iff 0 \leq \delta_4 - \delta_2 \leq \delta_3 - \delta_1. \end{aligned}$$

The corresponding piece is therefore given by

$$\Delta_{q01} \equiv \left\{\delta \in \mathbb{R}^4 : 0 \leq \delta_1 \leq \delta_3,\ w_0 \delta_2 + w_1 \delta_4 = 0,\ 0 \leq \delta_4 - \delta_2 \leq \delta_3 - \delta_1\right\}.$$

Next, suppose $p_0 < q \leq p_1$. Then $\theta = w_1 y_1 + w_0 y_q$, so

$$w_0 \delta_2 + w_1 \delta_4 = w_0 y_0 + w_1 y_1 - \theta = -w_0 (y_q - y_0)$$

and $\delta_4 = y_1 - \theta = w_0 (y_1 - y_q)$, meaning

$$\begin{aligned} 0 \leq y_q - y_0 \leq q - p_0 &\iff w_0 \delta_1 \leq w_0 \delta_2 + w_1 \delta_4 \leq 0, \\ 0 \leq y_1 - y_q \leq p_1 - q &\iff 0 \leq \delta_4 \leq w_0 \delta_3. \end{aligned}$$

The corresponding piece is therefore given by

$$\Delta_{0q1} \equiv \left\{\delta \in \mathbb{R}^4 : \delta_1 < 0 \leq \delta_3,\ w_0 \delta_1 \leq w_0 \delta_2 + w_1 \delta_4 \leq 0,\ 0 \leq \delta_4 \leq w_0 \delta_3\right\}.$$

Finally, suppose $p_0 \leq p_1 < q$. Then $\theta = y_q$, so that $\delta_2 = y_0 - y_q$ and $\delta_4 = y_1 - y_q$, and

$$\begin{aligned} 0 \leq y_1 - y_0 \leq p_1 - p_0 &\iff 0 \leq \delta_4 - \delta_2 \leq \delta_3 - \delta_1, \\ 0 \leq y_q - y_1 \leq q - p_1 &\iff \delta_3 \leq \delta_4 \leq 0. \end{aligned}$$

The corresponding piece is therefore given by

$$\Delta_{01q} \equiv \left\{\delta \in \mathbb{R}^4 : \delta_1 \leq \delta_3 < 0,\ 0 \leq \delta_4 - \delta_2 \leq \delta_3 - \delta_1,\ \delta_3 \leq \delta_4 \leq 0\right\}.$$

If instead $p_1 \leq p_0$, the analogous orderings swap (δ_1, δ_2) with (δ_3, δ_4) and w_0 with w_1 in the above, yielding Δ_{q10}, Δ_{1q0}, and Δ_{10q}. Putting the pieces together, then, (1) holds with

$$\Delta \equiv \Delta_{q01} \cup \Delta_{0q1} \cup \Delta_{01q} \cup \Delta_{q10} \cup \Delta_{1q0} \cup \Delta_{10q},$$

the union of six polyhedra, one for each possible ordering.

A.4 Union Bounds

Bei (2025) studies inference on θ when its identified set takes the form

$$\Theta(\mu) \equiv \big[\min_{b \leq B} A_{\ell,b}\mu, \max_{b \leq B} A_{u,b}\mu\big],$$

where A_ℓ and A_u are known $B \times k_\mu$ matrices with bth rows $A_{\ell,b}$ and $A_{u,b}$, respectively. Let $A \equiv \left[\begin{smallmatrix} A_\ell \\ A_u \end{smallmatrix}\right]$, and suppose that A has full column rank with $\mathbf{1}_{2B} \in \text{range}(A)$. Then (1) holds for

$$K \equiv (A'A)^{-1}A'\mathbf{1}_{2B}, \quad l \equiv 1, \quad \text{and} \quad \Delta \equiv \Big\{\delta \in \mathbb{R}^{k_\mu} : \min_{b \leq B} A_{\ell,b}\delta \leq 0 \leq \max_{b \leq B} A_{u,b}\delta\Big\}.$$

The desired condition on the matrix A holds in every application Bei (2025) considers. The leading case is $A_\ell = A_u = \mathbf{I}_{k_\mu}$, which corresponds to $\mathcal{J}_\ell = \mathcal{J}_u = \{1, \ldots, k_\mu\}$ in Example 2.4. That is, each moment is both a candidate lower bound and a candidate upper bound. Just as in Bei (2025), the moments cannot simply be redefined as $A\mu$, since the matrix $A\Sigma A'$ will typically be singular, violating Assumption 2.4.

A.5 Normalization and Informativeness

A.6 Normalization and Informativeness

The normalization $\|l\| = 1$ in Assumption 2.1 is without loss of generality. To see this, let $c > 0$ and consider writing (1) with cl in place of l. The rescaled target $(cl)'\tau = c\theta$ is simply θ in different units, and since c is known, reporting the confidence set A for θ is equivalent to reporting cA for $c\theta$. Every decision problem I consider takes the form $\inf_{S \in \mathcal{S}_\alpha} \Psi(S)$, where the objective $\Psi : \mathcal{S} \to \overline{\mathbb{R}}_+$ aggregates risk across the parameter space. Write $\mathcal{S}_{\alpha,c}$ for the class of confidence procedures with correct coverage when (1) is written with cl in place of l, and Ψ_c for the objective of the corresponding decision problem. The following proposition verifies that a change of units transfers the solutions of one problem to the other.

Proposition A.1 (Normalization)**.** *Under Assumptions 2.1 to 2.4, let $c > 0$ and suppose that Ψ_c and Ψ satisfy $\Psi_c(cS) = c\,\Psi(S)$ for every $S \in \mathcal{S}$. Then*

$$c \inf_{S \in \mathcal{S}_\alpha} \Psi(S) = \inf_{S \in \mathcal{S}_{\alpha,c}} \Psi_c(S),$$

and $S^\star$ solves the left-hand side if and only if $cS^\star$ solves the right-hand side.

All of the decision problems I study satisfy the hypothesis of Proposition A.1, as verified at the end of its proof. Hence S is optimal for the normalized target parameter θ if and only if cS is optimal for the rescaled target parameter $c\theta$. The researcher may therefore solve the normalized problem and rescale the resulting confidence set by c.

Turning to informativeness, recall from Section 2.4 that if $Kl \in V$, then $\Theta(\mu) = \mathbb{R}$ for all $\mu \in \mathcal{M}$. The following proposition establishes that the reverse direction also holds.

Proposition A.2 (Informativeness). *Under Assumption 2.1, $Kl \notin V$ if and only if for some $\mu \in \mathcal{M}$, $\Theta(\mu) \neq \mathbb{R}$. Moreover, $Kl \notin V$ implies that l lies in the row space of K.*

Finally, consider the claim from Example 2.1 that $V = \{(0, s, 0)' : s \in \mathbb{R}\}$ for

$$\mathcal{M}(0) = \{(\mu_2, \mu_1, \mu_{-1})' : |\mu_2| \leq 2\bar{M}|\mu_{-1}|\}.$$

To verify this, note first that the inequality defining $\mathcal{M}(0)$ does not involve μ_1, so for $v = (0, s, 0)'$ and any $t \in \mathbb{R}$, the translation $\mu \mapsto \mu + tv$ leaves membership in $\mathcal{M}(0)$ unchanged, meaning $tv + \mathcal{M}(0) = \mathcal{M}(0)$ as desired. Conversely, let $v = (v_2, v_1, v_{-1})' \in V$, so $\mu + tv \in \mathcal{M}(0)$ for each $\mu \in \mathcal{M}(0)$ and $t \in \mathbb{R}$. Suppose for the sake of contradiction that $v_{-1} \neq 0$. Consider

$$\mu^+ = (\bar{M}, 0, 1)', \quad \mu^- = (-\bar{M}, 0, 1)', \quad \text{and} \quad t = -1/v_{-1}.$$

Then $\mu^+, \mu^- \in \mathcal{M}(0)$, so $\mu^+ + tv \in \mathcal{M}(0) \implies v_2/v_{-1} = \bar{M}$, while $\mu^- + tv \in \mathcal{M}(0) \implies v_2/v_{-1} = -\bar{M}$, which is not possible since $\bar{M} > 0$ by assumption. Hence $v_{-1} = 0$ by contradiction. Since $0 \in \mathcal{M}(0)$, taking $t = 1$ implies that $v \in \mathcal{M}(0)$, so $|v_2| \leq 2\bar{M}|v_{-1}| = 0$. Therefore $v \in \{(0, s, 0)' : s \in \mathbb{R}\}$ as claimed.

A.7 Indicator

Let $\lambda_{S(x,u)}$ denote the restriction of the Lebesgue measure λ to $S(x, u)$, and consider the Lebesgue density $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda(t)$, which exists and is equal to zero or one for almost every $t \in \mathbb{R}$ by Lebesgue's density theorem (Stein and Shakarchi, 2005). The indicator $(t, x, u) \mapsto \mathbb{1}\{t \in S(x, u)\}$ may be rigorously defined as a measurable map $I_S : \mathbb{R} \times \mathbb{R}^{k_\mu} \times [0, 1] \to [0, 1]$ such that for every (x, u), the slice $I_S(\,\cdot\,, x, u)$ is a version of $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$. Such a map exists by Part (iii) of Lemma B.1, and is determined up to a.e. equivalence in t for each (x, u) by definition.

A.8 Other Loss Functions and Decision Criteria

Under length loss, reporting $A \in \mathcal{A}$ incurs a loss of $\lambda(A)$, so the risk of a confidence procedure is given by its expected length, $R_\lambda(\mu, S) \equiv \mathbb{E}_{P_{\mu,U}}[\lambda(S(X, U))]$. The risk of a test follows from simply dropping the indicator from the definition of $R(\nu_{-1}, \varphi)$,

$$R_\lambda(\nu_{-1}, \varphi) \equiv \int \mathbb{E}_{P_\nu}[1 - \varphi(Y)]\, \mathrm{d}\nu_1,$$

which integrates acceptance probabilities over both the null and the alternative. The following corollary extends Theorem 4.1 to the case of length loss.

Corollary A.1 (Length Loss). *Under Assumptions 2.1 to 2.4, there exists a solution to*

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R_\lambda(\nu_{-1}, \varphi).$$

Moreover, if $\varphi^\star$ solves the above problem, then $S_{\varphi^\star}$ has minimax expected length, meaning it solves

$$\inf_{S\in\mathcal{S}_\alpha}\ \sup_{\mu\in\mathcal{M}}\ R_\lambda(\mu,S).$$

Theorem 4.1 also extends to the other two decision criteria from Section 3.4. Under minimax regret, risk is measured net of the risk of an oracle who knows μ but is subject to the same coverage requirement,

$$R^\star(\mu)\equiv\inf_{S\in\mathcal{S}_\alpha}R(\mu,S).$$

Under Bayes risk, the researcher specifies a prior $\pi\in\Pi$ over ν_{-1}, which determines the Bayes risk $R(\pi,\varphi)$ of a test inversion confidence procedure as defined in Section 4.3. The risk of a general confidence procedure also depends on the remaining directions in moment space, over which the researcher specifies no prior. I therefore consider the worst-case Bayes risk over the set Π_π of probability distributions on $\mathcal{M}$ under which $\nu_{-1}=Q_{-1}\mu$ is distributed according to π.

Corollary A.2 (Minimax Regret and Bayes Risk)**.** *Under Assumptions 2.1 to 2.4, the following hold.*

(i) Suppose $R^\star(\mu)<\infty$ for every $\mu\in\mathcal{M}$. Then $R^\star(\mu)$ depends on μ only through $\nu_{-1}=Q_{-1}\mu$, and writing $R^\star(\nu_{-1})$ accordingly, there exists a solution to

$$\inf_{\varphi\in\Phi_\alpha}\ \sup_{\nu_{-1}\in Q_{-1}\mathcal{M}}\big[R(\nu_{-1},\varphi)-R^\star(\nu_{-1})\big].$$

Moreover, if $\varphi^\star$ solves the above problem, then $S_{\varphi^\star}$ has minimax regret, meaning it solves

$$\inf_{S\in\mathcal{S}_\alpha}\ \sup_{\mu\in\mathcal{M}}\ \big[R(\mu,S)-R^\star(\mu)\big].$$

(ii) Let $\pi\in\Pi$. There exists a solution to $\inf_{\varphi\in\Phi_\alpha}R(\pi,\varphi)$. For any such solution $\varphi^\star$, $S_{\varphi^\star}$ solves

$$\inf_{S\in\mathcal{S}_\alpha}\ \sup_{\rho\in\Pi_\pi}\ \int R(\mu,S)\,\rho(\mathrm{d}\mu).$$

A.9 Unbiased Risk Estimator

Write $p_{\nu_{-1}}$ for the density of $Y_{-1}\sim\mathcal{N}(\nu_{-1},Q_{-1}\Sigma Q_{-1}')$ and $p_{\nu_1|\nu_{-1}}(\,\cdot\mid y_{-1})$ for the conditional density of Y_1 given $Y_{-1}=y_{-1}$, so that $p_\nu(y)=p_{\nu_{-1}}(y_{-1})\,p_{\nu_1|\nu_{-1}}(y_1\mid y_{-1})$. Substituting this factorization into the definition of risk and exchanging the order of integration,

$$\begin{aligned}R(\nu_{-1},\varphi)&=\iint(1-\varphi(y))\,p_\nu(y)\,\mathbb{1}\{\nu\notin Q\mathcal{M}(0)\}\,\mathrm{d}y\,\mathrm{d}\nu_1\\&=\int p_{\nu_{-1}}(y_{-1})\int(1-\varphi(y_1,y_{-1}))\underbrace{\int p_{\nu_1|\nu_{-1}}(y_1\mid y_{-1})\,\mathbb{1}\{\nu\notin Q\mathcal{M}(0)\}\,\mathrm{d}\nu_1}_{\equiv\,\psi_{\nu_{-1}}(y_1,y_{-1})}\,\mathrm{d}y_1\,\mathrm{d}y_{-1}.\end{aligned}$$

Viewed as a function of ν_1, $p_{\nu_1|\nu_{-1}}(y_1 \mid y_{-1})$ represents the posterior density of ν_1 upon observing $Y = y$, under a flat prior on ν_1 with ν_{-1} known. So, $\psi_{\nu_{-1}}(y)$ represents the posterior probability that $\nu \notin Q\mathcal{M}(0)$. Given $Y_{-1} \sim \mathcal{N}(\nu_{-1}^{(l)}, Q_{-1}\Sigma Q_{-1}')$, the above expression for risk suggests the following unbiased estimator of $R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi},\tilde{\Gamma}})$,

$$\int \big(1 - \varphi_{\tilde{\pi},\tilde{\Gamma}}(y_1, Y_{-1})\big)\, \psi_{\nu_{-1}^{(l)}}(y_1, Y_{-1})\, \mathrm{d}y_1. \tag{11}$$

However, computing (11) would require locating the (potentially many) intervals on which the integrand is nonzero, which amounts to inverting the test $\varphi_{\tilde{\pi},\tilde{\Gamma}}$. I avoid this by approximating (11) via Monte Carlo integration, as described in the following paragraphs.

For the remainder of this section, I consider the leading case in which the identified set $\Theta(\mu)$ is a bounded interval for each $\mu \in \mathcal{M}$, as in all of the examples from Section 2.2. In particular, this implies that the slices $\{\nu_1 : (\nu_1, \nu_{-1}^{(l)}) \in Q\mathcal{M}(0)\}$ of the null set at the risk grid points are bounded intervals, which I denote by $[\underline{\nu}_1^{(l)}, \overline{\nu}_1^{(l)}]$. As a result, we may express $\psi_{\nu_{-1}^{(l)}}$ in closed form. To see this, note that the conditional distribution of Y_1 given $Y_{-1} = y_{-1}$ is $\mathcal{N}(\nu_1 + \beta'(y_{-1} - \nu_{-1}), \sigma^2)$, where β and σ^2 are the coefficient vector and residual variance in the population regression of Y_1 on Y_{-1},

$$\beta \equiv (Q_{-1}\Sigma Q_{-1}')^{-1} Q_{-1}\Sigma Q_1', \qquad \sigma^2 \equiv Q_1\Sigma Q_1' - Q_1\Sigma Q_{-1}'\beta,$$

with $\sigma^2 > 0$ since $Q\Sigma Q' \succ 0$ by Part (ii) of Lemma B.3. The posterior of ν_1 is therefore $\mathcal{N}(y_1 - \beta'(y_{-1} - \nu_{-1}^{(l)}), \sigma^2)$, so writing Φ for the standard normal CDF,

$$\psi_{\nu_{-1}^{(l)}}(y) = \Phi\bigg(\frac{\underline{\nu}_1^{(l)} - y_1 + \beta'(y_{-1} - \nu_{-1}^{(l)})}{\sigma}\bigg) + \Phi\bigg(\frac{y_1 - \beta'(y_{-1} - \nu_{-1}^{(l)}) - \overline{\nu}_1^{(l)}}{\sigma}\bigg).$$

To approximate (11) via Monte Carlo integration, suppose for now that the integrand is equal to zero outside of a bounded interval $B_{\tilde{\pi},\tilde{\Gamma}}(Y_{-1})$. Then for a chosen number $J \geq 1$ of auxiliary draws, let $\tilde{Y}_1^{(1)}, \ldots, \tilde{Y}_1^{(J)}$ be i.i.d. according to the density on $B_{\tilde{\pi},\tilde{\Gamma}}(Y_{-1})$ proportional to $\psi_{\nu_{-1}^{(l)}}(\,\cdot\,, Y_{-1})$, and set

$$\hat{R}(Y_{-1}; \nu_{-1}^{(l)}, \tilde{\pi}, \tilde{\Gamma}) \equiv \bigg(\int_{B_{\tilde{\pi},\tilde{\Gamma}}(Y_{-1})} \psi_{\nu_{-1}^{(l)}}(y_1, Y_{-1})\, \mathrm{d}y_1\bigg) \frac{1}{J} \sum_{j=1}^{J} \big(1 - \varphi_{\tilde{\pi},\tilde{\Gamma}}(\tilde{Y}_1^{(j)}, Y_{-1})\big).$$

Conditional on Y_{-1}, $\hat{R}$ has expectation (11) by construction, meaning $\hat{R}$ is unbiased for $R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi},\tilde{\Gamma}})$. In addition, both the integral in the above expression as well as the CDF of $\tilde{Y}_1^{(j)}$ are available in closed form. The auxiliary draws can therefore be obtained efficiently via inverse transform sampling. Similarly to the type I error rate estimator, the main computational burden is evaluating the test $\varphi_{\tilde{\pi},\tilde{\Gamma}}$ itself, which cannot be avoided.

To implement the proposed estimator, I must first compute a bounded interval $B_{\tilde{\pi},\tilde{\Gamma}}(Y_{-1})$ that contains the support of the integrand in (11). Ideally, we would like this interval to be as short as possible in order to increase the precision of $\hat{R}$. Of course, the choice of interval will also be subject to computational trade-offs. My preferred interval can be computed via the following algorithm.

Algorithm 2 Bounding interval $B_{\tilde{\pi},\tilde{\Gamma}}(y_{-1})$

Require: y_{-1}; $\tilde{\pi}$ and $\tilde{\Gamma}$; risk and null grids; β and σ

for $l = 1, \dots, L$ **do**
  $u_l \leftarrow \tilde{\pi}_l \, p_{\nu_{-1}^{(l)}}(y_{-1})$; $\quad f_l \leftarrow 2\Phi\Big(-\frac{\overline{\nu}_1^{(l)} - \underline{\nu}_1^{(l)}}{2\sigma}\Big)$
  $\underline{c}_l \leftarrow \underline{\nu}_1^{(l)} + \beta'(y_{-1} - \nu_{-1}^{(l)})$; $\quad \overline{c}_l \leftarrow \overline{\nu}_1^{(l)} + \beta'(y_{-1} - \nu_{-1}^{(l)})$
end for
for $m = 1, \dots, M$ **do**
  $v_m \leftarrow \tilde{\Gamma}_m \, p_{\nu_{-1}^{(m)}}(y_{-1})$; $\quad c_m \leftarrow \nu_1^{(m)} + \beta'(y_{-1} - \nu_{-1}^{(m)})$
end for
$\overline{B} \leftarrow \textsc{UpperBound}(c, \underline{c}, \overline{c})$
$\underline{B} \leftarrow -\textsc{UpperBound}(-c, -\overline{c}, -\underline{c})$ ▷ reflection $y_1 \mapsto -y_1$
return $[\underline{B}, \overline{B}]$

function $\textsc{UpperBound}(c, \underline{c}, \overline{c})$ ▷ u, v, f and σ as above
  for $m = 1, \dots, M$ **do**
    $a \leftarrow \sum_{l=1}^{L} u_l \max\{f_l, \ \frac{1}{2}\mathbb{1}\{\overline{c}_l \le c_m\}\}$
    $g \leftarrow \sum_{m' : c_{m'} \le c_m} v_{m'}$; $\quad h \leftarrow \sum_{m' : c_{m'} > c_m} v_{m'}$
    if $\sqrt{2\pi}\,\sigma a > h$ **then**
      $b_m \leftarrow c_m + \sigma\sqrt{2\max\{0, \ \log\big(\frac{g}{\sqrt{2\pi}\,\sigma a - h}\big)\}}$
    else
      $b_m \leftarrow +\infty$
    end if
  end for
  return $\min_{m \le M} b_m$
end function

Proposition A.3 (Bounding Interval)**.** *Suppose* $\tilde{\Gamma} > 0$*, and let* $[\underline{B}, \overline{B}]$ *be the output of Algorithm 2 at* y_{-1}*. Then* $\underline{B}$ *and* $\overline{B}$ *are finite, and* $\varphi_{\tilde{\pi},\tilde{\Gamma}}(y_1, y_{-1}) = 1$ *for every* $y_1 \notin [\underline{B}, \overline{B}]$*.*

A.10 Simulation Study Specification Details

A.10.1 Papers

Table 2 lists the six papers used in the simulation study of Section 6, along with the number of pre-periods and the conditional standard error of $\hat{\mu}_{-1}$.

Table 2: Papers in the Simulation Study

Paper	$\underline{T}$	$\tilde{\sigma}_{-1}$
Kuziemko et al. (2018)	2	0.62
Fitzpatrick and Lovenheim (2014)	3	0.83
He and Wang (2017)	3	0.79
Markevich and Zhuravskaya (2018)	3	1.19
Deryugina (2017)	4	0.71
Ujhelyi (2014)	4	0.27

A.10.2 Units of μ_{-1}

To motivate plotting performance against $\mu_{-1}/\tilde{\sigma}_{-1}$, write $\hat{\mu}^{\text{pre}} \equiv (\hat{\mu}_{-1}, \ldots, \hat{\mu}_{-\underline{T}})'$ for the estimate of μ^{pre} and $\Sigma^{\text{pre}} \equiv \text{Var}(\hat{\mu}^{\text{pre}})$ for the corresponding block of Σ. Then $\tilde{\sigma}_{-1}^2 = 1/[(\Sigma^{\text{pre}})^{-1}]_{11}$, so $\tilde{\sigma}_{-1}$ is at most the standard error of $\hat{\mu}_{-1}$, with equality if and only if $\hat{\mu}_{-1}$ is uncorrelated with the remaining pre-period estimates. By the Neyman–Pearson lemma, the most powerful test of $H_0 : \mu^{\text{pre}} = 0$ against $H_1 : \mu^{\text{pre}} = \mu_{-1} e_1$ with $\mu_{-1} > 0$ rejects for large values of

$$\frac{e_1'(\Sigma^{\text{pre}})^{-1}\hat{\mu}^{\text{pre}}}{\sqrt{[(\Sigma^{\text{pre}})^{-1}]_{11}}} \sim \mathcal{N}\Big(\frac{\mu_{-1}}{\tilde{\sigma}_{-1}}, 1\Big),$$

Its power at level α is $\Phi(\mu_{-1}/\tilde{\sigma}_{-1} - z_{1-\alpha})$, as claimed in Section 6.1.

A.10.3 The Reduced Problem

Let $\bar{M} > 0$. First, I characterize the set $\mathcal{M}(0)$ under $\Delta^{\text{RM}}(\bar{M})$ and $\Delta^{\text{SDRM}}(\bar{M})$. Notice that $\theta = \tau_1 = 0$ is consistent with μ if and only if there exists $\delta \in \Delta$ with $(\delta_1, \delta^{\text{pre}}) = (\mu_1, \mu^{\text{pre}})$. For each definition of Δ, the only constraint that matters is the one corresponding to $t = 0$, since $\delta_{\overline{T}}, \ldots, \delta_2$ can be freely chosen to satisfy the other constraints. In other words, recalling that $\mu_0 = 0$,

$$\mathcal{M}(0) = \begin{cases} \{\mu : |\mu_1| \le \bar{M} \max_{s<0} |\mu_{s+1} - \mu_s|\} & \text{under } \Delta^{\text{RM}}(\bar{M}), \\ \{\mu : |\mu_1 + \mu_{-1}| \le \bar{M} \max_{s<0} |(\mu_{s+1} - \mu_s) - (\mu_s - \mu_{s-1})|\} & \text{under } \Delta^{\text{SDRM}}(\bar{M}), \end{cases}$$

where the second case requires $\underline{T} \ge 2$.

Next, I construct the matrix Q in each case. Recall that the rows of Q must form a basis of $V^{\perp}$ with $QKl = e_1$. Under $\Delta^{\text{RM}}(\bar{M})$, the coordinates $\mu_{\overline{T}}, \ldots, \mu_2$ do not enter the above expression for $\mathcal{M}(0)$, and are therefore uninformative. Let V be given by their span, so $k_\nu = \underline{T} + 1$. Then we may specify the reduced

moments as

$$\nu_1(\tilde{t}) = \mu_1 - \tilde{t}, \qquad \nu_{-1} = \big(\mu_{s+1} - \mu_s\big)_{s=-1}^{-\underline{T}},$$

with Q defined accordingly. Here I denote the posited value of θ by $\tilde{t}$, so as to avoid confusion with the time index t.

Under $\Delta^{\mathrm{SDRM}}(\bar{M})$, there is an additional uninformative direction. Adding a linear trend to the moments, $\mu_t \mapsto \mu_t + ct$ for every t, keeps second differences the same, so it does not affect whether $\mu \in \mathcal{M}(0)$. Let V be spanned by the coordinates $\mu_{\overline{T}}, \ldots, \mu_2$ along with the linear trend $(\overline{T}, \ldots, 1, -1, \ldots, -\underline{T})'$, so $k_\nu = \underline{T}$. Then we may specify the reduced moments as

$$\nu_1(\tilde{t}) = \mu_1 + \mu_{-1} - \tilde{t}, \qquad \nu_{-1} = \big((\mu_{s+1} - \mu_s) - (\mu_s - \mu_{s-1})\big)_{s=-1}^{-\underline{T}+1},$$

with Q defined accordingly.

In both cases we have that

$$\begin{aligned} Q\mathcal{M}(0) &= \big\{\nu : |\nu_1| \leq \bar{M} \max_j |(\nu_{-1})_j|\big\}, \\ \Theta(\mu) &= \big[\nu_1(0) - \bar{M} \max_j |(\nu_{-1})_j|, \nu_1(0) + \bar{M} \max_j |(\nu_{-1})_j|\big], \end{aligned}$$

expressing the identified set in terms of the reduced moments. Since the identified set is never empty, neither model is testable, so $\mathcal{M} = \mathbb{R}^{k_\mu}$ and $Q_{-1}\mathcal{M} = \mathbb{R}^{k_\nu - 1}$. In the simulation design of Section 6.1, it follows that the identified set for $\theta = \tau_1$ under $\Delta^{\mathrm{RM}}(1)$ is given by

$$\Theta(\mu) = \big[\mu_1 - \mu_{-1}, \mu_1 + \mu_{-1}\big],$$

while under $\Delta^{\mathrm{SDRM}}(1)$ it is given by

$$\Theta(\mu) = \big[\mu_1 - \mu_{-1}, \mu_1 + 3\mu_{-1}\big],$$

since $\mu_{-1} \geq 0$ by convention.

A.10.4 Decomposition

In both cases, the reduced null set may be represented as a finite union of polyhedral cones,

$$Q\mathcal{M}(0) = \bigcup_{j=1}^{k_\nu - 1} \bigcup_{\sigma \in \{-1,1\}} \big\{\nu : \nu_1 - \sigma \bar{M} (\nu_{-1})_j \leq 0, -\nu_1 - \sigma \bar{M} (\nu_{-1})_j \leq 0\big\}.$$

The above decomposition corresponds to that of $\Delta^{\mathrm{RM}}(\bar{M})$ in Rambachan and Roth (2023, footnote 12).

B Proofs

B.1 Miscellaneous Results

Proposition A.1 (Normalization)**.** *Under Assumptions 2.1 to 2.4, let $c > 0$ and suppose that Ψ_c and Ψ satisfy $\Psi_c(cS) = c\,\Psi(S)$ for every $S \in \mathcal{S}$. Then*

$$c \inf_{S \in \mathcal{S}_\alpha} \Psi(S) = \inf_{S \in \mathcal{S}_{\alpha,c}} \Psi_c(S),$$

and $S^\star$ solves the left-hand side if and only if $cS^\star$ solves the right-hand side.

Proof. Throughout, mark with a subscript c the objects defined in Sections 2 to 3 for the model with cl in place of l, so that $\mathcal{M}_c(t)$ denotes its null parameter space, $\Theta_c(\mu)$ its identified set, and $R_c(\mu, S)$ its risk function. Since $ct = (cl)'\tau$ if and only if $t = l'\tau$, $\mathcal{M}_c(ct) = \mathcal{M}(t)$ for every $t \in \mathbb{R}$. Similarly, $\Theta_c(\mu) = \{ct \in \mathbb{R} : \mu \in \mathcal{M}_c(ct)\} = c\,\Theta(\mu)$ for every $\mu \in \mathcal{M}$, and the parameter space $\mathcal{M} = \text{range}(K) + \Delta$ is common to both models.

Let $S \in \mathcal{S}$ and let I_S be as in Part (iii) of Lemma B.1. For every (x, u) and $B \in \mathcal{B}$, the change of variables $t = cs$ yields

$$\int_B I_S(t/c, x, u)\,\mathrm{d}t = c \int_{B/c} I_S(s, x, u)\,\mathrm{d}s = c\,\lambda(S(x, u) \cap B/c) = \lambda_{cS(x,u)}(B),$$

so $(t, x, u) \mapsto I_S(t/c, x, u)$ satisfies the conditions of Part (iii) for cS. It follows that $C(ct, \mu, cS) = C(t, \mu, S)$ for every $t \in \mathbb{R}$ and $\mu \in \mathcal{M}$, so since $\mathcal{M}_c(ct) = \mathcal{M}(t)$,

$$\inf_{\mu \in \mathcal{M}_c(ct)} C(ct, \mu, cS) = \inf_{\mu \in \mathcal{M}(t)} C(t, \mu, S) \quad \text{for every } t \in \mathbb{R}.$$

Since $t \mapsto ct$ preserves null sets, taking essential infima over t establishes that the coverage value of cS in the rescaled model equals that of S, which is well-defined by Lemma B.5. Hence $cS \in \mathcal{S}_{\alpha,c}$ if and only if $S \in \mathcal{S}_\alpha$, so since $S \mapsto cS$ is a bijection from $\mathcal{S}$ onto itself, it must map $\mathcal{S}_\alpha$ onto $\mathcal{S}_{\alpha,c}$.

The claims of the proposition now follow:

$$c \inf_{S \in \mathcal{S}_\alpha} \Psi(S) = \inf_{S \in \mathcal{S}_\alpha} \Psi_c(cS) = \inf_{S \in \mathcal{S}_{\alpha,c}} \Psi_c(S),$$

where the first equality is the hypothesis on Ψ_c and Ψ, and the second holds because the bijection maps $\mathcal{S}_\alpha$ onto $\mathcal{S}_{\alpha,c}$. Since $c > 0$, $S^\star$ attains the first infimum if and only if $cS^\star$ attains the last.

Finally, I verify that every decision problem considered in Sections A.8 and 3.4 satisfies the hypothesis, taking Ψ_c to be the objective of the corresponding problem for the rescaled model. Expanding the expression for risk,

$$R_c(\mu, cS) = \mathbb{E}_{P_{\mu,U}}\big[\lambda\big(cS(X, U) \setminus c\,\Theta(\mu)\big)\big] = c\,\mathbb{E}_{P_{\mu,U}}\big[\lambda\big(S(X, U) \setminus \Theta(\mu)\big)\big] = c\,R(\mu, S)$$

for every $\mu \in \mathcal{M}$ and $S \in \mathcal{S}$, and likewise $R_\lambda(\mu, cS) = c\, R_\lambda(\mu, S)$ under length loss. Taking suprema over $\mu \in \mathcal{M}$, the maximum risk objectives under excess length and length loss satisfy the hypothesis. For maximum regret,

$$R_c^\star(\mu) = \inf_{S \in \mathcal{S}_{\alpha,c}} R_c(\mu, S) = \inf_{S \in \mathcal{S}_\alpha} R_c(\mu, cS) = c\, R^\star(\mu),$$

meaning $R_c(\mu, cS) - R_c^\star(\mu) = c\big[R(\mu, S) - R^\star(\mu)\big]$, so the hypothesis follows from taking the supremum over $\mu \in \mathcal{M}$. For worst-case Bayes risk, the two models share the lineality space V, since it is defined in terms of $\mathcal{M}(0) = \mathcal{M}_c(0)$, so the matrix Q for the rescaled model may be chosen by rescaling the first row of Q to Q_1/c. In particular, Q_{-1}, and hence the class Π_π, is common to both models. Then $\int R_c(\mu, cS)\, \rho(\mathrm{d}\mu) = c \int R(\mu, S)\, \rho(\mathrm{d}\mu)$ for every $\rho \in \Pi_\pi$, and the hypothesis follows from taking the supremum over $\rho \in \Pi_\pi$. □

Proposition A.2 (Informativeness)**.** *Under Assumption 2.1, $Kl \notin V$ if and only if for some $\mu \in \mathcal{M}$, $\Theta(\mu) \neq \mathbb{R}$. Moreover, $Kl \notin V$ implies that l lies in the row space of K.*

Proof. First, suppose $Kl \in V$. Then $tKl + \mathcal{M}(0) = \mathcal{M}(0)$ for all $t \in \mathbb{R}$, so by (3) $\mathcal{M}(t) = \mathcal{M}(0) = \mathcal{M}$ meaning $\Theta(\mu) = \mathbb{R}$ for all $\mu \in \mathcal{M}$. Now, suppose $Kl \notin V$, and fix $t \in \mathbb{R}$ with $\mathcal{M}(t) \neq \mathcal{M}(0)$. Then there exists μ belonging to exactly one of $\mathcal{M}(t)$ and $\mathcal{M}(0)$. Such a μ lies in $\mathcal{M}$, and $\Theta(\mu)$ contains exactly one of t and 0, so $\Theta(\mu) \neq \mathbb{R}$.

To verify the second claim, suppose l does not lie in the row space of K. Then l is not orthogonal to $\ker(K)$, so we may choose $v \in \ker(K)$ with $l'v = 1$. Since $l'(l - v) = 0$ by Assumption 2.1, the vector $l - v$ lies in $\ker(l')$, so $Kl = K(l - v) \in K\ker(l')$. But $K\ker(l') \subseteq V$ since $\mathcal{M}(0) = K\ker(l') + \Delta$, meaning $Kl \in V$. □

Lemma B.1 (Measurability Facts)**.** *Maintain Assumptions 2.1 to 2.4. Then*

(i) The sets $\mathcal{M}$, $\mathcal{M}(t)$ for each $t \in \mathbb{R}$, and $\Theta(\mu)$ for each $\mu \in \mathcal{M}$ are non-empty and σ-compact, hence Borel.

(ii) The map $S : \mathbb{R}^{k_\mu} \times [0,1] \to \mathcal{A}$ is measurable if and only if $(x, u) \mapsto \lambda(S(x, u) \cap B)$ is measurable for all $B \in \mathcal{B}$.

(iii) For each $S \in \mathcal{S}$, there exists a measurable map $I_S : \mathbb{R} \times \mathbb{R}^{k_\mu} \times [0,1] \to [0,1]$ such that for every (x, u), the slice $I_S(\,\cdot\,, x, u)$ is a version of $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$.

(iv) For each $S \in \mathcal{S}$, if I_S and I_S' both satisfy the conditions of Part (iii), then $I_S = I_S'$ for almost every (t, x, u). In particular, for each $\mu \in \mathcal{M}$,

$$\mathbb{E}_{P_{\mu,U}}[I_S(t, X, U)] = \mathbb{E}_{P_{\mu,U}}[I_S'(t, X, U)] \quad \text{for almost every } t \in \mathbb{R}.$$

(v) Say that $\overline{S} : \mathbb{R}^{k_\mu} \times [0,1] \to \mathcal{B}$ is a measurable representative of $S \in \mathcal{S}$ if $\overline{S}(x, u)$ belongs to the equivalence class $S(x, u)$ for every (x, u), and $(t, x, u) \mapsto \mathbb{1}\{t \in \overline{S}(x, u)\}$ is measurable. Then $S \in \mathcal{S}_\alpha$ if and only if S admits a measurable representative $\overline{S}$ such that

$$\inf_{\mu \in \mathcal{M}} \inf_{t \in \Theta(\mu)} \Pr_{P_{\mu,U}}(t \in \overline{S}(X, U)) \geq 1 - \alpha.$$

Proof of Lemma B.1. Part (i). First, note that images and preimages of σ-compact sets under continuous maps between Euclidean spaces are σ-compact, as are finite products of σ-compact sets, and finally that σ-compact sets are Borel. It therefore suffices to express each set in the statement as the image of a non-empty σ-compact set under a linear map.

First, $\mathcal{M}$ is the image of $\mathbb{R}^{k_\tau} \times \Delta$ under $(\tau, \delta) \mapsto K\tau + \delta$. Second, for each $t \in \mathbb{R}$, $\mathcal{M}(t)$ is the image of $\{\tau \in \mathbb{R}^{k_\tau} : l'\tau = t\} \times \Delta$ under the same map. Finally, for each $\mu \in \mathcal{M}$, we have $t \in \Theta(\mu)$ if and only if $t = l'\tau$ for some τ with $\mu - K\tau \in \Delta$, so $\Theta(\mu)$ is the image of $\{\tau \in \mathbb{R}^{k_\tau} : \mu - K\tau \in \Delta\}$ under $\tau \mapsto l'\tau$. Each domain is σ-compact by Assumption 2.3, since $\{\tau \in \mathbb{R}^{k_\tau} : l'\tau = t\}$ and $\{\tau \in \mathbb{R}^{k_\tau} : \mu - K\tau \in \Delta\}$ are the preimages of $\{t\}$ and Δ under the continuous maps $\tau \mapsto l'\tau$ and $\tau \mapsto \mu - K\tau$. Each domain is also non-empty: the first since Δ is non-empty by Assumption 2.3; the second since $l'(tl) = t$ by Assumption 2.1; and the third since $\mu \in \mathcal{M}$ implies that $\mu = K\tau + \delta$ for some $(\tau, \delta) \in \mathbb{R}^{k_\tau} \times \Delta$, so $\mu - K\tau \in \Delta$.

Part (ii). Any finite measure equivalent to λ induces the same topology on $\mathcal{A}$ (Le Maître, 2026, Lemma 2.1); for concreteness I use the Laplace probability measure $\rho(\mathrm{d}t) \equiv w(t)\,\mathrm{d}t$ with density $w(t) \equiv \frac{1}{2}e^{-|t|} > 0$, and metrize $\mathcal{A}$ with $d_\rho(A, A') \equiv \rho(A \triangle A')$.

Evaluation maps. I first verify that for each measurable $g : \mathbb{R} \to \overline{\mathbb{R}}_+$, the evaluation map $A \mapsto \int_A g\,\mathrm{d}\lambda$ is well-defined and measurable on $\mathcal{A}$. Suppose first that $\int g\,\mathrm{d}\lambda < \infty$, and let $\varepsilon > 0$. Since $w > 0$, we have $g \wedge nw \uparrow g$ pointwise as $n \to \infty$, so by monotone convergence we may choose n with $\int (g - g \wedge nw)\,\mathrm{d}\lambda \leq \varepsilon/2$. Then for all $A, A' \in \mathcal{A}$,

$$\Big| \int_A g\,\mathrm{d}\lambda - \int_{A'} g\,\mathrm{d}\lambda \Big| \leq \int_{A \triangle A'} g\,\mathrm{d}\lambda \leq \int_{A \triangle A'} g \wedge nw\,\mathrm{d}\lambda + \frac{\varepsilon}{2} \leq n d_\rho(A, A') + \frac{\varepsilon}{2},$$

where the last inequality follows from the fact that $g \wedge nw \leq nw$. The final expression is at most ε whenever $d_\rho(A, A') \leq \varepsilon/2n$, so the evaluation map of g is continuous. For general g, each $g \wedge nw$ satisfies $\int g \wedge nw\,\mathrm{d}\lambda \leq n$, so the corresponding evaluation maps are continuous, and by monotone convergence

$$\int_A g\,\mathrm{d}\lambda = \sup_{n \in \mathcal{N}} \int_A g \wedge nw\,\mathrm{d}\lambda \quad \text{for every } A \in \mathcal{A}.$$

As a pointwise supremum of continuous functions, the evaluation map of g is lower semicontinuous, hence measurable.

Necessity. Suppose S is measurable. For each $B \in \mathcal{B}$, the map $(x, u) \mapsto \lambda(S(x, u) \cap B) = \int_{S(x,u)} \mathbb{1}_B\,\mathrm{d}\lambda$ is the composition of S with the evaluation map of $\mathbb{1}_B$, hence measurable.

Sufficiency. Suppose $(x, u) \mapsto \lambda(S(x, u) \cap B)$ is measurable for all $B \in \mathcal{B}$. I first upgrade from indicators to all nonnegative g. For simple $s = \sum_i c_i \mathbb{1}_{B_i}$ with $c_i \geq 0$,

$$\int_{S(x,u)} s\,\mathrm{d}\lambda = \sum_i c_i\,\lambda(S(x, u) \cap B_i)$$

is measurable in (x, u) by hypothesis. For general measurable $g \geq 0$, choose simple $s_m \uparrow g$ pointwise; by monotone convergence, $\int_{S(x,u)} g\,\mathrm{d}\lambda = \lim_{m \to \infty} \int_{S(x,u)} s_m\,\mathrm{d}\lambda$ for each (x, u), so the limit is measurable as well. Now, by Le Maître (2026, Proposition 2.4), the finite unions $\mathcal{J}$ of open intervals with rational endpoints form

a countable dense subset of $\mathcal{A}$, so

$$\mathcal{B}(\mathcal{A}) = \sigma(\{A \mapsto d_\rho(A, J) : J \in \mathcal{J}\}),$$

and it suffices to show that $(x, u) \mapsto d_\rho(S(x, u), J)$ is measurable for each $J \in \mathcal{J}$. Expanding the symmetric difference,

$$d_\rho(S(x,u), J) = \rho(S(x,u)) + \rho(J) - 2\rho(S(x,u) \cap J) = \int_{S(x,u)} w \, \mathrm{d}\lambda + \rho(J) - 2\int_{S(x,u)} w \mathbb{1}_J \, \mathrm{d}\lambda,$$

which is measurable in (x, u) by the above, applied to $g = w$ and $g = w\mathbb{1}_J$.

Part (iii). Fix $S \in \mathcal{S}$. Since $\lambda_{S(x,u)}(B) = \lambda(S(x,u) \cap B) \le \lambda(B)$ for every $B \in \mathcal{B}$, the measure $\lambda_{S(x,u)}$ is absolutely continuous with respect to λ, with density $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda \in [0, 1]$. For $n \in \mathbb{N}$, define

$$I_{S,n}(t, x, u) \equiv \tfrac{n}{2} \, \lambda_{S(x,u)}\big((t - \tfrac{1}{n}, t + \tfrac{1}{n})\big),$$

which takes values in $[0, 1]$ since $\lambda_{S(x,u)} \le \lambda$. For fixed (x, u), the map $t \mapsto I_{S,n}(t, x, u)$ is continuous, since for any $B, B' \in \mathcal{B}$,

$$|\lambda_{S(x,u)}(B) - \lambda_{S(x,u)}(B')| \le \lambda_{S(x,u)}(B \triangle B') \le \lambda(B \triangle B'),$$

and as $t' \to t$ the symmetric difference of the intervals $(t' - \frac{1}{n}, t' + \frac{1}{n})$ and $(t - \frac{1}{n}, t + \frac{1}{n})$ has vanishing Lebesgue measure. For fixed t, the map $(x, u) \mapsto I_{S,n}(t, x, u)$ is measurable by Part (ii). Thus $I_{S,n}$ is a Carathéodory function, hence measurable in (t, x, u) (Aliprantis and Border, 2006, Lemma 4.51). Now define

$$I_S(t, x, u) \equiv \liminf_{n \to \infty} I_{S,n}(t, x, u),$$

which is measurable as a limit inferior of countably many measurable functions, with $0 \le I_S \le 1$. Finally, fix (x, u). The density $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$ is bounded, hence locally integrable, so by the Lebesgue differentiation theorem (Stein and Shakarchi, 2005), for almost every t,

$$\lim_{\varepsilon \downarrow 0} \tfrac{1}{2\varepsilon} \, \lambda_{S(x,u)}\big((t - \varepsilon, t + \varepsilon)\big) = \frac{\mathrm{d}\lambda_{S(x,u)}}{\mathrm{d}\lambda}(t).$$

At every such t, the limit along $\varepsilon = 1/n$ exists and coincides with the full limit, so $I_S(\,\cdot\,, x, u) = \mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$ almost everywhere. Hence $I_S(\,\cdot\,, x, u)$ is a version of $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$ for every (x, u).

Part (iv). Since I_S and I'_S both satisfy the conditions of Part (iii), for all (and in particular a.e.) (x, u), $I_S(t, x, u) = I'_S(t, x, u)$ for a.e. t, since both are versions of the same density $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$. By Tonelli, then, $I_S = I'_S$ for a.e. (t, x, u), and for each $\mu \in \mathcal{M}$,

$$\mathbb{E}_{P_{\mu,U}}[I_S(t, X, U)] = \mathbb{E}_{P_{\mu,U}}[I'_S(t, X, U)] \quad \text{for almost every } t \in \mathbb{R},$$

since $P_{\mu,U} = P_\mu \otimes \mathrm{Unif}[0, 1] \ll \lambda_{k_\mu+1}$, the Lebesgue measure on $\mathbb{R}^{k_\mu+1}$, by Assumption 2.4.

Part (v). First, note that if $\overline{S}$ is any measurable representative of $S \in \mathcal{S}$, then $(t,x,u) \mapsto \mathbb{1}\{t \in \overline{S}(x,u)\}$ satisfies the conditions of Part (iii): it is measurable and $[0,1]$-valued by definition, and for each (x,u) its slice is a version of $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$, since for every $B \in \mathcal{B}$,

$$\lambda_{S(x,u)}(B) = \lambda(\overline{S}(x,u) \cap B) = \int_B \mathbb{1}\{t \in \overline{S}(x,u)\}\,\mathrm{d}t.$$

By the proof of Lemma B.5, the coverage value of S may therefore be computed using this map,

$$\operatorname*{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,S) = \operatorname*{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} \Pr_{P_{\mu,U}}(t \in \overline{S}(X,U)). \tag{12}$$

Sufficiency. Since $\mu \in \mathcal{M}(t)$ if and only if $t \in \Theta(\mu)$, the right hand side of (12) is bounded below by

$$\inf_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} \Pr_{P_{\mu,U}}(t \in \overline{S}(X,U)) = \inf_{\mu\in\mathcal{M}} \inf_{t\in\Theta(\mu)} \Pr_{P_{\mu,U}}(t \in \overline{S}(X,U)) \geq 1-\alpha,$$

so $S \in \mathcal{S}_\alpha$ by the definition of $\mathcal{S}_\alpha$.

Necessity. Let $S \in \mathcal{S}_\alpha$ and I_S be as in Part (iii), and define $\widetilde{S}(x,u) \equiv \{t \in \mathbb{R} : I_S(t,x,u) = 1\}$, whose indicator $\mathbb{1}\{t \in \widetilde{S}(x,u)\}$ is measurable by construction. Fix (x,u) and let $B \in \mathcal{B}$ be any set in the class $S(x,u)$. Then both $\mathbb{1}_B$ and $I_S(\,\cdot\,,x,u)$ are versions of $\mathrm{d}\lambda_{S(x,u)}/\mathrm{d}\lambda$, so they agree almost everywhere. Hence $\widetilde{S}(x,u) \triangle B \subseteq \{t \in \mathbb{R} : I_S(t,x,u) \neq \mathbb{1}_B(t)\}$ is null, meaning $\widetilde{S}(x,u)$ belongs to $S(x,u)$, so $\widetilde{S}$ is a measurable representative.

Applying (12) with $\widetilde{S}$ in place of $\overline{S}$, there must exist a Borel null set $N \subseteq \mathbb{R}$ such that $\inf_{\mu\in\mathcal{M}(t)} \Pr_{P_{\mu,U}}(t \in \widetilde{S}(X,U)) \geq 1-\alpha$ for every $t \notin N$. Let $\overline{S}(x,u) \equiv \widetilde{S}(x,u) \cup N$, whose indicator $\max\{\mathbb{1}\{t \in \widetilde{S}(x,u)\}, \mathbb{1}\{t \in N\}\}$ is measurable. Since $\widetilde{S}(x,u) \triangle \overline{S}(x,u) \subseteq N$, the set $\overline{S}(x,u)$ also belongs to $S(x,u)$, so $\overline{S}$ is a measurable representative. Finally, let $\mu \in \mathcal{M}$ and $t \in \Theta(\mu)$, so that $\mu \in \mathcal{M}(t)$. If $t \in N$, then $\Pr_{P_{\mu,U}}(t \in \overline{S}(X,U)) = 1$; if $t \notin N$, then $t \in \overline{S}(x,u)$ if and only if $t \in \widetilde{S}(x,u)$, so $\Pr_{P_{\mu,U}}(t \in \overline{S}(X,U)) \geq 1-\alpha$. It follows that $\inf_{\mu\in\mathcal{M}} \inf_{t\in\Theta(\mu)} \Pr_{P_{\mu,U}}(t \in \overline{S}(X,U)) \geq 1-\alpha$. □

Lemma B.2 (Reparameterization)**.** *Suppose Assumptions 2.1 and 2.3 hold but Assumption 2.4 does not, meaning Σ is only positive semi-definite. If* $\mathrm{range}(K) \subseteq \mathrm{range}(\Sigma)$*, then the model admits an equivalent reparameterization under which Assumptions 2.1, 2.3 and 2.4 hold.*

Proof. Let the rows of R and S form orthonormal bases of $\mathrm{range}(\Sigma)$ and $\ker(\Sigma)$, respectively. Observing X is equivalent to observing the pair (RX, SX), with $SX = S\mu$ almost surely. I therefore condition on this event and treat $S\mu$ as known throughout the remainder of the proof. Fix $\tau \in \mathbb{R}^{k_\tau}$. The hypothesis $\mathrm{range}(K) \subseteq \mathrm{range}(\Sigma)$ implies that $SK = 0$, so

$$\mu = K\tau + \delta \iff \left[\begin{smallmatrix} R \\ S \end{smallmatrix}\right]\mu = \left[\begin{smallmatrix} R \\ S \end{smallmatrix}\right]K\tau + \left[\begin{smallmatrix} R \\ S \end{smallmatrix}\right]\delta \iff R\mu = RK\tau + R\delta \quad \text{and} \quad S\mu = S\delta.$$

This suggests the reparameterization

$$\widetilde{\mu} \equiv R\mu, \quad \widetilde{K} \equiv RK, \quad \widetilde{\Delta} \equiv \{R\delta : \delta \in \Delta, S\delta = S\mu\},$$

under which $\mu = K\tau + \delta$ for some $\delta \in \Delta$ if and only if $\widetilde{\mu} = \widetilde{K}\tau + \widetilde{\delta}$ for some $\widetilde{\delta} \in \widetilde{\Delta}$. Hence, the reparameterized model is equivalent to the original model in the sense that τ is consistent with $\widetilde{\mu}$ under the former if and only if it is consistent with μ under the latter. The value of l is identical in both models, so identified sets are preserved and Assumption 2.1 holds. Assumption 2.3 holds since $\widetilde{\Delta}$ is the image of the intersection of a σ-compact set Δ and a closed set $\{\delta : S\delta = S\mu\}$ under the linear map $\delta \mapsto R\delta$, which is non-empty since $\mu \in \mathcal{M}$. Finally, under the reparameterization, the researcher observes $\widetilde{X} \sim \mathcal{N}(\widetilde{\mu}, \widetilde{\Sigma})$ for $\widetilde{X} \equiv RX$ and $\widetilde{\Sigma} \equiv R\Sigma R' \succ 0$ by construction, so Assumption 2.4 holds.

The known moment directions $S\mu$ are absorbed into $\widetilde{\Delta}$, so the reparameterized model depends on the value of $SX = S\mu$ that is actually observed. In particular, it may be the case that $\Theta(m) = \mathbb{R}$ for all $m \in \mathcal{M}$ such that $Sm = S\mu$, even though $\Theta(m) \neq \mathbb{R}$ for some $m \in \mathcal{M}$. So by Proposition A.2, Assumption 2.2 may fail under the reparameterization even if it holds in the original model, which explains why it does not appear in the statement of this lemma. □

Lemma B.3 (Reduced Testing Problem)**.** *Maintain Assumptions 2.1 to 2.4, and let* $Q \equiv \left[\begin{smallmatrix} Q_1 \\ Q_{-1} \end{smallmatrix}\right]$ *be a matrix whose rows form a basis of* $V^\perp$*. Then*

(i) For all $t \in \mathbb{R}$ *and* $\mu \in \mathbb{R}^{k_\mu}$*,*

$$\mu \in \mathcal{M}(t) \iff \mu - tKl \in \mathcal{M}(0) \iff Q(\mu - tKl) \in Q\mathcal{M}(0).$$

(ii) The sets $Q\mathcal{M}(0)$ *and* $Q_{-1}\mathcal{M}$ *are non-empty and* σ*-compact, hence Borel, with* $Q\Sigma Q' \succ 0$ *and*

$$\{u \in \mathbb{R}^{k_\nu} : su + Q\mathcal{M}(0) = Q\mathcal{M}(0) \text{ for all } s \in \mathbb{R}\} = \{0\}.$$

(iii) The matrix Q *may be chosen so that* $QKl = e_1 \in \mathbb{R}^{k_\nu}$*.*

Proof. Since the rows of Q form a basis of $V^\perp$, the matrix Q has full row rank k_ν and $\ker(Q) = (V^\perp)^\perp = V$. In particular, Q is surjective and Q' is injective.

Part (i). Let $t \in \mathbb{R}$ and $\mu \in \mathbb{R}^{k_\mu}$. The first equivalence follows from (3). For the second, the forward direction is immediate. To see the reverse direction, suppose that $Q(\mu - tKl) = Q\mu'$ for some $\mu' \in \mathcal{M}(0)$. Then $\mu - tKl - \mu' \in \ker(Q) = V$, so $\mu - tKl = \mu' + (\mu - tKl - \mu') \in \mathcal{M}(0)$ by (4).

Part (ii). The sets $\mathcal{M}(0)$ and $\mathcal{M}$ are non-empty by Assumption 2.3 and σ-compact by Part (i) of Lemma B.1, so their images $Q\mathcal{M}(0)$ and $Q_{-1}\mathcal{M}$ under linear (hence continuous) maps are non-empty and σ-compact, hence Borel.

For $a \in \mathbb{R}^{k_\nu}$ we have $a'Q\Sigma Q'a = (Q'a)'\Sigma(Q'a) \geq 0$ by Assumption 2.4, with equality only if $Q'a = 0$, in which case $a = 0$ since Q' is injective. Hence $Q\Sigma Q' \succ 0$.

Finally, let $u \in \mathbb{R}^{k_\nu}$ be such that $su + Q\mathcal{M}(0) = Q\mathcal{M}(0)$ for all $s \in \mathbb{R}$. Since Q is surjective, we may choose $v \in \mathbb{R}^{k_\mu}$ with $Qv = u$. Let $\mu \in \mathcal{M}(0)$ and $s \in \mathbb{R}$. Then

$$Q(sv + \mu) = su + Q\mu \in Q\mathcal{M}(0),$$

so $sv + \mu \in \mathcal{M}(0)$ by the second equivalence in Part (i) with $t = 0$. Since $\mu \in \mathcal{M}(0)$ and $s \in \mathbb{R}$ were arbitrary, $sv + \mathcal{M}(0) \subseteq \mathcal{M}(0)$ for all $s \in \mathbb{R}$; applying this with $-s$ in place of s and adding sv to both sides yields $\mathcal{M}(0) \subseteq sv + \mathcal{M}(0)$, so $sv + \mathcal{M}(0) = \mathcal{M}(0)$. Hence $v \in V = \ker(Q)$ by (4), meaning $u = Qv = 0$.

Part (iii). Let Q_0 be any matrix whose rows form a basis of $V^\perp$, so that $\ker(Q_0) = V$ as shown above. Since $Kl \notin V$ by Assumption 2.2, the vector $a \equiv Q_0 Kl$ is nonzero, so it may be extended to a basis $a, a_2, \ldots, a_{k_\nu}$ of $\mathbb{R}^{k_\nu}$. Let A denote the inverse of the matrix with columns $a, a_2, \ldots, a_{k_\nu}$, so that $Aa = e_1$, and set $Q \equiv AQ_0$. Since A is invertible, the rows of Q span the same space as the rows of Q_0, so they form a basis of $V^\perp$, and $QKl = AQ_0Kl = e_1$. □

Lemma B.4 (Ties)**.** *Let $\pi \in \Pi$ and $\Gamma \in \mathbb{R}_+\Pi_0$. Then p_{ω_π} and p_Γ are real-analytic on $\mathbb{R}^{k_\nu}$, and the set $\{y \in \mathbb{R}^{k_\nu} : p_{\omega_\pi}(y) = p_\Gamma(y)\}$ has Lebesgue measure zero unless $\omega_\pi = 0$ and $\Gamma = 0$.*

Proof. *Analyticity.* Each p_ν is real-analytic, as the exponential of a polynomial. The difficulty lies in showing that mixing across ν according to the weights ω_π and Γ does not destroy this property. Complex analysis is a useful tool for this purpose, since the restriction of a holomorphic function to the reals is real-analytic, and holomorphy is easier to verify. With this in mind, view p_ν as a function from $\mathbb{C}^{k_\nu} \to \mathbb{C}$, and notice that for $z = a + ib$, $|p_\nu(z)| = \exp(\frac{1}{2}b'(Q\Sigma Q')^{-1}b)p_\nu(a)$. To show that $p_{\omega_\pi}(z) = \int p_\nu(z)\omega_\pi(\mathrm{d}\nu)$ is holomorphic, I verify the conditions of Mattner (2001).[17] First, $p_\nu(z)$ is continuous, hence measurable, in ν for every z. Second, p_ν is holomorphic as the exponential of a polynomial for each ν. Finally, $z \mapsto \int |p_\nu(z)|\omega_\pi(\mathrm{d}\nu)$ is locally bounded. To see this, for $z = a + ib$ write

$$\int |p_\nu(z)|\omega_\pi(\mathrm{d}\nu) = \exp(\tfrac{1}{2}b'(Q\Sigma Q')^{-1}b) \int p_\nu(a)\omega_\pi(\mathrm{d}\nu).$$

The claim then follows since $\int p_\nu(a)\omega_\pi(\mathrm{d}\nu)$ is bounded by the maximum value of the density of Y_{-1},

$$\int p_\nu(a)\omega_\pi(\mathrm{d}\nu) \leq \iint p_\nu(a)\,\mathrm{d}\nu_1\,\pi(\mathrm{d}\nu_{-1}) \leq (2\pi)^{-(k_\nu-1)/2} \det\left(Q_{-1}\Sigma Q_{-1}'\right)^{-1/2}.$$

Analogously, $z \mapsto \int |p_\nu(z)|\Gamma(\mathrm{d}\nu)$ is locally bounded since

$$\int p_\nu(a)\Gamma(\mathrm{d}\nu) \leq (2\pi)^{-k_\nu/2} \det\left(Q\Sigma Q'\right)^{-1/2}\Gamma(Q\mathcal{M}(0)).$$

Hence, both $p_{\omega_\pi}(z)$ and $p_\Gamma(z)$ are holomorphic by Mattner (2001, Part C_3), so their restrictions to the reals are real-analytic.

Ties. Suppose that $p_{\omega_\pi} = p_\Gamma$ on a set of positive Lebesgue measure. Then by Mityagin (2020), $p_{\omega_\pi} = p_\Gamma$ everywhere. Integrating the identity over $\mathbb{R}^{k_\nu}$ yields $\omega_\pi(\mathbb{R}^{k_\nu}) = \Gamma(\mathbb{R}^{k_\nu}) < \infty$, so ω_π is a finite measure. For $\xi \in \mathbb{R}^{k_\nu}$, equating characteristic functions and applying Fubini yields

$$\int e^{i\xi' y} p_{\omega_\pi}(y)\,\mathrm{d}y = \int e^{i\xi' y} p_\Gamma(y)\,\mathrm{d}y \implies e^{-\frac{1}{2}\xi' Q\Sigma Q'\xi} \int e^{i\xi'\nu}\,\omega_\pi(\mathrm{d}\nu) = e^{-\frac{1}{2}\xi' Q\Sigma Q'\xi} \int e^{i\xi'\nu}\,\Gamma(\mathrm{d}\nu).$$

[17] Strictly speaking, I require a mild generalization of the theorem in Mattner (2001) that allows for arguments $z \in \mathbb{C}^{k_\nu}$ rather than $z \in \mathbb{C}$. This follows easily from the argument in his second proof of conclusion C_3.

Since $e^{-\frac{1}{2}\xi' Q\Sigma Q'\xi} > 0$, it follows that ω_π and Γ share the same characteristic function and the same total mass, hence $\omega_\pi = \Gamma$ by the uniqueness theorem for characteristic functions. But $\omega_\pi(Q\mathcal{M}(0)) = 0$ whereas $\Gamma(Q\mathcal{M}(0)) = \Gamma(\mathbb{R}^{k_\nu})$, so $\Gamma = 0$ and therefore $\omega_\pi = 0$. □

Proposition A.3 (Bounding Interval)**.** *Suppose $\tilde{\Gamma} > 0$, and let $[\underline{B}, \overline{B}]$ be the output of Algorithm 2 at y_{-1}. Then $\underline{B}$ and $\overline{B}$ are finite, and $\varphi_{\tilde{\pi},\tilde{\Gamma}}(y_1, y_{-1}) = 1$ for every $y_1 \notin [\underline{B}, \overline{B}]$.*

Proof. Throughout, u_l, f_l, $\underline{c}_l$, $\overline{c}_l$, v_m and c_m are as computed in the first two loops of Algorithm 2. Fix y_1 and write $y = (y_1, y_{-1})$. By the definition of $p_{\omega_{\tilde{\pi}}}$, the factorization $p_\nu(y) = p_{\nu_{-1}}(y_{-1})\, p_{\nu_1|\nu_{-1}}(y_1 \mid y_{-1})$ and the closed form of $\psi_{\nu_{-1}^{(l)}}$,

$$p_{\omega_{\tilde{\pi}}}(y) = \sum_{l=1}^{L} \tilde{\pi}_l\, p_{\nu_{-1}^{(l)}}(y_{-1})\, \psi_{\nu_{-1}^{(l)}}(y) = \sum_{l=1}^{L} u_l \left[\Phi\left(\frac{\underline{c}_l - y_1}{\sigma}\right) + \Phi\left(\frac{y_1 - \overline{c}_l}{\sigma}\right)\right],$$

and, since $p_{\nu_1^{(m)}|\nu_{-1}^{(m)}}(\,\cdot \mid y_{-1})$ is the $\mathcal{N}(c_m, \sigma^2)$ density, writing ϕ for the standard normal density,

$$p_{\tilde{\Gamma}}(y) = \sum_{m=1}^{M} \tilde{\Gamma}_m\, p_{\nu_{-1}^{(m)}}(y_{-1})\, p_{\nu_1^{(m)}|\nu_{-1}^{(m)}}(y_1 \mid y_{-1}) = \sum_{m=1}^{M} v_m\, \frac{1}{\sigma}\, \phi\left(\frac{y_1 - c_m}{\sigma}\right).$$

Recall that $\varphi_{\tilde{\pi},\tilde{\Gamma}}(y) = \mathbb{1}\{p_{\omega_{\tilde{\pi}}}(y) > p_{\tilde{\Gamma}}(y)\}$.

Consider either call of UpperBound, and write $(c, \underline{c}, \overline{c})$ for its inputs, which are the quantities above or their reflections; in both cases $\overline{c}_l - \underline{c}_l = \overline{\nu}_1^{(l)} - \underline{\nu}_1^{(l)}$. I claim that the value returned is finite, and that for every t exceeding it,

$$\sum_{l=1}^{L} u_l \left[\Phi\left(\frac{\underline{c}_l - t}{\sigma}\right) + \Phi\left(\frac{t - \overline{c}_l}{\sigma}\right)\right] > \sum_{m=1}^{M} v_m\, \frac{1}{\sigma}\, \phi\left(\frac{t - c_m}{\sigma}\right). \tag{13}$$

The value is finite because, for the m with the largest c_m, $h = 0$ while $a \geq \sum_l u_l f_l > 0$, as $\tilde{\pi}$ has a positive entry, the marginal densities are positive and $\Phi > 0$. Since every t exceeding the returned value exceeds some finite b_m, it suffices to fix an m with $b_m < \infty$ and a $t > b_m$, and to prove (13) for this t. Put $w \equiv t - c_m$, so that $w > \sigma\sqrt{2\max\{0, \log(g/(\sqrt{2\pi}\,\sigma a - h))\}} \geq 0$, where $g \geq v_m > 0$ because $\tilde{\Gamma} > 0$.

The bracket in (13) is the probability that a $\mathcal{N}(t, \sigma^2)$ random variable falls outside $[\underline{c}_l, \overline{c}_l]$. Since the normal density is symmetric and unimodal, this probability is smallest when t is the midpoint of the interval, where it equals f_l. Moreover, its second term is at least $\Phi(0) = \frac{1}{2}$ whenever $t \geq \overline{c}_l$, which holds for every l with $\overline{c}_l \leq c_m$, since $t \geq c_m$. The left-hand side of (13) is therefore at least a. On the right-hand side, $\phi \leq \phi(0) = 1/\sqrt{2\pi}$, and for m' with $c_{m'} \leq c_m$ we have $t - c_{m'} \geq w \geq 0$, so $\phi((t - c_{m'})/\sigma) \leq \phi(w/\sigma)$, as ϕ is decreasing on $[0, \infty)$. Applying the first bound to the m' with $c_{m'} > c_m$, whose weights sum to h, and the second to the rest, whose weights sum to g, the right-hand side of (13) is at most $(e^{-w^2/2\sigma^2} g + h)/(\sqrt{2\pi}\,\sigma)$. Finally, if $\log(g/(\sqrt{2\pi}\,\sigma a - h)) \geq 0$, then $e^{-w^2/2\sigma^2} < (\sqrt{2\pi}\,\sigma a - h)/g$; otherwise $g < \sqrt{2\pi}\,\sigma a - h$ and $e^{-w^2/2\sigma^2} \leq 1$. In either case $e^{-w^2/2\sigma^2} g + h < \sqrt{2\pi}\,\sigma a$, so the right-hand side of (13) is less than a, proving the claim.

In the first call, (13) at $t = y_1$ reads $p_{\omega_{\tilde{\pi}}}(y) > p_{\tilde{\Gamma}}(y)$, so the test rejects whenever $y_1 > \overline{B}$. In the second call, the inputs are $(-c, -\overline{c}, -\underline{c})$, and at $t = -y_1$ the bracket in (13) becomes $\Phi((y_1 - \overline{c}_l)/\sigma) + \Phi((\underline{c}_l - y_1)/\sigma)$,

while $\phi((-y_1 + c_m)/\sigma) = \phi((y_1 - c_m)/\sigma)$ by the symmetry of ϕ. So (13) again reads $p_{\omega_{\tilde{\pi}}}(y) > p_{\tilde{\Gamma}}(y)$, and the test rejects whenever $-y_1$ exceeds the value returned by the second call, that is, whenever $y_1 < \underline{B}$. Both endpoints are finite by the claim. □

B.2 Proof of Theorem 4.1

Theorem 4.1. *Under Assumptions 2.1 to 2.4, there exists a solution to the optimal testing problem (7),*

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi).$$

Moreover, if $\varphi^\star$ solves (7), then the test inversion confidence procedure $S_{\varphi^\star}$ has minimax expected excess length, meaning it solves decision problem (6),

$$\inf_{S \in \mathcal{S}_\alpha} \sup_{\mu \in \mathcal{M}} R(\mu, S).$$

Finally, the two problems share the same value, that is, $\sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi^\star) = \sup_{\mu \in \mathcal{M}} R(\mu, S_{\varphi^\star})$.

I will begin by providing a high-level overview of the proof. First, I show that it is without loss to work with acceptance functions, rather than confidence procedures. An acceptance function $d \in L^\infty(\mathbb{R}^{1+k_\mu})$ outputs the probability $d(t, x)$ that t is included in the confidence set (i.e. that $H_0 : \theta = t$ is accepted) upon observing $X = x$. Second, I show that decision problem (6) is well-behaved in the weak-$*$ topology when written in terms of acceptance functions: the set of acceptance functions with $1 - \alpha$ coverage (the feasible set) is convex and weak-$*$ compact, and the maximum risk functional (the objective) is convex and weak-$*$ lower semicontinuous. In addition, both the feasible set and the objective are invariant to a group of translations motivated by (3) and (4) that send $d(t, x) \mapsto d(t + s, x + sKl + v)$ for $(s, v) \in \mathbb{R} \times V$. Third, I apply Orbanz (2024, Theorem 6) to conclude that there exists a translation-invariant acceptance function such that the corresponding confidence procedure solves decision problem (6). Finally, I establish that translation-invariant acceptance functions with $1 - \alpha$ coverage are precisely those formed by inverting tests $\varphi \in \Phi_\alpha$, completing the proof.

B.2.1 Acceptance Functions

It is possible to go back and forth between confidence procedures and acceptance functions as follows. Given a confidence procedure $S \in \mathcal{S}$, define the corresponding acceptance function

$$d_S(t, x) \equiv \mathbb{E}_{\mathrm{Unif}[0,1]}[I_S(t, x, U)].$$

By Part (iv) of Lemma B.1 and Tonelli, d_S is measurable, determined up to a.e. (t, x) equivalence, and takes values in $[0, 1]$, so it is an element of

$$\mathcal{D} \equiv \{d \in L^\infty(\mathbb{R}^{1+k_\mu}) : 0 \leq d \leq 1\}.$$

Conversely, given a representative of an acceptance function $d \in \mathcal{D}$, define the corresponding confidence procedure

$$S_d(x,u) \equiv \{t \in \mathbb{R} : u \le d(t,x)\}.$$

Then $S_d \in \mathcal{S}$ by Part (ii) of Lemma B.1, since for each $B \in \mathcal{B}$ the map

$$(x,u) \mapsto \lambda_{S_d(x,u)}(B) = \int_B \mathbb{1}\{u \le d(t,x)\}\,\mathrm{d}t$$

is measurable by Tonelli, since the indicator $(t,x,u) \mapsto \mathbb{1}\{u \le d(t,x)\}$ is measurable. This same indicator also satisfies the conditions of Part (iii) of Lemma B.1, since it is a version of the density $\mathrm{d}\lambda_{S_d(x,u)}/\mathrm{d}\lambda$ by the above.

The coverage probability function of a confidence procedure may be expressed in terms of its acceptance function by the law of iterated expectations,

$$C(t,\mu,S) = \mathbb{E}_{P_{\mu,U}}[I_S(t,X,U)] = \mathbb{E}_{P_\mu}[\mathbb{E}_{\mathrm{Unif}[0,1]}[I_S(t,X,U)]] = \mathbb{E}_{P_\mu}[d_S(t,X)].$$

To reformulate decision problem (6) in terms of acceptance functions, then, define

$$\begin{aligned} C(t,\mu,d) &\equiv \mathbb{E}_{P_\mu}[d(t,X)], \\ \mathcal{D}_\alpha &\equiv \{d \in \mathcal{D} : \operatorname*{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d) \ge 1-\alpha\}, \\ R(\mu,d) &\equiv \int_{\mathbb{R}\setminus\Theta(\mu)} C(t,\mu,d)\,\mathrm{d}t, \end{aligned}$$

and consider solving

$$\inf_{d\in\mathcal{D}_\alpha} \sup_{\mu\in\mathcal{M}} R(\mu,d). \tag{14}$$

The above reformulation is only possible because risk $R(\mu,d)$ is expressible in terms of coverage probabilities $C(\,\cdot\,,\mu,d)$ by the Ghosh–Pratt identity in (5). Before establishing that this reformulation is equivalent to the original problem, I first verify that both problems are in fact well-posed.

Let $\mu \in \mathcal{M}$, $S \in \mathcal{S}$, and $d \in \mathcal{D}$. Then $R(\mu,S)$ is well-defined, since $\mathbb{R}\setminus\Theta(\mu)$ is Borel by Part (i) of Lemma B.1, so

$$(x,u) \mapsto L(\mu,S(x,u)) = \lambda(S(x,u) \cap (\mathbb{R}\setminus\Theta(\mu)))$$

is a non-negative measurable function by Part (ii) of Lemma B.1. In addition, $R(\mu,d)$ is well-defined as the integral of $d(t,x)\mathbb{1}\{t \notin \Theta(\mu)\}p_\mu(x)$ over $\mathbb{R}^{1+k_\mu}$, where p_μ denotes the Lebesgue density of P_μ, which exists by Assumption 2.4. The integrand is measurable and non-negative a.e., so its integral exists in $\overline{\mathbb{R}}_+$ and does not depend on the choice of representative of d. Finally, the feasible sets $\mathcal{S}_\alpha$ and $\mathcal{D}_\alpha$ are well-defined.

Lemma B.5 (Feasible Sets)**.** *$\mathcal{S}_\alpha$ and $\mathcal{D}_\alpha$ are well-defined subsets of $\mathcal{S}$ and $\mathcal{D}$, respectively. Moreover, $S \in \mathcal{S}$ shares the same coverage value as $d_S \in \mathcal{D}$.*

PROOF. Fix a representative of $d \in \mathcal{D}$ taking values in $[0,1]$. I first establish that the coverage value $\operatorname{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d)$ is well-defined for this representative. I then show that it does not depend on the choice of representative.

With the representative fixed, the difficulty lies in showing that the map $t \mapsto \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d)$ is measurable. To this end, let $\{h_n\}_{n\in\mathbb{N}}$ be a countable dense subset of $\mathcal{M}(0)$, and let $\mu_n(t) \equiv tKl + h_n$ so that $\{\mu_n(t)\}_{n\in\mathbb{N}}$ is a dense subset of $\mathcal{M}(t)$ by (3). For each $t \in \mathbb{R}$, the map $\mu \mapsto C(t,\mu,d) = \langle d(t,\,\cdot\,), p_\mu\rangle$ is continuous since as $\mu' \to \mu$,

$$|\langle d(t,\,\cdot\,), p_{\mu'}\rangle - \langle d(t,\,\cdot\,), p_\mu\rangle| \le \|p_{\mu'} - p_\mu\|_1 = 2(2\Phi(\|\Sigma^{-1/2}(\mu'-\mu)\|/2) - 1) \to 0,$$

where the inequality holds because $0 \le d(t,\,\cdot\,) \le 1$ by assumption, Φ denotes the standard normal CDF, and $\Sigma^{-1/2} \succ 0$ is well-defined by Assumption 2.4. Since a continuous function has the same infimum over a set as over a dense subset, $\inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d) = \inf_{n\in\mathbb{N}} C(t,\mu_n(t),d)$ for each $t \in \mathbb{R}$. Each map $t \mapsto C(t,\mu_n(t),d)$ is measurable by Tonelli, so the right-hand side of the equality is measurable as a countable infimum of measurable functions. Hence, $t \mapsto \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d)$ is measurable, so the coverage value $\operatorname{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d)$ is well-defined for the chosen representative.

Now, let d' be another representative, so $d' = d$ a.e. (t,x). By Tonelli, for a.e. t, $d'(t,\,\cdot\,) = d(t,\,\cdot\,)$ a.e. x. For each such t, $\inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d') = \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d)$, since $P_\mu \ll \lambda_{k_\mu}$ for all $\mu \in \mathcal{M}(t)$ by Assumption 2.4. Since two functions that agree a.e. have the same essential infimum, we may conclude that the coverage value does not depend on the choice of representative, so $\mathcal{D}_\alpha$ is a well-defined subset of $\mathcal{D}$.

To see that $\mathcal{S}_\alpha$ is a well-defined subset of $\mathcal{S}$, let $S \in \mathcal{S}$ and let I_S be as in Part (iii) of Lemma B.1, which fixes a representative $d_S(t,x) = \mathbb{E}_{\mathrm{Unif}[0,1]}[I_S(t,x,U)]$ taking values in $[0,1]$. By the law of iterated expectations, $C(t,\mu,S) = \mathbb{E}_{P_\mu}[d_S(t,X)] = C(t,\mu,d_S)$ for every $t \in \mathbb{R}$ and $\mu \in \mathcal{M}$, so for this choice of I_S the coverage value of S exists and equals that of d_S. Another choice I'_S satisfying the conditions of Part (iii) of Lemma B.1 fixes another representative d'_S, which does not affect the coverage value as previously established. Hence $\mathcal{S}_\alpha$ is well-defined, and S shares the same coverage value as d_S. □

Establishing equivalence of decision problems (6) and (14) is now straightforward. The first step is to show that $\{d_S : S \in \mathcal{S}_\alpha\} = \mathcal{D}_\alpha$. If $S \in \mathcal{S}_\alpha$, then d_S shares the same coverage value by Lemma B.5, so $d_S \in \mathcal{D}_\alpha$. Conversely, if $d \in \mathcal{D}_\alpha$, fix a representative and consider the confidence procedure $S_d \in \mathcal{S}$. Then S_d shares the same coverage value as

$$d_{S_d}(t,x) = \mathbb{E}_{\mathrm{Unif}[0,1]}[I_{S_d}(t,x,U)] = \mathbb{E}_{\mathrm{Unif}[0,1]}[\mathbb{1}\{U \le d(t,x)\}] = d(t,x) \quad \text{a.e. } (t,x),$$

so $S_d \in \mathcal{S}_\alpha$ and $d = d_{S_d}$ as an element of $L^\infty(\mathbb{R}^{1+k_\mu})$. Now, let $\Psi(\,\cdot\,) \equiv \sup_{\mu\in\mathcal{M}} R(\mu,\,\cdot\,)$. Then since $\Psi(S) = \Psi(d_S)$ by the Ghosh–Pratt identity in (5),

$$\inf_{S\in\mathcal{S}_\alpha} \Psi(S) = \inf_{S\in\mathcal{S}_\alpha} \Psi(d_S) = \inf_{d\in\mathcal{D}_\alpha} \Psi(d),$$

so the two problems share the same value $v^\star$. If $S^\star$ solves (6), then $d_{S^\star} \in \mathcal{D}_\alpha$ with $\Psi(d_{S^\star}) = \Psi(S^\star) = v^\star$, so $d_{S^\star}$ solves (14). If $d^\star$ solves (14), then for any choice of representative, $S_{d^\star} \in \mathcal{S}_\alpha$ with $\Psi(S_{d^\star}) = \Psi(d_{S_{d^\star}}) = \Psi(d^\star) = v^\star$, so $S_{d^\star}$ solves (6).

B.2.2 Convexity and Weak-$*$ Regularity

Let $\mathcal{E}$ denote $L^\infty(\mathbb{R}^{1+k_\mu})$ equipped with the weak-$*$ topology, under which it is a locally convex Hausdorff topological vector space (Aliprantis and Border, 2006, Section 5.14), and view $\mathcal{D}$ as a subset of $\mathcal{E}$. As in the previous section, let $\Psi : \mathcal{D} \to \overline{\mathbb{R}}_+$ denote the maximum risk functional $\Psi(d) \equiv \sup_{\mu \in \mathcal{M}} R(\mu, d)$. Then decision problem (14) satisfies the following regularity conditions.

Lemma B.6 (Regularity)**.** *$\mathcal{D}_\alpha$ is convex and weak-$*$ compact, and Ψ is convex and weak-$*$ lower semicontinuous.*

Proof. *$\mathcal{D}_\alpha$ is convex and weak-$*$ compact.* As in the proof of Lemma B.5, let $\{h_n\}_{n \in \mathbb{N}}$ be a countable dense subset of $\mathcal{M}(0)$ and let $\mu_n(t) \equiv tKl + h_n$. For $t_0 \in \mathbb{R}$, $\varepsilon > 0$, and $n \in \mathbb{N}$, define

$$\psi_{t_0,\varepsilon,n}(t, x) \equiv \mathbb{1}\{|t - t_0| < \varepsilon\} p_{\mu_n(t)}(x),$$

which is non-negative and measurable, with $\|\psi_{t_0,\varepsilon,n}\|_1 = 2\varepsilon$ by Tonelli. So, $e \mapsto \langle e, \psi_{t_0,\varepsilon,n} \rangle$ is a weak-$*$ continuous linear functional on $\mathcal{E}$. I will now verify that

$$\mathcal{D}_\alpha = \mathcal{D} \cap \bigcap_{t_0 \in \mathbb{R},\ \varepsilon > 0,\ n \in \mathbb{N}} \{d \in \mathcal{E} : \langle d, \psi_{t_0,\varepsilon,n} \rangle \geq 2\varepsilon(1 - \alpha)\}. \tag{15}$$

Since $\mathcal{D}_\alpha \subseteq \mathcal{D}$, it suffices to show that $d \in \mathcal{D}$ belongs to $\mathcal{D}_\alpha$ if and only if $\langle d, \psi_{t_0,\varepsilon,n} \rangle \geq 2\varepsilon(1 - \alpha)$ for all $t_0 \in \mathbb{R}$, $\varepsilon > 0$, and $n \in \mathbb{N}$. To this end, let $d \in \mathcal{D}$ and fix a representative taking values in $[0, 1]$. By Tonelli, the map $t \mapsto C(t, \mu_n(t), d)$, which takes values in $[0, 1]$, is measurable with

$$\langle d, \psi_{t_0,\varepsilon,n} \rangle = \int_{t_0 - \varepsilon}^{t_0 + \varepsilon} C(t, \mu_n(t), d)\, \mathrm{d}t.$$

By the proof of Lemma B.5, $d \in \mathcal{D}_\alpha$ if and only if $\operatorname{ess\,inf}_{t \in \mathbb{R}} \inf_{n \in \mathbb{N}} C(t, \mu_n(t), d) \geq 1 - \alpha$, meaning for a.e. $t \in \mathbb{R}$, $C(t, \mu_n(t), d) \geq 1 - \alpha$ for all $n \in \mathbb{N}$.

First, suppose $d \in \mathcal{D}_\alpha$, and let $t_0 \in \mathbb{R}$, $\varepsilon > 0$, and $n \in \mathbb{N}$. Then $C(t, \mu_n(t), d) \geq 1 - \alpha$ for a.e. $t \in \mathbb{R}$, so integrating over $(t_0 - \varepsilon, t_0 + \varepsilon)$ yields $\langle d, \psi_{t_0,\varepsilon,n} \rangle \geq 2\varepsilon(1 - \alpha)$.

Conversely, suppose $\langle d, \psi_{t_0,\varepsilon,n} \rangle \geq 2\varepsilon(1 - \alpha)$ for all $t_0 \in \mathbb{R}$, $\varepsilon > 0$, and $n \in \mathbb{N}$, and let $n \in \mathbb{N}$. Since $t \mapsto C(t, \mu_n(t), d)$ is locally integrable, by the Lebesgue differentiation theorem (Stein and Shakarchi, 2005, Chapter 3, Theorem 1.4),

$$C(t_0, \mu_n(t_0), d) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \langle d, \psi_{t_0,\varepsilon,n} \rangle \geq 1 - \alpha \quad \text{for a.e. } t_0 \in \mathbb{R}.$$

Hence, for all $n \in \mathbb{N}$, $C(t, \mu_n(t), d) \geq 1 - \alpha$ for a.e. $t \in \mathbb{R}$. Since countable unions of null sets are null, it follows that for a.e. $t \in \mathbb{R}$, $C(t, \mu_n(t), d) \geq 1 - \alpha$ for all $n \in \mathbb{N}$, so $d \in \mathcal{D}_\alpha$.

Finally, notice that $\mathcal{D}$ may be represented as an intersection of weak-$*$ closed half-spaces,

$$\mathcal{D} = \{d \in \mathcal{E} : 0 \leq \langle d, g \rangle \leq \|g\|_1 \text{ for all } g \in L^1_+(\mathbb{R}^{1+k_\mu})\}.$$

Hence, $\mathcal{D}_\alpha$ is an intersection of weak-$*$ closed half-spaces, and is therefore convex and weak-$*$ closed. It is contained in the closed unit ball of $\mathcal{E}$, which is weak-$*$ compact (Aliprantis and Border, 2006, Theorem 5.105), so $\mathcal{D}_\alpha$ is weak-$*$ compact.

Ψ is convex and weak-$*$ lower semicontinuous. For $\mu \in \mathcal{M}$ and $n \in \mathbb{N}$, define

$$w_\mu^{(n)}(t, x) \equiv \mathbb{1}\{t \in [-n, n] \setminus \Theta(\mu)\} p_\mu(x),$$

which is measurable since $\Theta(\mu)$ is Borel by Part (i) of Lemma B.1, with $\|w_\mu^{(n)}\|_1 = \lambda([-n, n] \setminus \Theta(\mu)) \leq 2n$. So, $e \mapsto \langle e, w_\mu^{(n)} \rangle$ is a weak-$*$ continuous linear functional on $\mathcal{E}$. Let $d \in \mathcal{D}$, and fix a representative taking values in $[0, 1]$. Then by the monotone convergence theorem and Tonelli,

$$R(\mu, d) = \int_{\mathbb{R} \setminus \Theta(\mu)} C(t, \mu, d) \, \mathrm{d}t = \sup_{n \in \mathbb{N}} \int_{[-n,n] \setminus \Theta(\mu)} C(t, \mu, d) \, \mathrm{d}t = \sup_{n \in \mathbb{N}} \langle d, w_\mu^{(n)} \rangle.$$

A pointwise supremum of weak-$*$ continuous linear functionals is convex and weak-$*$ lower semicontinuous, so $R(\mu, \cdot)$ has both properties, and so does $\Psi = \sup_{\mu \in \mathcal{M}} R(\mu, \cdot)$ in turn. □

B.2.3 Invariance

For $(s, v) \in \mathbb{R} \times V$, define the translation $g_{s,v} : \mathcal{E} \to \mathcal{E}$ as

$$(g_{s,v} e)(t, x) \equiv e(t + s, x + sKl + v).$$

Notice that $g_{0,0}$ is the identity map and $g_{s,v} \circ g_{s',v'} = g_{s+s',\, v+v'}$, so the translations define an action on $\mathcal{E}$ of the additive group $G \equiv \mathbb{R} \times V$. Decision problem (14) is then G-invariant in the following sense.

Lemma B.7 (Invariance)**.** *Let $(s, v) \in G$. Then $\Psi(g_{s,v} d) = \Psi(d)$ for all $d \in \mathcal{D}$, and $g_{s,v} \mathcal{D}_\alpha = \mathcal{D}_\alpha$.*

Proof. I first record several translation symmetries of the model. By properties (3) and (4), for all $t \in \mathbb{R}$,

$$sKl + v + \mathcal{M}(t) = (t + s)Kl + v + \mathcal{M}(0) = (t + s)Kl + \mathcal{M}(0) = \mathcal{M}(t + s).$$

Taking the union over t, then, $sKl + v + \mathcal{M} = \mathcal{M}$. In addition, for all $t \in \mathbb{R}$ and $\mu \in \mathcal{M}$,

$$\mu + sKl + v \in \mathcal{M}(t) \iff \mu + sKl + v \in sKl + v + \mathcal{M}(t - s) \iff \mu \in \mathcal{M}(t - s),$$

meaning $\Theta(\mu + sKl + v) = s + \Theta(\mu)$ by the definition of the identified set. Finally, let $d \in \mathcal{D}$ and fix a representative taking values in $[0,1]$, so that $(t,x) \mapsto d(t+s, x+sKl+v)$ is a representative of $g_{s,v}d$ taking values in $[0,1]$. Then for all $t \in \mathbb{R}$ and $\mu \in \mathcal{M}$,

$$C(t,\mu,g_{s,v}d) = \mathbb{E}_{P_\mu}[d(t+s, X+sKl+v)] = \mathbb{E}_{P_{\mu+sKl+v}}[d(t+s,X)] = C(t+s,\mu+sKl+v,d)$$

since $X + sKl + v \sim \mathcal{N}(\mu + sKl + v, \Sigma) = P_{\mu+sKl+v}$ under P_μ.

Invariance of Ψ. Let $\mu \in \mathcal{M}$. Then

$$\begin{aligned} R(\mu, g_{s,v}d) &= \int_{\mathbb{R}\setminus\Theta(\mu)} C(t,\mu,g_{s,v}d)\,\mathrm{d}t \\ &= \int_{\mathbb{R}\setminus\Theta(\mu)} C(t+s,\mu+sKl+v,d)\,\mathrm{d}t \\ &= \int_{\mathbb{R}\setminus\Theta(\mu+sKl+v)} C(t,\mu+sKl+v,d)\,\mathrm{d}t \\ &= R(\mu+sKl+v,d), \end{aligned}$$

where the third equality follows from the change of variables $t \mapsto t+s$, which preserves Lebesgue measure and maps $\mathbb{R}\setminus\Theta(\mu)$ onto $\mathbb{R}\setminus\Theta(\mu+sKl+v)$ since $\Theta(\mu+sKl+v) = s+\Theta(\mu)$. Since $sKl+v+\mathcal{M} = \mathcal{M}$, taking the supremum over $\mu \in \mathcal{M}$ yields

$$\Psi(g_{s,v}d) = \sup_{\mu\in\mathcal{M}} R(\mu+sKl+v,d) = \sup_{\mu\in\mathcal{M}} R(\mu,d) = \Psi(d).$$

Invariance of $\mathcal{D}_\alpha$. Let $t \in \mathbb{R}$. Then for the chosen representatives,

$$\inf_{\mu\in\mathcal{M}(t)} C(t,\mu,g_{s,v}d) = \inf_{\mu\in\mathcal{M}(t)} C(t+s,\mu+sKl+v,d) = \inf_{\mu\in\mathcal{M}(t+s)} C(t+s,\mu,d).$$

Since this holds for all $t \in \mathbb{R}$ and $t \mapsto t+s$ preserves Lebesgue measure, taking essential infima yields

$$\operatorname*{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,g_{s,v}d) = \operatorname*{ess\,inf}_{t\in\mathbb{R}} \inf_{\mu\in\mathcal{M}(t)} C(t,\mu,d).$$

That is, $g_{s,v}d$ and d share the same coverage value, so $g_{s,v}\mathcal{D}_\alpha \subseteq \mathcal{D}_\alpha$. Conversely, if $d \in \mathcal{D}_\alpha$, then $d = (g_{s,v} \circ g_{-s,-v})d$ with $g_{-s,-v}d \in \mathcal{D}_\alpha$ by the same argument applied to $(-s,-v) \in G$, so $d \in g_{s,v}\mathcal{D}_\alpha$. We may conclude that $g_{s,v}\mathcal{D}_\alpha = \mathcal{D}_\alpha$. □

B.2.4 Existence of an Invariant Solution

The fact that both the objective and the feasible set in decision problem (14) are G-invariant motivates restricting attention to acceptance functions that share this property. In particular, define the set of G-invariant acceptance functions as

$$\mathcal{D}^{\text{inv}} \equiv \{d \in \mathcal{D} : g_{s,v}d = d \text{ for all } (s,v) \in G\},$$

and denote those with correct coverage by $\mathcal{D}_\alpha^{\text{inv}} \equiv \mathcal{D}_\alpha \cap \mathcal{D}^{\text{inv}}$. I establish that one such acceptance function solves decision problem (14) by verifying the conditions of Orbanz (2024, Theorem 6), an abstract result regarding the existence of invariant minimizers.

Lemma B.8 (Generalized Hunt–Stein)**.** *There exists a solution $d^\star \in \mathcal{D}_\alpha^{\text{inv}}$ to decision problem (14), meaning*

$$\sup_{\mu\in\mathcal{M}} R(\mu, d^\star) = \inf_{d\in\mathcal{D}_\alpha^{\text{inv}}} \sup_{\mu\in\mathcal{M}} R(\mu, d) = \inf_{d\in\mathcal{D}_\alpha} \sup_{\mu\in\mathcal{M}} R(\mu, d).$$

Proof. I proceed by enumerating the conditions of Orbanz (2024, Theorem 6), verifying each one in turn. Note that the map $((s, v), e) \mapsto g_{s,v}e$ is not jointly continuous; I show that it is separately continuous, which is all that is required by inspection of the proof of Orbanz (2024, Theorem 6) and personal communication with the author.

1. *G is locally compact, Polish, and amenable.* G is a closed subset of $\mathbb{R}^{1+k_\mu}$, so it is locally compact and Polish. G is abelian since $(s, v) + (s', v') = (s', v') + (s, v)$, hence amenable.

2. *For each $(s, v) \in G$, the translation $g_{s,v} : \mathcal{E} \to \mathcal{E}$ is linear and weak-$*$ continuous.* Fix $(s, v) \in G$. Linearity follows from the definition of $g_{s,v}$. To establish continuity, let $g^*_{s,v} : L^1(\mathbb{R}^{1+k_\mu}) \to L^1(\mathbb{R}^{1+k_\mu})$ be given by
$$(g^*_{s,v} f)(t, x) \equiv f(t + s, x + sKl + v).$$
The change of variables $(t', x') = (t + s, x + sKl + v)$ then yields $\langle g_{s,v}e, f\rangle = \langle e, g^*_{-s,-v} f\rangle$ for all $e \in \mathcal{E}$ and $f \in L^1(\mathbb{R}^{1+k_\mu})$. For each $f \in L^1(\mathbb{R}^{1+k_\mu})$, we know that $g^*_{-s,-v} f \in L^1(\mathbb{R}^{1+k_\mu})$ since $\|g^*_{-s,-v} f\|_1 = \|f\|_1 < \infty$, so the map $e \mapsto \langle e, g^*_{-s,-v} f\rangle$ is weak-$*$ continuous. It follows that the map $e \mapsto \langle g_{s,v}e, f\rangle$ is weak-$*$ continuous for all $f \in L^1(\mathbb{R}^{1+k_\mu})$, so the map $e \mapsto g_{s,v}e$ is weak-$*$ continuous.

3. *For each $e \in \mathcal{E}$, the map $(s, v) \mapsto g_{s,v}e$ is weak-$*$ continuous.* Fix $e \in \mathcal{E}$ and let $(s, v) \to (s_0, v_0)$. For each $f \in L^1(\mathbb{R}^{1+k_\mu})$, Hölder's inequality yields
$$\begin{aligned} |\langle g_{s,v}e, f\rangle - \langle g_{s_0,v_0}e, f\rangle| &= |\langle e, g^*_{-s,-v} f - g^*_{-s_0,-v_0} f\rangle| \\ &\le \|e\|_\infty \, \|g^*_{-s,-v} f - g^*_{-s_0,-v_0} f\|_1 \to 0, \end{aligned}$$
since $\|e\|_\infty < \infty$, translation is strongly continuous on $L^1(\mathbb{R}^{1+k_\mu})$, and the shift vector $(-s, -sKl - v)$ depends continuously on (s, v). Hence $\langle g_{s,v}e, f\rangle \to \langle g_{s_0,v_0}e, f\rangle$ for all $f \in L^1(\mathbb{R}^{1+k_\mu})$, meaning $g_{s,v}e \to g_{s_0,v_0}e$ weak-$*$ as $(s, v) \to (s_0, v_0)$.

4. *The set $\mathcal{D}_\alpha$ is convex, weak-$*$ compact, and G-invariant.* This follows from Lemma B.6 and Lemma B.7.

5. *The function Ψ is convex, weak-$*$ lower semicontinuous, and G-invariant.* This follows from Lemma B.6 and Lemma B.7.

By Orbanz (2024, Theorem 6(iii)), then,

$$\inf_{d\in\mathcal{D}_\alpha\cap\mathcal{E}_G} \Psi(d) = \inf_{d\in\mathcal{D}_\alpha} \Psi(d),$$

where $\mathcal{E}_G$ is the set of G-invariant elements of $\mathcal{E}$, a weak-$*$ closed linear subspace. By definition, $\mathcal{D}^{\text{inv}} = \mathcal{D} \cap \mathcal{E}_G$, so $\mathcal{D}_\alpha^{\text{inv}} = \mathcal{D}_\alpha \cap \mathcal{D}^{\text{inv}} = \mathcal{D}_\alpha \cap \mathcal{E}_G$. It remains to show that the above infimum is attained at some $d^\star \in \mathcal{D}_\alpha^{\text{inv}}$. But this follows immediately from the Weierstrass extreme value theorem, since $\mathcal{D}_\alpha^{\text{inv}}$ is weak-$*$ compact (it is the intersection of the weak-$*$ compact $\mathcal{D}_\alpha$ with a weak-$*$ closed set), nonempty (the constant function $d \equiv 1$ lies in $\mathcal{D}_\alpha$ and is G-invariant), and Ψ is weak-$*$ lower semicontinuous. □

B.2.5 Test Inversion

Recall the reduced testing problem from Section 4.1, and adopt the notation developed there. For $\varphi \in \Phi$, let $d_\varphi \in \mathcal{D}$ denote the acceptance function

$$d_\varphi(t, x) \equiv 1 - \varphi(Q(x - tKl)),$$

so that the test inversion confidence procedure S_φ is given by S_{d_φ} as defined in Section B.2.1. The following lemma establishes that acceptance functions of this form are precisely the G-invariant ones, and that those with $1 - \alpha$ coverage are precisely those obtained by inverting tests $\varphi \in \Phi_\alpha$.

Lemma B.9 (Test Inversion)**.** *The map $\varphi \mapsto d_\varphi$ is a bijection from Φ onto $\mathcal{D}^{\text{inv}}$, which restricts to a bijection from Φ_α onto $\mathcal{D}_\alpha^{\text{inv}}$.*

Proof. The proof rests on a linear change of coordinates under which the translations $g_{s,v}$ act by shifting a single block of coordinates. G-invariance of an acceptance function then reduces to the statement that it depends only on the complementary block $Q(x - tKl)$, which is precisely the form of d_φ.

Coordinates. Fix a matrix $Q_V \in \mathbb{R}^{\dim(V) \times k_\mu}$ whose rows form a basis of V, and let the linear map $\Xi : \mathbb{R}^{1+k_\mu} \to \mathbb{R}^{1+k_\mu}$ be given by

$$\Xi(t, x) \equiv \begin{bmatrix} t \\ Q_V(x - tKl) \\ Q(x - tKl) \end{bmatrix}.$$

If $\Xi(t, x) = 0$, then $t = 0$, $Q_V x = 0$, and $Qx = 0$. Since $\ker(Q_V) = V^\perp$ and $\ker(Q) = V$ by the proof of Lemma B.3, it follows that $x \in V^\perp \cap V = \{0\}$, so Ξ is a linear isomorphism. In particular, Ξ maps null sets to null sets in both directions, so for $d \in \mathcal{D}$, the composition $d \circ \Xi^{-1} \in \mathcal{D}$ is well-defined. I refer to this composition as the block form of d, and view it as a function of $(w, y) \in \mathbb{R}^{1+\dim(V)} \times \mathbb{R}^{k_\nu}$. Two acceptance functions coincide if and only if their block forms do, and the block form of d_φ is simply $(w, y) \mapsto 1 - \varphi(y)$.

Now, let $d \in \mathcal{D}$ and denote its block form by $b \in \mathcal{D}$. For $(s, v) \in G$, the block form of $g_{s,v} d$ is $(w, y) \mapsto b(w + [\,{}^{s}_{Q_V v}\,], y)$, since

$$\Xi(t + s, x + sKl + v) = \begin{bmatrix} t+s \\ Q_V(x - tKl) + Q_V v \\ Q(x - tKl) + Qv \end{bmatrix} = \Xi(t, x) + \begin{bmatrix} s \\ Q_V v \\ 0 \end{bmatrix},$$

where $Qv = 0$ because $v \in V = \ker(Q)$. The map $(s, v) \mapsto [\,{}^{s}_{Q_V v}\,]$ is a bijection from G onto $\mathbb{R}^{1+\dim(V)}$, so $d \in \mathcal{D}^{\text{inv}}$ if and only if for every $w' \in \mathbb{R}^{1+\dim(V)}$,

$$b(w + w', y) = b(w, y) \quad \text{for a.e. } (w, y).$$

Injectivity. Fix representatives of $\varphi, \varphi' \in \Phi$. The block forms of d_φ and $d_{\varphi'}$ disagree exactly on $\mathbb{R}^{1+\dim(V)} \times \{\varphi \neq \varphi'\}$, which is null if and only if $\{\varphi \neq \varphi'\}$ is null by Tonelli. Hence $d_\varphi = d_{\varphi'}$ in $\mathcal{D}$ if and only if $\varphi = \varphi'$ in Φ, so the map $\varphi \mapsto d_\varphi$ is well-defined and injective.

Range. Since $0 \leq d_\varphi \leq 1$ and its block form does not depend on w, every d_φ is G-invariant, so the map $\varphi \mapsto d_\varphi$ sends Φ into $\mathcal{D}^{\mathrm{inv}}$. Conversely, let $d \in \mathcal{D}^{\mathrm{inv}}$, and fix a representative of its block form b taking values in $[0,1]$. Since d is G-invariant, for every $w' \in \mathbb{R}^{1+\dim(V)}$ we have $b(w+w', y) = b(w, y)$ for a.e. (w, y). By Tonelli, then, for a.e. (w, y) we have $b(w+w', y) = b(w, y)$ for a.e. w', or equivalently, since $w' \mapsto w + w'$ preserves null sets, $b(w', y) = b(w, y)$ for a.e. w'. Averaging over the unit cube, then, the map

$$\bar{b}(y) \equiv \int_{[0,1]^{1+\dim(V)}} b(w', y)\, \mathrm{d}w'$$

satisfies $\bar{b}(y) = b(w, y)$ for a.e. (w, y). Since $\bar{b}$ is measurable by Tonelli and takes values in $[0,1]$, we have $\varphi \equiv 1 - \bar{b} \in \Phi$, and the block form of d_φ is $(w, y) \mapsto 1 - \varphi(y) = \bar{b}(y)$, which agrees with b for a.e. (w, y). Hence $d_\varphi = d$, so the map $\varphi \mapsto d_\varphi$ is a bijection from Φ onto $\mathcal{D}^{\mathrm{inv}}$.

Restriction. Fix a representative of $\varphi \in \Phi$ taking values in $[0,1]$, so that $(t, x) \mapsto 1 - \varphi(Q(x - tKl))$ is a representative of d_φ taking values in $[0,1]$. Since $Y = Q(X - tKl) \sim P_{\nu(t)}$ under P_μ,

$$C(t, \mu, d_\varphi) = \mathbb{E}_{P_\mu}[1 - \varphi(Q(X - tKl))] = 1 - \mathbb{E}_{P_{\nu(t)}}[\varphi(Y)]$$

for every $t \in \mathbb{R}$ and $\mu \in \mathcal{M}$. By (3), as μ ranges over $\mathcal{M}(t) = tKl + \mathcal{M}(0)$, the reduced moments $\nu(t)$ range over all of $Q\mathcal{M}(0)$. Taking the infimum over $\mu \in \mathcal{M}(t)$ therefore yields

$$\inf_{\mu \in \mathcal{M}(t)} C(t, \mu, d_\varphi) = 1 - \sup_{\nu \in Q\mathcal{M}(0)} \mathbb{E}_{P_\nu}[\varphi(Y)] \quad \text{for every } t \in \mathbb{R}.$$

The right-hand side does not depend on t, so it coincides with the essential infimum of the left-hand side over $t \in \mathbb{R}$, which is the coverage value of d_φ. Hence $d_\varphi \in \mathcal{D}_\alpha$ if and only if $\sup_{\nu \in Q\mathcal{M}(0)} \mathbb{E}_{P_\nu}[\varphi(Y)] \leq \alpha$, which is to say $\varphi \in \Phi_\alpha$. The bijection therefore restricts to a bijection from Φ_α onto $\mathcal{D}_\alpha \cap \mathcal{D}^{\mathrm{inv}} = \mathcal{D}^{\mathrm{inv}}_\alpha$. □

To complete the proof of the theorem, it remains to assemble Lemmas B.8 and B.9 and express the risk derivation from Section 4.1 in terms of acceptance functions. For $\mu \in \mathcal{M}$ and $\varphi \in \Phi$,

$$\begin{aligned} R(\mu, d_\varphi) &= \int_{\mathbb{R} \setminus \Theta(\mu)} C(t, \mu, d_\varphi)\, \mathrm{d}t \\ &= \int \mathbb{1}\{\nu(t) \notin Q\mathcal{M}(0)\} \mathbb{E}_{P_{\nu(t)}}[1 - \varphi(Y)]\, \mathrm{d}t \\ &= \int \mathbb{1}\{\nu \notin Q\mathcal{M}(0)\} \mathbb{E}_{P_\nu}[1 - \varphi(Y)]\, \mathrm{d}\nu_1 \\ &= R(\nu_{-1}, \varphi), \end{aligned}$$

where the second equality follows from Part (i) of Lemma B.3 and the coverage identity in the proof of Lemma B.9, and the third equality follows from the change of variables $\nu_1 = Q_1\mu - t$ under the convention

that $QKl = e_1$, just as in Section 4.1. Recall that $\nu_{-1} = Q_{-1}\mu$, so taking the supremum over $\mu \in \mathcal{M}$,

$$\Psi(d_\varphi) = \sup_{\mu \in \mathcal{M}} R(\mu, d_\varphi) = \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi).$$

The optimal testing problem (7) may therefore be expressed as $\inf_{\varphi \in \Phi_\alpha} \Psi(d_\varphi)$. A solution $\varphi^\star$ must exist, since Lemma B.8 provides $d^\star$ that solves $\inf_{d \in \mathcal{D}_\alpha^{\text{inv}}} \Psi(d)$. By Lemma B.9, then, $d^\star = d_{\varphi^\star}$ for some $\varphi^\star \in \Phi_\alpha$, which therefore solves (7). Now, suppose $\varphi^\star \in \Phi_\alpha$ solves (7). Then by Lemma B.9, $d_{\varphi^\star}$ solves $\inf_{d \in \mathcal{D}_\alpha^{\text{inv}}} \Psi(d)$, so $d_{\varphi^\star}$ also solves decision problem (14) by Lemma B.8. Finally, by the equivalence result from Section B.2.1, for any representative of $\varphi^\star$ the confidence procedure

$$S_{d_{\varphi^\star}}(x, u) = \{t \in \mathbb{R} : u \leq 1 - \varphi^\star(Q(x - tKl))\}$$

solves decision problem (6), and the same result implies that $\Psi(S_{d_{\varphi^\star}}) = \Psi(d_{\varphi^\star})$. Since $\Psi(d_{\varphi^\star}) = \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi^\star)$ as derived above, the two problems share the same value, completing the proof.

B.3 Proofs of Corollary A.1 and Corollary A.2

Corollary A.1 (Length Loss). *Under Assumptions 2.1 to 2.4, there exists a solution to*

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R_\lambda(\nu_{-1}, \varphi).$$

Moreover, if $\varphi^\star$ solves the above problem, then $S_{\varphi^\star}$ has minimax expected length, meaning it solves

$$\inf_{S \in \mathcal{S}_\alpha} \sup_{\mu \in \mathcal{M}} R_\lambda(\mu, S).$$

Proof. The proof of Theorem 4.1 uses the form of the risk function in four places: in Section B.2.1, where the identity $R(\mu, S) = R(\mu, d_S)$ transfers decision problem (6) to acceptance functions; in Lemma B.6, where the maximum risk functional $\Psi(d) = \sup_{\mu \in \mathcal{M}} R(\mu, d)$ is shown to be convex and weak-$*$ lower semicontinuous; in Lemma B.7, where Ψ is shown to be G-invariant; and in the final step, where $\Psi(d_\varphi)$ is expressed in terms of φ. Every other step concerns only the feasible set $\mathcal{D}_\alpha$ or the group action, neither of which depends on the loss function. I verify these four properties under length loss, after which the proof of Theorem 4.1 applies with R_λ in place of R.

Under length loss, the derivation of the Ghosh–Pratt identity (5) applies with $\mathbb{R}$ in place of $\mathbb{R} \setminus \Theta(\mu)$, so $R_\lambda(\mu, S) = \int_{\mathbb{R}} C(t, \mu, S)\, \mathrm{d}t$ is well-defined by Part (ii) of Lemma B.1 with $B = \mathbb{R}$. Accordingly, define $R_\lambda(\mu, d) \equiv \int_{\mathbb{R}} C(t, \mu, d)\, \mathrm{d}t$ for $d \in \mathcal{D}$, which is well-defined by the argument given for $R(\mu, d)$ in Section B.2.1.

Transfer. Since $C(t, \mu, S) = C(t, \mu, d_S)$ by Section B.2.1, $R_\lambda(\mu, S) = R_\lambda(\mu, d_S)$.

Regularity. The proof of Lemma B.6 applies with $w_\mu^{(n)}(t, x) \equiv \mathbb{1}\{t \in [-n, n]\} p_\mu(x)$, for which $\|w_\mu^{(n)}\|_1 = 2n$ and $R_\lambda(\mu, d) = \sup_{n \in \mathbb{N}} \langle d, w_\mu^{(n)} \rangle$.

Invariance. The proof of Lemma B.7 applies with $\mathbb{R}$ as the domain of integration, which the change of variables $t \mapsto t + s$ maps onto itself, so $R_\lambda(\mu, g_{s,v}d) = R_\lambda(\mu + sKl + v, d)$ without appealing to the identity $\Theta(\mu + sKl + v) = s + \Theta(\mu)$.

Reduction. By the coverage identity in the proof of Lemma B.9 and the change of variables $\nu_1 = Q_1\mu - t$,

$$R_\lambda(\mu, d_\varphi) = \int \mathbb{E}_{P_{\nu(t)}}[1 - \varphi(Y)]\,\mathrm{d}t = \int \mathbb{E}_{P_\nu}[1 - \varphi(Y)]\,\mathrm{d}\nu_1 = R_\lambda(\nu_{-1}, \varphi),$$

so taking the supremum over $\mu \in \mathcal{M}$ yields $\Psi(d_\varphi) = \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R_\lambda(\nu_{-1}, \varphi)$. □

Corollary A.2 (Minimax Regret and Bayes Risk)**.** *Under Assumptions 2.1 to 2.4, the following hold.*

(i) Suppose $R^\star(\mu) < \infty$ for every $\mu \in \mathcal{M}$. Then $R^\star(\mu)$ depends on μ only through $\nu_{-1} = Q_{-1}\mu$, and writing $R^\star(\nu_{-1})$ accordingly, there exists a solution to

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} \big[R(\nu_{-1}, \varphi) - R^\star(\nu_{-1})\big].$$

Moreover, if $\varphi^\star$ solves the above problem, then $S_{\varphi^\star}$ has minimax regret, meaning it solves

$$\inf_{S \in \mathcal{S}_\alpha} \sup_{\mu \in \mathcal{M}} \big[R(\mu, S) - R^\star(\mu)\big].$$

(ii) Let $\pi \in \Pi$. There exists a solution to $\inf_{\varphi \in \Phi_\alpha} R(\pi, \varphi)$. For any such solution $\varphi^\star$, $S_{\varphi^\star}$ solves

$$\inf_{S \in \mathcal{S}_\alpha} \sup_{\rho \in \Pi_\pi} \int R(\mu, S)\,\rho(\mathrm{d}\mu).$$

Proof. As in the proof of Corollary A.1, it suffices to redefine the maximum risk functional Ψ for each criterion and verify the four properties listed there, after which the proof of Theorem 4.1 applies verbatim. Both criteria involve the coordinates $\nu_{-1} = Q_{-1}\mu$, so I first record that for $\mu, \mu' \in \mathbb{R}^{k_\mu}$,

$$Q_{-1}\mu' = Q_{-1}\mu \iff \mu' = \mu + sKl + v \text{ for some } (s, v) \in G. \tag{16}$$

Under the convention $QKl = e_1$, and since $\ker(Q) = V$ by the proof of Lemma B.3, $Q(sKl + v) = se_1$ for every $(s, v) \in G$, which implies the reverse direction. Conversely, if $Q_{-1}\mu' = Q_{-1}\mu$, then $Q(\mu' - \mu) = se_1 = Q(sKl)$ for $s \equiv Q_1(\mu' - \mu)$, so $\mu' - \mu - sKl \in \ker(Q) = V$.

Minimax regret. Define $\Psi(d) \equiv \sup_{\mu \in \mathcal{M}}[R(\mu, d) - R^\star(\mu)]$ for $d \in \mathcal{D}$.

Transfer. By Section B.2.1, $\{d_S : S \in \mathcal{S}_\alpha\} = \mathcal{D}_\alpha$ and $R(\mu, S) = R(\mu, d_S)$ for every $\mu \in \mathcal{M}$ and $S \in \mathcal{S}$. Hence $R^\star(\mu) = \inf_{d \in \mathcal{D}_\alpha} R(\mu, d)$, and $\Psi(S) \equiv \sup_{\mu \in \mathcal{M}}[R(\mu, S) - R^\star(\mu)]$ satisfies $\Psi(S) = \Psi(d_S)$.

Regularity. Since $R^\star(\mu)$ is finite and $R(\mu, \cdot)$ is convex and weak-$*$ lower semicontinuous by the proof of Lemma B.6, so is $d \mapsto R(\mu, d) - R^\star(\mu)$, and so is Ψ as a pointwise supremum.

Invariance. Let $(s,v) \in G$. By Lemma B.7 and its proof, $g_{s,v}\mathcal{D}_\alpha = \mathcal{D}_\alpha$ and $R(\mu, g_{s,v}d) = R(\mu + sKl + v, d)$ for all $\mu \in \mathcal{M}$ and $d \in \mathcal{D}$. Taking the infimum over $d \in \mathcal{D}_\alpha$,

$$R^\star(\mu + sKl + v) = \inf_{d \in \mathcal{D}_\alpha} R(\mu, g_{s,v}d) = \inf_{d \in g_{s,v}\mathcal{D}_\alpha} R(\mu, d) = R^\star(\mu).$$

Hence $R(\mu, g_{s,v}d) - R^\star(\mu) = R(\mu + sKl + v, d) - R^\star(\mu + sKl + v)$, and taking the supremum over $\mu \in \mathcal{M} = sKl + v + \mathcal{M}$ yields $\Psi(g_{s,v}d) = \Psi(d)$.

Reduction. By the invariance of $R^\star$ just established and (16), $R^\star(\mu)$ depends on μ only through ν_{-1}, which is the first claim of Part (i). Since $R(\mu, d_\varphi) = R(\nu_{-1}, \varphi)$ by the final step of the proof of Theorem 4.1, taking the supremum over $\mu \in \mathcal{M}$ yields

$$\Psi(d_\varphi) = \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} \big[R(\nu_{-1}, \varphi) - R^\star(\nu_{-1})\big].$$

Bayes risk. For $d \in \mathcal{D}$ with a fixed representative and $\mu \in \mathcal{M}$, $R(\mu, d)$ is the integral over (t,x) of $\mathbb{1}\{\mu - tKl \notin \mathcal{M}(0)\} d(t,x) p_\mu(x)$, which is measurable in (μ, t, x) since $\mathcal{M}(0)$ is Borel by Part (i) of Lemma B.1. By Tonelli, $\mu \mapsto R(\mu, d)$ is measurable, so $\Psi(d) \equiv \sup_{\rho \in \Pi_\pi} \int R(\mu, d)\, \rho(\mathrm{d}\mu)$ is well-defined. Moreover, Π_π is non-empty. To see this, let $\nu_{-1} \sim \pi$ and set $\mu \equiv Q'_{-1}(Q_{-1}Q'_{-1})^{-1}\nu_{-1}$, so that $Q_{-1}\mu = \nu_{-1} \in Q_{-1}\mathcal{M}$. By (16), μ differs from an element of $\mathcal{M}$ by a vector $sKl + v$ with $(s,v) \in G$, so $\mu \in sKl + v + \mathcal{M} = \mathcal{M}$ by the proof of Lemma B.7. The law of μ therefore belongs to Π_π.

Transfer. Since $R(\mu, S) = R(\mu, d_S)$ for every $\mu \in \mathcal{M}$, $\Psi(S) \equiv \sup_{\rho \in \Pi_\pi} \int R(\mu, S)\, \rho(\mathrm{d}\mu)$ is well-defined and satisfies $\Psi(S) = \Psi(d_S)$.

Regularity. Let $\rho \in \Pi_\pi$ and $n \in \mathbb{N}$, and let $w_\mu^{(n)}$ be as in the proof of Lemma B.6, which is measurable in (μ, t, x) by the argument just given. The function $w_\rho^{(n)}(t,x) \equiv \int w_\mu^{(n)}(t,x)\, \rho(\mathrm{d}\mu)$ is therefore non-negative and measurable, with $\|w_\rho^{(n)}\|_1 \le 2n$ by Tonelli, so $e \mapsto \langle e, w_\rho^{(n)} \rangle$ is a weak-$*$ continuous linear functional on $\mathcal{E}$. For $d \in \mathcal{D}$ with a representative taking values in $[0,1]$, the proof of Lemma B.6 implies that $R(\mu, d) = \sup_{n \in \mathbb{N}} \langle d, w_\mu^{(n)} \rangle$, where the terms increase in n, so by the monotone convergence theorem and Tonelli,

$$\int R(\mu, d)\, \rho(\mathrm{d}\mu) = \sup_{n \in \mathbb{N}} \int \langle d, w_\mu^{(n)} \rangle\, \rho(\mathrm{d}\mu) = \sup_{n \in \mathbb{N}} \langle d, w_\rho^{(n)} \rangle.$$

Hence $d \mapsto \int R(\mu, d)\, \rho(\mathrm{d}\mu)$ is convex and weak-$*$ lower semicontinuous, and so is Ψ as a pointwise supremum.

Invariance. Let $(s,v) \in G$ and $\rho \in \Pi_\pi$, and let $\rho_{s,v}$ denote the law of $\mu + sKl + v$ when $\mu \sim \rho$. Then $\rho_{s,v} \in \Pi_\pi$, since $sKl + v + \mathcal{M} = \mathcal{M}$ and $Q_{-1}(\mu + sKl + v) = Q_{-1}\mu$ by (16). The map $\rho \mapsto \rho_{s,v}$ is a bijection of Π_π, with inverse $\rho \mapsto \rho_{-s,-v}$. By the proof of Lemma B.7,

$$\int R(\mu, g_{s,v}d)\, \rho(\mathrm{d}\mu) = \int R(\mu + sKl + v, d)\, \rho(\mathrm{d}\mu) = \int R(\mu, d)\, \rho_{s,v}(\mathrm{d}\mu),$$

and taking the supremum over $\rho \in \Pi_\pi$ yields $\Psi(g_{s,v}d) = \Psi(d)$.

Reduction. Let $\rho \in \Pi_\pi$. Since $R(\mu, d_\varphi) = R(\nu_{-1}, \varphi)$ with $\nu_{-1} = Q_{-1}\mu$ for every $\mu \in \mathcal{M}$, and $Q_{-1}\mu \sim \pi$ when $\mu \sim \rho$,

$$\int R(\mu, d_\varphi)\, \rho(\mathrm{d}\mu) = \int R(\nu_{-1}, \varphi)\, \pi(\mathrm{d}\nu_{-1}) = R(\pi, \varphi),$$

where $\nu_{-1} \mapsto R(\nu_{-1}, \varphi)$ is measurable by the proof of Lemma 4.1. As Π_π is non-empty, $\Psi(d_\varphi) = R(\pi, \varphi)$.

In both cases, the proof of Theorem 4.1 now yields a solution to $\inf_{\varphi \in \Phi_\alpha} \Psi(d_\varphi)$, and shows that $S_{\varphi^\star}$ solves $\inf_{S \in \mathcal{S}_\alpha} \Psi(S)$ whenever $\varphi^\star$ does. □

B.4 Proof of Lemma 4.1

Lemma 4.1 (Bayes-Minimax)**.** *The optimal testing problem (7) satisfies*

$$\inf_{\varphi \in \Phi_\alpha} \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi) = \inf_{\varphi \in \Phi_\alpha} \sup_{\pi \in \Pi} R(\pi, \varphi) = \sup_{\pi \in \Pi} \inf_{\varphi \in \Phi_\alpha} R(\pi, \varphi).$$

Suppose $\pi^\star$ is least favorable, meaning it attains the right-hand side supremum. Then if $\varphi^\star$ solves (7), it also solves $\inf_{\varphi \in \Phi_\alpha} R(\pi^\star, \varphi)$.

Proof. Write p_ν for the Lebesgue density of P_ν, which exists since $Q\Sigma Q' \succ 0$ by Part (ii) of Lemma B.3. Since $(\nu, y) \mapsto p_\nu(y)$ is continuous and $Q\mathcal{M}(0)$ is Borel by Part (ii) of Lemma B.3, the integrand in

$$R(\nu_{-1}, \varphi) = \iint \mathbb{1}\{\nu \notin Q\mathcal{M}(0)\}\, p_\nu(y)\, (1 - \varphi(y))\, \mathrm{d}y\, \mathrm{d}\nu_1$$

is measurable in (ν, y), so $\nu_{-1} \mapsto R(\nu_{-1}, \varphi)$ is measurable by Tonelli and $R(\pi, \varphi) \in \overline{\mathbb{R}}_+$ is well-defined for every $\pi \in \Pi$.

First equality. The point mass at any $\nu_{-1} \in Q_{-1}\mathcal{M}$ belongs to Π and has Bayes risk $R(\nu_{-1}, \varphi)$, while $R(\pi, \varphi) \leq \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi)$ for every $\pi \in \Pi$ because π is a probability measure. Hence $\sup_{\pi \in \Pi} R(\pi, \varphi) = \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi)$ for every $\varphi \in \Phi$, and taking the infimum over $\varphi \in \Phi_\alpha$ yields the first equality.

Second equality. I verify the conditions of Evans et al. (2005, Theorem 4).

1. Π *is convex and* $R(\,\cdot\,, \varphi)$ *is affine.* Convexity is immediate, and $R(\,\cdot\,, \varphi)$ is affine as the integral of $\nu_{-1} \mapsto R(\nu_{-1}, \varphi)$ against π.

2. Φ_α *is convex and weak-$*$ compact.* Arguing as in the proof of Lemma B.6, Φ is an intersection of weak-$*$ closed half-spaces contained in the closed unit ball of $L^\infty(\mathbb{R}^{k_\nu})$. Since $\mathbb{E}_{P_\nu}[\varphi(Y)] = \langle \varphi, p_\nu \rangle$ with $p_\nu \in L^1(\mathbb{R}^{k_\nu})$,

$$\Phi_\alpha = \Phi \cap \bigcap_{\nu \in Q\mathcal{M}(0)} \{\varphi \in L^\infty(\mathbb{R}^{k_\nu}) : \langle \varphi, p_\nu \rangle \leq \alpha\}$$

is likewise an intersection of weak-$*$ closed half-spaces contained in the closed unit ball, so it is convex and weak-$*$ compact (Aliprantis and Border, 2006, Theorem 5.105).

3. *$R(\pi, \cdot)$ is convex and weak-$*$ lower semicontinuous.* For $\pi \in \Pi$ and $n \in \mathbb{N}$, define

$$w_\pi^{(n)}(y) \equiv \iint \mathbb{1}\{|\nu_1| \le n,\ \nu \notin Q\mathcal{M}(0)\}\, p_\nu(y)\, \mathrm{d}\nu_1\, \pi(\mathrm{d}\nu_{-1}),$$

which is non-negative and measurable by Tonelli, with $\|w_\pi^{(n)}\|_1 \le 2n$. By Tonelli and the monotone convergence theorem,

$$\begin{aligned} R(\pi, \varphi) &= \sup_{n\in\mathbb{N}} \iint \mathbb{1}\{|\nu_1| \le n,\ \nu \notin Q\mathcal{M}(0)\}\, \mathbb{E}_{P_\nu}[1 - \varphi(Y)]\, \mathrm{d}\nu_1\, \pi(\mathrm{d}\nu_{-1}) \\ &= \sup_{n\in\mathbb{N}} \big(\|w_\pi^{(n)}\|_1 - \langle \varphi, w_\pi^{(n)} \rangle\big), \end{aligned}$$

a pointwise supremum of weak-$*$ continuous affine functionals, so $R(\pi, \cdot)$ is convex and weak-$*$ lower semicontinuous.

By Evans et al. (2005, Theorem 4), then, the second equality holds.

Least favorable priors. Let $\pi^\star \in \Pi$ be least favorable, and let $\varphi^\star$ solve (7). Writing $v^\star$ for the common value in the statement of this lemma,

$$R(\pi^\star, \varphi^\star) \le \sup_{\nu_{-1} \in Q_{-1}\mathcal{M}} R(\nu_{-1}, \varphi^\star) = v^\star = \inf_{\varphi \in \Phi_\alpha} R(\pi^\star, \varphi) \le R(\pi^\star, \varphi^\star),$$

where the first inequality holds because $\pi^\star$ is a probability measure on $Q_{-1}\mathcal{M}$, the first equality because $\varphi^\star$ solves (7), the second equality because $\pi^\star$ is least favorable, and the last inequality because $\varphi^\star \in \Phi_\alpha$. Hence $\varphi^\star$ solves $\inf_{\varphi\in\Phi_\alpha} R(\pi^\star, \varphi)$. □

B.5 Proof of Lemma 5.1

Lemma 5.1 (Duality). *The optimal testing problem (7) shares its value with the dual problem*

$$\sup_{\pi\in\Pi} \sup_{\Gamma\in\mathcal{G}} R(\pi, \varphi_{\pi,\Gamma}) + \int \big(\mathbb{E}_{P_\nu}[\varphi_{\pi,\Gamma}(Y)] - \alpha\big)\, \Gamma(\mathrm{d}\nu), \tag{8}$$

where $\varphi_{\pi,\Gamma} \equiv \mathbb{1}\{p_{\omega_\pi} > p_\Gamma\}$. Moreover, if this value is positive and $(\pi^\star, \Gamma^\star)$ solves (8), then $\varphi_{\pi^\star,\Gamma^\star}$ is the unique solution to (7).

Proof. *Strong duality.* Write $\mathcal{L}_\pi(\Gamma, \varphi)$ for the Lagrangian, so that the objective of (8) is $\mathcal{L}_\pi(\Gamma, \varphi_{\pi,\Gamma})$. The proof amounts to verifying that for each $\pi \in \Pi$, the following chain of equalities holds,

$$\sup_{\Gamma\in\mathcal{G}} \mathcal{L}_\pi(\Gamma, \varphi_{\pi,\Gamma}) = \sup_{\Gamma\in\mathcal{G}} \inf_{\varphi\in\Phi} \mathcal{L}_\pi(\Gamma, \varphi) = \inf_{\varphi\in\Phi} \sup_{\Gamma\in\mathcal{G}} \mathcal{L}_\pi(\Gamma, \varphi) = \inf_{\varphi\in\Phi_\alpha} R(\pi, \varphi).$$

For the first equality, recall the expression for the Lagrangian derived in the main text,

$$\mathcal{L}_\pi(\Gamma, \varphi) = \int \big[(1 - \varphi(y))p_{\omega_\pi}(y) + \varphi(y)p_\Gamma(y)\big]\, \mathrm{d}y - \alpha\Gamma(Q\mathcal{M}(0)).$$

The Neyman–Pearson test $\varphi_{\pi,\Gamma} = \mathbb{1}\{p_{\omega_\pi} > p_\Gamma\}$ solves $\inf_{\varphi\in\Phi} \mathcal{L}_\pi(\Gamma,\varphi)$ since it minimizes the above integrand over $[0,1]$ for each $y \in \mathbb{R}^{k_\nu}$. In particular, the objective of (8) is given by

$$\mathcal{L}_\pi(\Gamma, \varphi_{\pi,\Gamma}) = \int \min\{p_{\omega_\pi}(y), p_\Gamma(y)\}\,\mathrm{d}y - \alpha\Gamma(Q\mathcal{M}(0)). \tag{17}$$

To establish the second equality, I verify the conditions of Evans et al. (2005, Theorem 4) as in the proof of Lemma 4.1. The set $\mathcal{G}$ is convex and $\mathcal{L}_\pi(\,\cdot\,,\varphi)$ is affine, as the integral of $\nu \mapsto \mathbb{E}_{P_\nu}[\varphi(Y)] - \alpha$ against Γ plus a constant. The set Φ is convex and weak-$*$ compact by the proof of Lemma 4.1. Finally, $\mathcal{L}_\pi(\Gamma,\,\cdot\,)$ is convex and weak-$*$ lower semicontinuous, as the sum of $R(\pi,\,\cdot\,)$, which has both properties by the proof of Lemma 4.1, and the weak-$*$ continuous affine functional $\varphi \mapsto \langle\varphi, p_\Gamma\rangle - \alpha\Gamma(Q\mathcal{M}(0))$, since $\|p_\Gamma\|_1 = \Gamma(Q\mathcal{M}(0))$ is finite.

The third and final equality follows from noticing that if $\varphi \notin \Phi_\alpha$, then $\mathbb{E}_{P_\nu}[\varphi(Y)] > \alpha$ for some $\nu \in Q\mathcal{M}(0)$, meaning Γ can place as much mass at ν as we like, sending $\mathcal{L}_\pi(\Gamma,\varphi) \to \infty$. On the other hand, if $\varphi \in \Phi_\alpha$, then the integral against Γ in the Lagrangian is non-positive, so it is maximized at $\Gamma = 0$, meaning $\sup_{\Gamma\in\mathcal{G}} \mathcal{L}_\pi(\Gamma,\varphi) = R(\pi,\varphi)$.

Taking the supremum of the chain of equalities over $\pi \in \Pi$, the left-hand side is the value of (8), while the right-hand side is the value of (7) by Lemma 4.1. Denote this common value by $v^\star$.

Uniqueness. Suppose $v^\star > 0$ and $(\pi^\star, \Gamma^\star)$ solves (8). Then by the chain of equalities, $\pi^\star$ is least favorable in the sense of Lemma 4.1. Let φ° be any solution to (7), which exists by Theorem 4.1, so $R(\pi^\star,\varphi^\circ) = v^\star$ by Lemma 4.1. Since $\varphi^\circ \in \Phi_\alpha$, the integral against $\Gamma^\star$ in the Lagrangian is non-positive, so

$$v^\star = \mathcal{L}_{\pi^\star}(\Gamma^\star, \varphi_{\pi^\star,\Gamma^\star}) = \inf_{\varphi\in\Phi} \mathcal{L}_{\pi^\star}(\Gamma^\star,\varphi) \le \mathcal{L}_{\pi^\star}(\Gamma^\star,\varphi^\circ) \le R(\pi^\star,\varphi^\circ) = v^\star,$$

meaning φ° solves $\inf_{\varphi\in\Phi} \mathcal{L}_{\pi^\star}(\Gamma^\star,\varphi)$. Then by the expression for the Lagrangian and (17),

$$\int \left[(1-\varphi^\circ(y))p_{\omega_{\pi^\star}}(y) + \varphi^\circ(y)p_{\Gamma^\star}(y)\right]\mathrm{d}y = \int \min\{p_{\omega_{\pi^\star}}(y),\, p_{\Gamma^\star}(y)\}\,\mathrm{d}y,$$

where both sides are bounded above by $\Gamma^\star(Q\mathcal{M}(0)) < \infty$. As the integrand on the left is at least the integrand on the right, the two must agree for almost every y, so for any such y with $p_{\omega_{\pi^\star}}(y) \neq p_{\Gamma^\star}(y)$ we have $\varphi^\circ(y) = \mathbb{1}\{p_{\omega_{\pi^\star}}(y) > p_{\Gamma^\star}(y)\} = \varphi_{\pi^\star,\Gamma^\star}(y)$. Since $v^\star > 0$ it follows that $\Gamma^\star \neq 0$, so Lemma B.4 implies that $p_{\omega_{\pi^\star}} \neq p_{\Gamma^\star}$ almost everywhere. Consequently, $\varphi^\circ = \varphi_{\pi^\star,\Gamma^\star}$ as elements of $L^\infty(\mathbb{R}^{k_\nu})$. □

B.6 Proof of Lemma 5.2 and Proposition 5.1

Lemma 5.2 (Discretization)**.** *There exists a solution $(\tilde\pi^\star, \tilde\Gamma^\star)$ to the discretized dual problem (9), which shares its value with (10). Moreover, if this value is positive, then $\varphi_{\tilde\pi^\star,\tilde\Gamma^\star}$ is the unique solution to (10).*

Proof. For $\tilde\pi$ and $\tilde\Gamma$ in the feasible set of (9), let $\pi \in \Pi$ denote the prior that places mass $\tilde\pi_l$ on $\nu_{-1}^{(l)}$, and let $\Gamma \in \mathcal{G}$ denote the measure that places mass $\tilde\Gamma_m$ on $\nu^{(m)}$. Then $p_{\omega_{\tilde\pi}}$, $p_{\tilde\Gamma}$, $\varphi_{\tilde\pi,\tilde\Gamma}$, and $\mathcal{L}_{\tilde\pi}(\tilde\Gamma,\varphi)$ are understood to

mean p_{ω_π}, p_Γ, $\varphi_{\pi,\Gamma}$, and $\mathcal{L}_\pi(\Gamma,\varphi)$, so by (17) the objective of (9) is

$$\mathcal{L}_{\tilde\pi}(\tilde\Gamma,\varphi_{\tilde\pi,\tilde\Gamma}) = \int \min\{p_{\omega_{\tilde\pi}}(y),\, p_{\tilde\Gamma}(y)\}\,\mathrm{d}y - \alpha\mathbf{1}'\tilde\Gamma,$$

where

$$p_{\omega_{\tilde\pi}}(y) = \sum_{l=1}^{L}\tilde\pi_l \int \mathbb{1}\big\{(\nu_1,\nu_{-1}^{(l)}) \notin Q\mathcal{M}(0)\big\}\, p_{(\nu_1,\nu_{-1}^{(l)})}(y)\,\mathrm{d}\nu_1 \quad\text{and}\quad p_{\tilde\Gamma}(y) = \sum_{m=1}^{M}\tilde\Gamma_m\, p_{\nu^{(m)}}(y).$$

Existence. I first show that the objective is negative whenever $\mathbf{1}'\tilde\Gamma$ is sufficiently large. By the proof of Lemma 5.1, $\varphi_{\tilde\pi,\tilde\Gamma}$ minimizes $\mathcal{L}_{\tilde\pi}(\tilde\Gamma,\,\cdot\,)$ over tests. In particular, for any $h>0$ with $\varphi_h(y)\equiv\mathbb{1}\{|y_1|>h\}$, the objective is at most $\mathcal{L}_{\tilde\pi}(\tilde\Gamma,\varphi_h)$. Inverting φ_h yields the interval $[Q_1X-h, Q_1X+h]$ of length $2h$, so $R(\nu_{-1}^{(l)},\varphi_h)\le 2h$ for each l. Moreover, since the null grid is finite, we may choose h large enough so that $\mathbb{E}_{P_{\nu^{(m)}}}[\varphi_h(Y)]\le\alpha/2$ for each m. Putting these bounds together,

$$\mathcal{L}_{\tilde\pi}(\tilde\Gamma,\varphi_h) = \sum_{l=1}^{L}\tilde\pi_l\, R(\nu_{-1}^{(l)},\varphi_h) + \sum_{m=1}^{M}\tilde\Gamma_m\big(\mathbb{E}_{P_{\nu^{(m)}}}[\varphi_h(Y)]-\alpha\big) \le 2h - \frac{\alpha}{2}\mathbf{1}'\tilde\Gamma,$$

which is negative whenever $\mathbf{1}'\tilde\Gamma > 4h/\alpha \equiv \overline{\gamma}$. As the objective is equal to zero at $\tilde\Gamma=0$, adding the constraint $\mathbf{1}'\tilde\Gamma\le\overline{\gamma}$ to (9) leaves its value unchanged and makes its feasible set compact. On this set, the integrand $\min\{p_{\omega_{\tilde\pi}}(y), p_{\tilde\Gamma}(y)\}$ is continuous in $(\tilde\pi,\tilde\Gamma)$ for each y and bounded by the integrable function $\overline{\gamma}\sum_m p_{\nu^{(m)}}(y)$, so the objective is continuous by dominated convergence. By the Weierstrass extreme value theorem, it attains its maximum at some $(\tilde\pi^\star,\tilde\Gamma^\star)$, which therefore solves (9).

Strong duality. Let $\tilde\Phi_\alpha$ denote the feasible set of (10), which is convex and weak-$*$ compact as the intersection of Φ with the M weak-$*$ closed half-spaces $\{\varphi : \langle\varphi, p_{\nu^{(m)}}\rangle\le\alpha\}$. The proof of Lemma 4.1 and the strong duality part of the proof of Lemma 5.1 go through with Φ_α replaced by $\tilde\Phi_\alpha$, and with Π and $\mathcal{G}$ replaced by the convex sets of priors and measures supported on the risk and null grids, respectively. In particular, the supremum over $Q_{-1}\mathcal{M}$ in Lemma 4.1 becomes a maximum over the risk grid, and a test outside $\tilde\Phi_\alpha$ violates the size constraint at some null grid point, at which Γ can place as much mass as we like. Hence (9) and (10) share the same value.

Uniqueness. Suppose this value is positive and $(\tilde\pi^\star,\tilde\Gamma^\star)$ solves (9). The uniqueness part of the proof of Lemma 5.1 applies with $\tilde\pi^\star$, $\tilde\Gamma^\star$, $\tilde\Phi_\alpha$, (9), and (10) in place of $\pi^\star$, $\Gamma^\star$, Φ_α, (8), and (7), except that a solution to (10) exists by the Weierstrass extreme value theorem rather than Theorem 4.1, since $\tilde\Phi_\alpha$ is non-empty and weak-$*$ compact, and $\varphi\mapsto\max_l R(\nu_{-1}^{(l)},\varphi)$ is weak-$*$ lower semicontinuous by the proof of Lemma 4.1. Hence $\varphi_{\tilde\pi^\star,\tilde\Gamma^\star}$ is the unique solution to (10). □

Proposition 5.1 (Concave Program)**.** *The objective of the discretized dual problem* (9) *is jointly concave in* $(\tilde\pi,\tilde\Gamma)$*. For some finite* $\overline{\gamma}>0$*, it is without loss to impose the constraint* $\mathbf{1}'\tilde\Gamma\le\overline{\gamma}$*, so the feasible set*

$$\{(\tilde\pi,\tilde\Gamma)\in\mathbb{R}^{L+M} : (\tilde\pi,\tilde\Gamma)\ge 0, \mathbf{1}'\tilde\pi = 1, \mathbf{1}'\tilde\Gamma\le\overline{\gamma}\}$$

is convex and compact. Finally, the gradient of the objective at $(\tilde{\pi}, \tilde{\Gamma}) > 0$ *is given by*

$$\big(R(\nu_{-1}^{(1)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}), \dots, R(\nu_{-1}^{(L)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}), \mathbb{E}_{P_{\nu^{(1)}}}[\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y)] - \alpha, \dots, \mathbb{E}_{P_{\nu^{(M)}}}[\varphi_{\tilde{\pi},\tilde{\Gamma}}(Y)] - \alpha\big).$$

Proof. Write $f(\tilde{\pi}, \tilde{\Gamma})$ for the objective of (9). As in the proof of Lemma 5.2, (17) implies that

$$f(\tilde{\pi}, \tilde{\Gamma}) = \int \min\{p_{\omega_{\tilde{\pi}}}(y),\, p_{\tilde{\Gamma}}(y)\}\, \mathrm{d}y - \alpha \mathbf{1}'\tilde{\Gamma}$$

where $p_{\omega_{\tilde{\pi}}} = \sum_{l=1}^{L} \tilde{\pi}_l\, p_{\omega_{e_l}}$ and $p_{\tilde{\Gamma}} = \sum_{m=1}^{M} \tilde{\Gamma}_m\, p_{\nu^{(m)}}$, with e_l denoting the lth unit vector in $\mathbb{R}^L$. Both sides are defined for all $(\tilde{\pi}, \tilde{\Gamma}) \geq 0$ and agree there, since the Neyman–Pearson test minimizes the integrand of the Lagrangian pointwise, so I regard f as a function on $\mathbb{R}_+^{L+M}$.

Concavity. For each y, the densities $p_{\omega_{\tilde{\pi}}}(y)$ and $p_{\tilde{\Gamma}}(y)$ are linear in $(\tilde{\pi}, \tilde{\Gamma})$, so their minimum is concave. Integrating over y preserves concavity, and the integral is finite since it is at most $\int p_{\tilde{\Gamma}}(y)\, \mathrm{d}y = \mathbf{1}'\tilde{\Gamma}$. Subtracting the linear term $\alpha \mathbf{1}'\tilde{\Gamma}$, it follows that f is finite and concave.

Compactness. By the proof of Lemma 5.2, there is a finite $\overline{\gamma} > 0$ such that f is negative at every feasible point of (9) with $\mathbf{1}'\tilde{\Gamma} > \overline{\gamma}$. Since f vanishes at $\tilde{\Gamma} = 0$, imposing the constraint $\mathbf{1}'\tilde{\Gamma} \leq \overline{\gamma}$ changes neither the value of (9) nor its solutions. The resulting feasible set is an intersection of closed half-spaces, hence convex and closed, and it is bounded since $0 \leq \tilde{\pi}_l \leq 1$ and $0 \leq \tilde{\Gamma}_m \leq \overline{\gamma}$, hence compact.

Gradient. Fix $(\tilde{\pi}, \tilde{\Gamma}) > 0$ and denote the proposed gradient by $\nabla f(\tilde{\pi}, \tilde{\Gamma})$. First, I verify that all of its components are finite. The excess type I error rates lie in $[-\alpha, 1 - \alpha]$ by inspection. For the risks, by (17) we have that $f(\tilde{\pi}, \tilde{\Gamma}) \leq (1 - \alpha)\mathbf{1}'\tilde{\Gamma}$, with the excess type I error rates contributing at least $-\alpha \mathbf{1}'\tilde{\Gamma}$ to the objective. It follows that $\sum_{l=1}^{L} \tilde{\pi}_l R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}) \leq \mathbf{1}'\tilde{\Gamma}$, so in particular $R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi},\tilde{\Gamma}}) \leq \mathbf{1}'\tilde{\Gamma}/\tilde{\pi}_l < \infty$ for each l since $\tilde{\pi}_l > 0$.

Now, observe that $f(\tilde{\pi}, \tilde{\Gamma}) = \nabla f(\tilde{\pi}, \tilde{\Gamma})'(\tilde{\pi}, \tilde{\Gamma}) \leq \nabla f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ)'(\tilde{\pi}, \tilde{\Gamma})$ for any $(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) > 0$, where the inequality follows from the fact that the Neyman–Pearson test $\varphi_{\tilde{\pi},\tilde{\Gamma}}$ minimizes the Lagrangian at $(\tilde{\pi}, \tilde{\Gamma})$. Adding and subtracting $f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) = \nabla f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ)'(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ)$ and applying the same reasoning with the roles of $(\tilde{\pi}, \tilde{\Gamma})$ and $(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ)$ reversed, we get that

$$\begin{aligned} f(\tilde{\pi}, \tilde{\Gamma}) &\leq f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) + \nabla f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ)'[(\tilde{\pi}, \tilde{\Gamma}) - (\tilde{\pi}^\circ, \tilde{\Gamma}^\circ)], \\ f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) &\leq f(\tilde{\pi}, \tilde{\Gamma}) + \nabla f(\tilde{\pi}, \tilde{\Gamma})'[(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) - (\tilde{\pi}, \tilde{\Gamma})]. \end{aligned}$$

Rearranging terms bounds the error of the proposed local linear approximation,

$$[\nabla f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) - \nabla f(\tilde{\pi}, \tilde{\Gamma})]'[(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) - (\tilde{\pi}, \tilde{\Gamma})] \leq f(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) - f(\tilde{\pi}, \tilde{\Gamma}) - \nabla f(\tilde{\pi}, \tilde{\Gamma})'[(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) - (\tilde{\pi}, \tilde{\Gamma})] \leq 0.$$

To show that f is differentiable at $(\tilde{\pi}, \tilde{\Gamma})$ with gradient $\nabla f(\tilde{\pi}, \tilde{\Gamma})$, then, it suffices to show that ∇f is continuous at $(\tilde{\pi}, \tilde{\Gamma})$, so the approximation error is of smaller order than $\|(\tilde{\pi}^\circ, \tilde{\Gamma}^\circ) - (\tilde{\pi}, \tilde{\Gamma})\|$ by Cauchy–Schwarz.

To this end, let $(\tilde{\pi}_n, \tilde{\Gamma}_n) > 0$ converge to $(\tilde{\pi}, \tilde{\Gamma})$ and write $\varphi_n \equiv \varphi_{\tilde{\pi}_n, \tilde{\Gamma}_n}$. Then $\nabla f(\tilde{\pi}_n, \tilde{\Gamma}_n)$ is bounded, since the risks are at most $\mathbf{1}'\tilde{\Gamma}_n/\tilde{\pi}_{n,l} \to \mathbf{1}'\tilde{\Gamma}/\tilde{\pi}_l < \infty$, so by the continuity of the concave function f on the open orthant,

$$\mathcal{L}_{\tilde{\pi}}(\tilde{\Gamma}, \varphi_n) = \nabla f(\tilde{\pi}_n, \tilde{\Gamma}_n)'(\tilde{\pi}, \tilde{\Gamma}) = f(\tilde{\pi}_n, \tilde{\Gamma}_n) + \nabla f(\tilde{\pi}_n, \tilde{\Gamma}_n)'\big[(\tilde{\pi}, \tilde{\Gamma}) - (\tilde{\pi}_n, \tilde{\Gamma}_n)\big] \to f(\tilde{\pi}, \tilde{\Gamma}).$$

Since Φ is weak-$*$ compact, (φ_n) has a weak-$*$ cluster point, and any such point φ° satisfies $\mathcal{L}_{\tilde{\pi}}(\tilde{\Gamma}, \varphi^\circ) \leq f(\tilde{\pi}, \tilde{\Gamma})$, as $\mathcal{L}_{\tilde{\pi}}(\tilde{\Gamma}, \cdot\,)$ is weak-$*$ lower semicontinuous by the proof of Lemma 5.1. Hence φ° minimizes $\mathcal{L}_{\tilde{\pi}}(\tilde{\Gamma}, \cdot\,)$ and therefore equals $\varphi_{\tilde{\pi}, \tilde{\Gamma}}$ by the proof of Lemma 5.1, applied after dividing $(\tilde{\pi}, \tilde{\Gamma})$ by $\mathbf{1}'\tilde{\pi} > 0$. A sequence in a compact space with a single cluster point converges to it, so $\varphi_n \to \varphi_{\tilde{\pi}, \tilde{\Gamma}}$ weak-$*$. The type I error rates of φ_n therefore converge to those of $\varphi_{\tilde{\pi}, \tilde{\Gamma}}$, meaning that $\sum_l \tilde{\pi}_l R(\nu_{-1}^{(l)}, \varphi_n) \to \sum_l \tilde{\pi}_l R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi}, \tilde{\Gamma}})$ since $\mathcal{L}_{\tilde{\pi}}(\tilde{\Gamma}, \varphi_n) \to f(\tilde{\pi}, \tilde{\Gamma}) = \mathcal{L}_{\tilde{\pi}}(\tilde{\Gamma}, \varphi_{\tilde{\pi}, \tilde{\Gamma}})$. Since each risk is weak-$*$ lower semicontinuous by the proof of Lemma 4.1 and each $\tilde{\pi}_l > 0$, it follows that $R(\nu_{-1}^{(l)}, \varphi_n) \to R(\nu_{-1}^{(l)}, \varphi_{\tilde{\pi}, \tilde{\Gamma}})$ for each l. Hence $\nabla f(\tilde{\pi}_n, \tilde{\Gamma}_n) \to \nabla f(\tilde{\pi}, \tilde{\Gamma})$. □

B.7 Proof of Lemma 5.3

Lemma 5.3 (Convergence)**.** *Let $\overline{\gamma}$ be as in Proposition 5.1, and suppose that there exists $\overline{r} > 0$ such that for every $(\tilde{\pi}, \tilde{\Gamma}) > 0$ in the feasible set of Proposition 5.1, the unbiased risk estimator in line 4 of Algorithm 1 computed at $(\tilde{\pi}, \tilde{\Gamma})$ satisfies*

$$\sqrt{\mathbb{E}\Big[\tfrac{1}{4}\big(\max_{l \leq L} \hat{R}_l - \min_{l \leq L} \hat{R}_l\big)^2\Big]} \leq \overline{r},$$

where the expectation is taken over Monte Carlo simulations. Write f for the objective of (9) and $f^\star$ for its optimal value. Then if $\alpha \leq \frac{1}{2}$, the output of Algorithm 1 with step sizes

$$\eta_L = \frac{1}{\overline{r}}\sqrt{\frac{2\log(L)}{T}} \quad \textit{and} \quad \eta_M = \frac{1}{1-\alpha}\sqrt{\frac{2\max\{1,\, 1+\log(\overline{\gamma}),\, \log(M)\}}{T}}$$

satisfies

$$f^\star - \mathbb{E}\big[f(\hat{\pi}_T, \hat{\Gamma}_T)\big] \leq \overline{r}\sqrt{\frac{2\log(L)}{T}} + (1-\alpha)\,\overline{\gamma}\sqrt{\frac{2\max\{1,\, 1+\log(\overline{\gamma}),\, \log(M)\}}{T}}.$$

Proof. The proof rests on a fundamental inequality for the method of mirror ascent that holds along an arbitrary sequence of vectors. Let $\mathbf{\Phi}(x) \equiv \sum_i x_i \log(x_i)$ denote the negative entropy of a finite-dimensional vector $x > 0$, which is a mirror map (Bubeck, 2015, §4.3) with Bregman divergence $D(x, y) = \sum_i [x_i \log(x_i/y_i) - x_i + y_i]$. Given a compact convex set $\mathcal{X}$, a step size $\eta > 0$, and vectors $g_1, \ldots, g_T$, mirror ascent begins at $x_1 = \arg\min_{x \in \mathcal{X}, x > 0} \mathbf{\Phi}(x)$ and sets $x_{t+1} = \arg\max_{x \in \mathcal{X}, x > 0} \eta g_t' x - D(x, x_t)$. If $\mathbf{\Phi}$ is ρ-strongly convex with respect to $\|\cdot\|_1$ over $\mathcal{X} \cap \{x > 0\}$, then for every $x \in \mathcal{X}$ with $x > 0$,

$$\frac{1}{T}\sum_{t=1}^{T} g_t'(x - x_t) \leq \frac{D(x, x_1)}{\eta T} + \frac{\eta}{2\rho T}\sum_{t=1}^{T} \|g_t\|_\infty^2 \tag{18}$$

by the proof of Bubeck (2015, Theorem 4.2).[18] We may view Algorithm 1 as performing mirror ascent twice: once for each coordinate block of $(\tilde{\pi}, \tilde{\Gamma})$. By Bubeck (2015, §4.3), lines 1 and 6 perform mirror ascent on the simplex $\mathcal{X} = \{\tilde{\pi} \geq 0 : \mathbf{1}'\tilde{\pi} = 1\}$ with step size η_L along $\hat{R}_1, \ldots, \hat{R}_T$. By Aradillas Fernández et al. (2025, Theorem 1, Parts 2 and 3), lines 1 and 7 perform mirror ascent on the capped orthant $\mathcal{X} = \{\tilde{\Gamma} \geq 0 : \mathbf{1}'\tilde{\Gamma} \leq \overline{\gamma}\}$ with step size η_M along $\hat{\alpha}_1 - \alpha, \ldots, \hat{\alpha}_T - \alpha$.[19] I will proceed by bounding the right-hand side of (18) for each block in turn.

By Shalev-Shwartz (2012, Example 2.5), $\rho = 1$ over the simplex. Since $\tilde{\pi}_1$ is uniform, $D(\tilde{\pi}, \tilde{\pi}_1) = \log(L) + \sum_l \tilde{\pi}_l \log(\tilde{\pi}_l) \leq \log(L)$. Now, notice that each $\hat{R}_t$ can be replaced by $\hat{R}_t + s_t\mathbf{1}$ for any $s_t \in \mathbb{R}$ without affecting the left-hand side of (18): the iterates $\tilde{\pi}_t$ are unchanged due to the normalization in line 6, and the summands $(\hat{R}_t + s_t\mathbf{1})'(\tilde{\pi} - \tilde{\pi}_t)$ are unchanged since $\mathbf{1}'(\tilde{\pi} - \tilde{\pi}_t) = 0$. Hence, for each t, s_t can be chosen so that $\|\hat{R}_t + s_t\mathbf{1}\|_\infty = \min_{s\in\mathbb{R}} \|\hat{R}_t + s\mathbf{1}\|_\infty = \frac{1}{2}(\max_{l\leq L} \hat{R}_{t,l} - \min_{l\leq L} \hat{R}_{t,l})$. By (18), then, for every $\tilde{\pi} > 0$ in the simplex,

$$\frac{1}{T}\sum_{t=1}^{T} \hat{R}_t'(\tilde{\pi} - \tilde{\pi}_t) \leq \frac{\log(L)}{\eta_L T} + \frac{\eta_L}{2T}\sum_{t=1}^{T} \tfrac{1}{4}\big(\max_{l\leq L} \hat{R}_{t,l} - \min_{l\leq L} \hat{R}_{t,l}\big)^2. \tag{19}$$

By Shalev-Shwartz (2012, Example 2.5), $\rho = 1/\overline{\gamma}$ over the capped orthant. To bound $D(\tilde{\Gamma}, \tilde{\Gamma}_1)$, note that $D(\,\cdot\,, \tilde{\Gamma}_1)$ is convex, so its maximum on the capped orthant is attained at a vertex: either 0, or $\overline{\gamma}e_m$ for some $m \leq M$. Plugging in, we get that $D(0, \tilde{\Gamma}_1) = \min\{M\exp(-1), \overline{\gamma}\} \leq \overline{\gamma}$ and

$$D(\overline{\gamma}e_m, \tilde{\Gamma}_1) = \overline{\gamma}\max\{1 + \log(\overline{\gamma}), \log(M)\} - \overline{\gamma} + \min\{M\exp(-1), \overline{\gamma}\} \leq \overline{\gamma}\max\{1 + \log(\overline{\gamma}), \log(M)\}.$$

It follows that $D(\tilde{\Gamma}, \tilde{\Gamma}_1) \leq \overline{\gamma}\max\{1, 1 + \log(\overline{\gamma}), \log(M)\}$. Finally, $\|\hat{\alpha}_t - \alpha\|_\infty \leq 1 - \alpha$ since $\hat{\alpha}_t \in [0,1]^M$ and $\alpha \leq 1/2$. By (18), then, for every $\tilde{\Gamma} > 0$ in the capped orthant,

$$\frac{1}{T}\sum_{t=1}^{T} (\hat{\alpha}_t - \alpha)'(\tilde{\Gamma} - \tilde{\Gamma}_t) \leq \frac{\overline{\gamma}\max\{1, 1 + \log(\overline{\gamma}), \log(M)\}}{\eta_M T} + \frac{\eta_M \overline{\gamma}(1-\alpha)^2}{2}. \tag{20}$$

Now, fix $(\tilde{\pi}, \tilde{\Gamma}) > 0$ in the feasible set of Proposition 5.1. Then

$$\begin{aligned}
f(\tilde{\pi}, \tilde{\Gamma}) - \mathbb{E}\big[f(\hat{\pi}_T, \hat{\Gamma}_T)\big] &\leq \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[f(\tilde{\pi}, \tilde{\Gamma}) - f(\tilde{\pi}_t, \tilde{\Gamma}_t)\big] \\
&\leq \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\nabla f(\tilde{\pi}_t, \tilde{\Gamma}_t)'(\tilde{\pi} - \tilde{\pi}_t, \tilde{\Gamma} - \tilde{\Gamma}_t)\big] \\
&= \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\hat{R}_t'(\tilde{\pi} - \tilde{\pi}_t) + (\hat{\alpha}_t - \alpha)'(\tilde{\Gamma} - \tilde{\Gamma}_t)\big].
\end{aligned}$$

The first inequality holds by concavity of f, since $(\hat{\pi}_T, \hat{\Gamma}_T)$ is the average of the iterates. The second holds because each iterate $(\tilde{\pi}_t, \tilde{\Gamma}_t)$ is positive, so the gradient of the concave function f exists and is a supergradient by Proposition 5.1. The equality follows from the law of iterated expectations, since $(\hat{R}_t, \hat{\alpha}_t - \alpha)$ is unbiased

[18] See also Bubeck (2015, Equation (4.10)) and the bound on Aradillas Fernández et al. (2025, Equation (41)).

[19] Aradillas Fernández et al. (2025) use the bound $\mathbf{1}'\tilde{\Gamma} \leq 1/\alpha$, which need not be valid in the current setting. Transferring their results therefore requires replacing $1/\alpha$ by $\overline{\gamma}$.

for the gradient conditional on $(\tilde{\pi}_t, \tilde{\Gamma}_t)$. The last line is the expectation of the sum of the left-hand sides of (19) and (20), so it is at most the expectation of the sum of their right-hand sides. In addition, the inequality $\mathbb{E}\Big[\frac{1}{4}\big(\max_{l\le L}\hat{R}_{t,l} - \min_{l\le L}\hat{R}_{t,l}\big)^2\Big] \le \overline{r}^2$ holds for all $t \le T$ by the law of iterated expectations, since $\overline{r}^2$ is a valid bound for the expectation conditional on $(\tilde{\pi}_t, \tilde{\Gamma}_t)$ by assumption. Hence,

$$f(\tilde{\pi}, \tilde{\Gamma}) - \mathbb{E}\big[f(\hat{\pi}_T, \hat{\Gamma}_T)\big] \le \frac{\log(L)}{\eta_L T} + \frac{\eta_L \overline{r}^2}{2} + \frac{\overline{\gamma}\max\{1, 1+\log(\overline{\gamma}), \log(M)\}}{\eta_M T} + \frac{\eta_M \overline{\gamma}(1-\alpha)^2}{2}.$$

This bound holds for all $(\tilde{\pi}, \tilde{\Gamma}) > 0$ in the feasible set of Proposition 5.1, so we may take the supremum of the left-hand side over such points to get $f^\star - \mathbb{E}\big[f(\hat{\pi}_T, \hat{\Gamma}_T)\big]$, since f is continuous on the feasible set by the proof of Lemma 5.2 and the constraint $\mathbf{1}'\tilde{\Gamma} \le \overline{\gamma}$ is without loss by Proposition 5.1. Finally, with the step sizes in the statement, the two terms from the $\tilde{\pi}$ block both equal $\overline{r}\sqrt{\log(L)/(2T)}$, and the two terms from the $\tilde{\Gamma}$ block both equal $(1-\alpha)\overline{\gamma}\sqrt{\max\{1, 1+\log(\overline{\gamma}), \log(M)\}/(2T)}$. Adding them yields the bound. □

C Local Asymptotic Optimality

This section makes precise the sense in which (2) arises as a limit experiment under local asymptotics. Moreover, it transfers optimality in the limit experiment to local asymptotic optimality of a feasible confidence procedure. The main result closely mirrors classical local asymptotic minimax theorems (van der Vaart, 1998, Theorem 25.21). In particular, the data are taken to be a random sample from a semiparametric model, confidence procedures are required to be asymptotically valid, and the local asymptotic maximum risk of any valid procedure is bounded below by the minimax value $v^\star$ of the limit experiment. This value is then attained by inverting the optimal test of Theorem 4.1 at an efficient estimator of the moments.

C.1 Sampling

The researcher observes a random sample $W_1, \ldots, W_n$ from a distribution P in a set $\mathcal{P}$ of probability measures on a measurable space $\mathcal{W}$, and the moments are a functional $\mu : \mathcal{P} \to \mathbb{R}^{k_\mu}$. A priori restrictions on the moments, such as the requirement that the moments in Example 2.2 be bounded between zero and one, are restrictions on $\mathcal{P}$. For $P \in \mathcal{P}$, write P^n for the joint law of $(W_1, \ldots, W_n, U)$ when the W_i are i.i.d. P and $U \sim \text{Unif}[0,1]$ is the independent auxiliary variable of Section 3.1. Asymptotics are local to a fixed $P_0 \in \mathcal{P}$, with $\mu_0 \equiv \mu(P_0)$. Paths $s \mapsto P_s$ in $\mathcal{P}$ through P_0 at $s = 0$, differentiability in quadratic mean, score functions, and differentiability of μ at P_0 are identical to the definitions given in van der Vaart (1998, Section 25.3). Finally, the tangent set $\dot{\mathcal{P}}_{P_0}$ consists of the score functions of all paths in $\mathcal{P}$ that are differentiable in quadratic mean at P_0, and $\tilde{\mu}_{P_0}$ denotes the efficient influence function of μ.

Assumption C.1 (Sampling)**.**

(i) μ is differentiable at P_0 relative to $\dot{\mathcal{P}}_{P_0}$ with efficient influence function $\tilde{\mu}_{P_0}$, and $\Sigma \equiv P_0 \tilde{\mu}_{P_0} \tilde{\mu}'_{P_0}$ is nonsingular.

(ii) For every $\mu \in \mathbb{R}^{k_\mu}$, there is a path $s \mapsto P_{s,\mu}$ in $\mathcal{P}$ that is differentiable in quadratic mean at $s = 0$ with score function $\mu' \Sigma^{-1} \tilde{\mu}_{P_0}$ and satisfies $\mu(P_{s,\mu}) = \mu_0 + s\mu$ for all sufficiently small $s \geq 0$.

Part (i) is the differentiability hypothesis of the convolution and local asymptotic minimax theorems for μ. The covariance matrix Σ is the semiparametric efficiency bound, and its nonsingularity is Assumption 2.4 for the limit experiment. Part (ii) asks for paths along which the moments are exactly linear, in the direction of the least favorable score function for μ. In particular, it requires that a priori restrictions on the moments do not bind at P_0. Part (ii) holds whenever $\mathcal{P}$ is sufficiently rich, in the sense of Assumption C.1 of Armstrong and Kolesár (2021); see their Appendix C and Lemma E.1.

Write $\mathbb{E}_{n,\mu}$ for expectation under $P^n_{1/\sqrt{n},\mu}$, and let $X_n \equiv n^{-1/2} \sum_{i=1}^n \tilde{\mu}_{P_0}(W_i)$, so that $X_n \xrightarrow{d} \mathcal{N}(0, \Sigma)$ under P_0^n by the central limit theorem. Then by Lemma 25.14 of van der Vaart (1998),

$$\log \frac{\mathrm{d}P^n_{1/\sqrt{n},\mu}}{\mathrm{d}P_0^n} = \mu' \Sigma^{-1} X_n - \tfrac{1}{2} \mu' \Sigma^{-1} \mu + o_{P_0}(1).$$

By Theorem 9.4 of van der Vaart (1998), the sequence of experiments $(P^n_{1/\sqrt{n},\mu} : \mu \in \mathbb{R}^{k_\mu})$ converges to the Gaussian shift experiment $(\mathcal{N}(\mu, \Sigma) : \mu \in \mathbb{R}^{k_\mu})$. This is the experiment (2) with covariance matrix Σ, where the mean is the local value $\sqrt{n}(\mu(P) - \mu_0)$ of the moments, equal to μ along $P_{1/\sqrt{n},\mu}$ for all sufficiently large n.

C.2 Model at Sample Size n

A pair $(\theta, P) \in \mathbb{R} \times \mathcal{P}$ satisfies the model at sample size n if

$$\begin{aligned} \mu(P) &= K\tau + n^{-1/2}\delta \\ \theta &= l'\tau \end{aligned} \quad \text{for some} \quad (\tau, \delta) \in \mathcal{T} \times \Delta, \tag{21}$$

where $\mathcal{T} \subseteq \mathbb{R}^{k_\tau}$ is an a priori restriction on the structural parameters, and

$$\Theta_n(P) \equiv \{\theta \in \mathbb{R} : (\theta, P) \text{ satisfies (21)}\}$$

is the identified set for θ under P at sample size n. The factor $n^{-1/2}$ places the nuisance parameters on the same scale as sampling error. In Examples 2.2 to 2.4, Δ is a cone, so the scaling is immaterial and the model does not depend on n.

Assumption C.2 (Localization)**.** $\mathcal{T}$ is σ-compact, and $\mu_0 = K\tau_0$ for some τ_0 in the interior of $\mathcal{T}$.

By Proposition A.2, l lies in the row space of K, so $l'\tau_0$ does not depend on the above choice of τ_0. Write $\theta_0 \equiv l'\tau_0$ for this common value. Interiority is what causes the a priori restrictions to disappear in the limit.

C.3 Feasible Procedures

A confidence procedure at sample size n is a measurable map $S_n : \mathcal{W}^n \times [0,1] \to \mathcal{A}$, where $w = (w_1, \ldots, w_n) \in \mathcal{W}^n$ denotes a realization of the sample. A measurable representative of S_n is a map $\overline{S}_n : \mathcal{W}^n \times [0,1] \to \mathcal{B}$ with the properties of Part (v) of Lemma B.1, and $P^n(\theta \in \overline{S}_n)$ denotes the probability that $\theta \in \overline{S}_n(W_1, \ldots, W_n, U)$.

Let $\hat{\mu}_n : \mathcal{W}^n \to \mathbb{R}^{k_\mu}$ be an estimator of $\mu(P)$ and $\hat{\Sigma}_n : \mathcal{W}^n \to \{\Sigma \succ 0\}$ an estimator of the asymptotic covariance matrix of $\sqrt{n}(\hat{\mu}_n - \mu(P))$. To accommodate substituting in an estimated covariance matrix, I make explicit how the primitives from Section 4 depend on it. In particular, let $\Phi_\alpha(\Sigma)$ denote the set of level-α tests when the experiment has covariance matrix Σ. A *family of tests* $\{\varphi_\Sigma : \Sigma \succ 0\}$ associates a test with each covariance matrix, where $(\Sigma, y) \mapsto \varphi_\Sigma(y)$ is a fixed measurable function from $\{\Sigma \succ 0\} \times \mathbb{R}^{k_\nu}$ to $[0,1]$. The *feasible procedure* based on such a family inverts the test at the estimated covariance matrix. For $(w, u) \in \mathcal{W}^n \times [0,1]$, let

$$\overline{S}_n(w,u) \equiv \big\{\theta \in \mathbb{R} : u \leq 1 - \varphi_{\hat{\Sigma}_n(w)}\big(\sqrt{n}\, Q(\hat{\mu}_n(w) - \theta K l)\big)\big\}, \tag{22}$$

and let $S_n(w,u)$ be the equivalence class of $\overline{S}_n(w,u)$ in $\mathcal{A}$. By Lemma C.6, S_n is a confidence procedure at sample size n with measurable representative $\overline{S}_n$. For each w, $\overline{S}_n(w,\cdot)$ is the test inversion procedure S_φ of

Section 4.1 evaluated at $x = \hat{\mu}_n(w)$, with $\varphi(y) \equiv \varphi_{\hat{\Sigma}_n(w)}(\sqrt{n}\, y)$. In practice, the idea is to use a family of solutions to (7), one for each covariance matrix, computed as in Section 5. The following results are stated for a general family, placing assumptions on its continuity in Σ.

C.4 Local Asymptotic Minimax Theorem

Definition C.1 (Asymptotic Validity)**.** A sequence of confidence procedures $\{S_n\}$ is asymptotically valid at P_0 if each S_n admits a measurable representative $\overline{S}_n$ such that

$$\liminf_{n\to\infty} \; P^n_{1/\sqrt{n}}\big(\theta_0 + n^{-1/2}t \in \overline{S}_n\big) \geq 1-\alpha$$

for every path $s \mapsto P_s$ in $\mathcal{P}$ that is differentiable in quadratic mean at P_0 and every $t \in \mathbb{R}$ such that $\theta_0 + n^{-1/2}t \in \Theta_n(P_{1/\sqrt{n}})$ for all sufficiently large n.

This is the coverage requirement of Section 3.2 imposed along each local sequence. I purposefully do not impose a uniform coverage requirement here, since this would needlessly restrict the class of confidence procedures considered in the following theorem; see Section D for a discussion of uniformity.

Theorem C.1 (Local Asymptotic Minimax)**.** *Let Assumptions 2.1 to 2.3 and Assumptions C.1 and C.2 hold, and let $v^\star$ be the value of decision problem (6) for the experiment with covariance matrix Σ. If $\{S_n\}$ is asymptotically valid at P_0, then*

$$\sup_{\bar{t}>0} \; \sup_{F} \; \liminf_{n\to\infty} \; \max_{\mu\in F} \; \sqrt{n}\,\mathbb{E}_{n,\mu}\Big[\lambda\Big(\big(S_n \setminus \Theta_n(P_{1/\sqrt{n},\mu})\big) \cap \big[\theta_0 - \bar{t}/\sqrt{n},\; \theta_0 + \bar{t}/\sqrt{n}\big]\Big)\Big] \geq v^\star,$$

where the second supremum is over finite subsets F of $\mathcal{M}$.

The above theorem is the analogue for confidence procedures of van der Vaart (1998, Theorem 25.21). The quantity inside the expectation is the excess length of S_n truncated to the window $[\theta_0 \pm \bar{t}/\sqrt{n}]$. The truncation is a technical device necessary to ensure boundedness. Expectations are taken along the least favorable paths $P_{1/\sqrt{n},\mu}$ of Part (ii), whereas Theorem 25.21 takes the supremum over finite sets of arbitrary tangent directions. I expect that, under additional regularity conditions, the bound continues to hold in that stronger form; I leave this as an area for future work.

Proposition C.1 (Attainment)**.** *Let Assumptions 2.1 to 2.3 and Assumptions C.1 and C.2 hold, and let $\{S_n\}$ be the feasible procedure of Section C.3 based on a family of tests $\{\varphi_\Sigma\}$ and estimators $\hat{\mu}_n$, $\hat{\Sigma}_n$ such that*

(i) $\sqrt{n}(\hat{\mu}_n - \mu_0) = X_n + o_{P_0}(1)$ *and* $\hat{\Sigma}_n = \Sigma + o_{P_0}(1)$,

(ii) φ_Σ *solves the optimal testing problem (7) for the experiment with covariance matrix Σ, and*

(iii) $(\Sigma', y) \mapsto \varphi_{\Sigma'}(y)$ *is continuous at (Σ, y) for almost every $y \in \mathbb{R}^{k_\nu}$.*

Then $\{S_n\}$ is asymptotically valid at P_0, and the left side of Theorem C.1 evaluated at $\{S_n\}$ equals $v^\star$.

Part (i) says that $\hat{\mu}_n$ is asymptotically efficient for μ at P_0 and $\hat{\Sigma}_n$ is consistent. The continuity assumption of Part (iii) is what licenses plugging in the consistent estimator $\hat{\Sigma}_n$.

C.5 Proofs

The proofs use the objects of the proof of Theorem 4.1 at the covariance matrix Σ of Assumption C.1. That is, $P_\mu \equiv \mathcal{N}(\mu, \Sigma)$ with density p_μ, the sets of acceptance functions $\mathcal{D}$ and $\mathcal{D}_\alpha$, the coverage probability function $C(t, \mu, d) = \mathbb{E}_{P_\mu}[d(t, X)]$, and the risk function $R(\mu, d)$, with $v^\star = \inf_{d \in \mathcal{D}_\alpha} \sup_{\mu \in \mathcal{M}} R(\mu, d)$ by the equivalence established in Section B.2.1.

Lemma C.1 (Local Model)**.** *Let Assumptions 2.1 and C.2 hold, and let $\mu \in \mathbb{R}^{k_\mu}$ and $t \in \mathbb{R}$.*

(a) If $P \in \mathcal{P}$ satisfies $\mu(P) = \mu_0 + n^{-1/2}\mu$ and $\theta_0 + n^{-1/2}t \in \Theta_n(P)$, then $t \in \Theta(\mu)$.

(b) If $t \in \Theta(\mu)$, there is n_0 such that $\theta_0 + n^{-1/2}t \in \Theta_n(P)$ for every $n \geq n_0$ and every $P \in \mathcal{P}$ with $\mu(P) = \mu_0 + n^{-1/2}\mu$.

Proof. *Part (a).* By (21), $\mu_0 + n^{-1/2}\mu = K\tau + n^{-1/2}\delta$ and $\theta_0 + n^{-1/2}t = l'\tau$ for some $\tau \in \mathcal{T}$ and $\delta \in \Delta$. Substituting $\mu_0 = K\tau_0$ and $\theta_0 = l'\tau_0$ and multiplying by $\sqrt{n}$,

$$\mu = K\sqrt{n}(\tau - \tau_0) + \delta \quad \text{and} \quad t = l'\sqrt{n}(\tau - \tau_0).$$

Thus $\mu = K\tau' + \delta$ and $t = l'\tau'$ for $\tau' \equiv \sqrt{n}(\tau - \tau_0) \in \mathbb{R}^{k_\tau}$ and $\delta \in \Delta$, which is the definition of $\mu \in \mathcal{M}(t)$, i.e. $t \in \Theta(\mu)$.

Part (b). Suppose $t \in \Theta(\mu)$, so $\mu = K\tau^\circ + \delta$ and $t = l'\tau^\circ$ for some $(\tau^\circ, \delta) \in \mathbb{R}^{k_\tau} \times \Delta$. By Assumption C.2, $\mathcal{T}$ contains a ball of radius $r > 0$ around τ_0; let n_0 be such that $n^{-1/2}\|\tau^\circ\| < r$ for all $n \geq n_0$. For such n, let $\tau \equiv \tau_0 + n^{-1/2}\tau^\circ$, and consider any $P \in \mathcal{P}$ with $\mu(P) = \mu_0 + n^{-1/2}\mu$. Then $\tau \in \mathcal{T}$, and

$$K\tau + n^{-1/2}\delta = K\tau_0 + n^{-1/2}(K\tau^\circ + \delta) = \mu_0 + n^{-1/2}\mu = \mu(P),$$
$$l'\tau = \theta_0 + n^{-1/2}l'\tau^\circ = \theta_0 + n^{-1/2}t,$$

so $(\theta_0 + n^{-1/2}t, P)$ satisfies (21) with $(\tau, \delta) \in \mathcal{T} \times \Delta$, which is to say $\theta_0 + n^{-1/2}t \in \Theta_n(P)$. □

Lemma C.2 (Excess Length)**.** *Let Assumptions 2.1 to 2.3 and Assumption C.2 hold, let $P \in \mathcal{P}$ and $\bar{t} > 0$, and let S_n be a confidence procedure with measurable representative $\overline{S}_n$. Then*

$$\sqrt{n}\,\mathbb{E}_{P^n}\Big[\lambda\Big(\big(S_n \setminus \Theta_n(P)\big) \cap \big[\theta_0 - \bar{t}/\sqrt{n},\ \theta_0 + \bar{t}/\sqrt{n}\big]\Big)\Big]$$
$$= \int_{-\bar{t}}^{\bar{t}} \mathbb{1}\big\{\theta_0 + n^{-1/2}t \notin \Theta_n(P)\big\}\, P^n\big(\theta_0 + n^{-1/2}t \in \overline{S}_n\big)\, \mathrm{d}t.$$

Proof. By Assumptions 2.3 and C.2, the set $\Theta_n(P)$ is σ-compact, hence Borel, as the image of $\{\tau \in \mathcal{T} : \sqrt{n}(\mu(P) - K\tau) \in \Delta\}$ under $\tau \mapsto l'\tau$. The identity then follows by the argument used to establish the

Ghosh–Pratt identity (5). Since $\overline{S}_n(w,u)$ differs from $S_n(w,u)$ by a Lebesgue-null set for every (w,u),

$$\lambda\Big(\big(S_n \setminus \Theta_n(P)\big) \cap \big[\theta_0 - \bar{t}/\sqrt{n},\ \theta_0 + \bar{t}/\sqrt{n}\big]\Big) = \int_{\theta_0-\bar{t}/\sqrt{n}}^{\theta_0+\bar{t}/\sqrt{n}} \mathbb{1}\{\theta \notin \Theta_n(P)\}\,\mathbb{1}\{\theta \in \overline{S}_n\}\,\mathrm{d}\theta,$$

and the change of variables $\theta = \theta_0 + n^{-1/2}t$ yields

$$n^{-1/2}\int_{-\bar{t}}^{\bar{t}} \mathbb{1}\{\theta_0 + n^{-1/2}t \notin \Theta_n(P)\}\,\mathbb{1}\{\theta_0 + n^{-1/2}t \in \overline{S}_n\}\,\mathrm{d}t.$$

The integrand is non-negative and jointly measurable in $(t, W_1, \ldots, W_n, U)$, since $\Theta_n(P)$ is Borel and $(t,w,u) \mapsto \mathbb{1}\{t \in \overline{S}_n(w,u)\}$ is measurable by Part (v) of Lemma B.1. Multiplying by $\sqrt{n}$, taking expectations under P^n, and exchanging the order of integration by Tonelli's theorem delivers the identity. □

Lemma C.3 (Limit Tests)**.** *Let Assumption C.1 hold, and for each n let ϕ_n be a $[0,1]$-valued measurable function of $(W_1, \ldots, W_n, U)$. Then every subsequence has a further subsequence along which, for some measurable $\phi : \mathbb{R}^{k_\mu} \to [0,1]$,*

$$\mathbb{E}_{n,\mu}[\phi_n] \to \mathbb{E}_{P_\mu}[\phi(X)] \quad \textit{for every } \mu \in \mathbb{R}^{k_\mu}.$$

Proof. The experiments $(P^n_{1/\sqrt{n},\mu} : \mu \in \mathbb{R}^{k_\mu})$ are locally asymptotically normal, as noted in Section C.1, so the argument given for this case in the proof of Theorem 15.1 of van der Vaart (1998) applies to the bounded statistics ϕ_n. In particular, every subsequence has a further subsequence along which ϕ_n converges in distribution under $P^n_{1/\sqrt{n},\mu}$ for every μ, meaning $\mathbb{E}_{n,\mu}[\phi_n]$ also converges to some limit for every μ since ϕ_n is bounded. Since the experiments converge to the Gaussian shift experiment $(P_\mu : \mu \in \mathbb{R}^{k_\mu})$, which is dominated by Lebesgue measure, Theorem 15.1 of van der Vaart (1998) provides a test ϕ in the limit experiment whose power function $\mu \mapsto \mathbb{E}_{P_\mu}[\phi(X)]$ is equal to this limit. □

For $\bar{t} > 0$, $\mu \in \mathcal{M}$, and $d \in \mathcal{D}$, let

$$R_{\bar{t}}(\mu, d) \equiv \int_{[-\bar{t},\bar{t}]\setminus\Theta(\mu)} C(t,\mu,d)\,\mathrm{d}t$$

denote the risk truncated to the window $[-\bar{t},\bar{t}]$, which increases to $R(\mu,d)$ as $\bar{t} \to \infty$ by monotone convergence.

In the proof of Theorem C.1, the limit experiment is reached one value of t at a time, which yields a limiting level-α test of $\mathcal{M}(t)$ for each t. The following lemma establishes that a single acceptance function can approximate the performance of each of these tests arbitrarily well, jointly over $t \in [-\bar{t},\bar{t}]$.

Lemma C.4 (Assembly)**.** *Maintain Assumptions 2.1 to 2.3, and let $F \subseteq \mathcal{M}$ denote the finite support of a probability measure ϖ. Then for any $\bar{t}, \varepsilon > 0$, there is a measurable $d : \mathbb{R}^{1+k_\mu} \to [0,1]$ in $\mathcal{D}_\alpha$ such that, for every $t \in [-\bar{t},\bar{t}]$ and every measurable $\phi_t : \mathbb{R}^{k_\mu} \to [0,1]$ with $\sup_{\mu\in\mathcal{M}(t)} \mathbb{E}_{P_\mu}[\phi_t(X)] \le \alpha$,*

$$\sum_{\mu\in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\}\,C(t,\mu,d) \le \sum_{\mu\in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\}\,\mathbb{E}_{P_\mu}[1-\phi_t(X)] + \varepsilon.$$

PROOF. First, notice that by (3), there is a one-to-one correspondence between level-α tests ϕ_t of $H_0 : \mu \in \mathcal{M}(t)$ and level-α tests ψ of $H_0 : \mu \in \mathcal{M}(0)$: for the forwards direction, let $\psi(x) = \phi_t(x + tKl)$, while for the reverse direction, let $\phi_t(x) = \psi(x - tKl)$. For any $E \subseteq F$ and level-α test ψ of $H_0 : \mu \in \mathcal{M}(0)$, write

$$b_t(E, \psi) \equiv \sum_{\mu \in E} \varpi_\mu \, \mathbb{E}_{P_{\mu - tKl}}[1 - \psi(X)] \quad \text{and} \quad F(t) \equiv \{\mu \in F : t \notin \Theta(\mu)\},$$

so that the right-hand side of the inequality in the statement is given by $b_t(F(t), \psi) + \varepsilon$. The quantity $b_t(E, \psi)$ is uniformly continuous in t, uniformly in E and ψ, since

$$|b_t(E, \psi) - b_{t'}(E, \psi)| \leq \max_{\mu \in E} \|p_{\mu - tKl} - p_{\mu - t'Kl}\|_1 = \|p_0 - p_{(t-t')Kl}\|_1,$$

which tends to zero as $t - t' \to 0$, just as in the proof of Lemma B.5.

Now, let m be a positive integer large enough so that $\|p_0 - p_{sKl}\|_1 \leq \varepsilon/3$ whenever $|s| \leq 2\bar{t}/m$, and for $j = 0, 1, \ldots, m$ define evenly-spaced grid points $t_j \equiv -\bar{t} + 2\bar{t}j/m$ on $[-\bar{t}, \bar{t}]$. Every $t \in [-\bar{t}, \bar{t}]$ lies within $2\bar{t}/m$ of the grid point $t_{j(t)}$, where $j(t) \equiv \lceil m(t + \bar{t})/(2\bar{t}) \rceil$. For each j and $E \subseteq F$, let $\psi_{j,E}$ denote a level-α test of $H_0 : \mu \in \mathcal{M}(0)$ with

$$b_{t_j}(E, \psi_{j,E}) \leq \inf_{\psi} b_{t_j}(E, \psi) + \varepsilon/3,$$

where the infimum is taken over level-α tests of $H_0 : \mu \in \mathcal{M}(0)$. Assemble these finitely many tests into

$$d(t, x) \equiv 1 - \psi_{j(t), F(t)}(x - tKl) \quad \text{for } |t| \leq \bar{t}, \qquad d(t, x) \equiv 1 \quad \text{for } |t| > \bar{t}.$$

Then d is measurable, takes values in $[0, 1]$, and belongs to $\mathcal{D}_\alpha$, since for $|t| \leq \bar{t}$ and $\mu \in \mathcal{M}(t)$,

$$C(t, \mu, d) = \mathbb{E}_{P_\mu}[d(t, X)] = \mathbb{E}_{P_{\mu - tKl}}[1 - \psi_{j(t), F(t)}(X)] \geq 1 - \alpha,$$

and $C(t, \mu, d) = 1$ for $|t| > \bar{t}$. In addition, the left-hand side of the inequality in the statement is given by

$$\sum_{\mu \in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\} C(t, \mu, d) = b_t(F(t), \psi_{j(t), F(t)})$$

Finally, for $t \in [-\bar{t}, \bar{t}]$, let ϕ_t be a level-α test of $H_0 : \mu \in \mathcal{M}(t)$, and let $\psi(x) = \phi_t(x + tKl)$. Then

$$b_t(F(t), \psi_{j(t), F(t)}) \leq b_{t_{j(t)}}(F(t), \psi_{j(t), F(t)}) + \tfrac{\varepsilon}{3} \leq b_{t_{j(t)}}(F(t), \psi) + \tfrac{2\varepsilon}{3} \leq b_t(F(t), \psi) + \varepsilon.$$

□

Lemma C.5 (Truncation)**.** *Let Assumptions 2.1 to 2.3 hold. Then*

$$v^\star = \sup_{\bar{t} > 0} \sup_{\varpi} \inf_{d \in \mathcal{D}_\alpha} \sum_{\mu} \varpi_\mu R_{\bar{t}}(\mu, d),$$

where the inner supremum is over finitely supported probability measures ϖ on $\mathcal{M}$.

PROOF. For $\mu \in \mathcal{M}$ and $\bar{t} > 0$, let $w_\mu^{(\bar{t})}(t,x) \equiv \mathbb{1}\{t \in [-\bar{t},\bar{t}] \setminus \Theta(\mu)\} p_\mu(x)$, as in the proof of Lemma B.6, so that $w_\mu^{(\bar{t})} \in L^1(\mathbb{R}^{1+k_\mu})$ and $\langle d, w_\mu^{(\bar{t})} \rangle = R_{\bar{t}}(\mu, d)$ for $d \in \mathcal{D}$ by Tonelli. Let $\mathcal{C}$ be the convex hull of $\{w_\mu^{(\bar{t})} : \mu \in \mathcal{M},\ \bar{t} > 0\}$. I apply Evans et al. (2005, Theorem 4) to $(d, w) \mapsto \langle d, w \rangle$ on $\mathcal{D}_\alpha \times \mathcal{C}$. To verify that the conditions of the theorem hold, note that $\mathcal{C}$ is convex and $\langle d, \cdot \rangle$ is linear, $\mathcal{D}_\alpha$ is convex and weak-$*$ compact by Lemma B.6, and $\langle \cdot, w \rangle$ is linear and weak-$*$ continuous for each $w \in \mathcal{C} \subseteq L^1(\mathbb{R}^{1+k_\mu})$. Therefore

$$\inf_{d \in \mathcal{D}_\alpha} \sup_{w \in \mathcal{C}} \langle d, w \rangle = \sup_{w \in \mathcal{C}} \inf_{d \in \mathcal{D}_\alpha} \langle d, w \rangle.$$

The left-hand side is equal to $v^\star$, since for $d \in \mathcal{D}_\alpha$,

$$\sup_{\mu \in \mathcal{M}} R(\mu, d) = \sup_{\mu \in \mathcal{M}, \bar{t} > 0} R_{\bar{t}}(\mu, d) = \sup_{\mu \in \mathcal{M}, \bar{t} > 0} \langle d, w_\mu^{(\bar{t})} \rangle = \sup_{w \in \mathcal{C}} \langle d, w \rangle.$$

Examining the right-hand side, let $w \in \mathcal{C}$. Then by definition, we may write w as a convex combination of finitely many weight functions, $w = \sum_{j=1}^m c_j w_{\mu_j}^{(\bar{t}_j)}$ with $c_j \geq 0$ and $\sum_{j=1}^m c_j = 1$. Let ϖ denote the probability measure on $\{\mu_1, \ldots, \mu_m\}$ with $\varpi_\mu = \sum_{j=1}^m \mathbb{1}\{\mu_j = \mu\} c_j$, and let $\bar{t} = \max_j \bar{t}_j$. Then

$$\langle d, w \rangle = \sum_{j=1}^m c_j R_{\bar{t}_j}(\mu_j, d) \leq \sum_\mu \varpi_\mu R_{\bar{t}}(\mu, d),$$

so in particular

$$v^\star = \inf_{d \in \mathcal{D}_\alpha} \sup_{w \in \mathcal{C}} \langle d, w \rangle = \sup_{w \in \mathcal{C}} \inf_{d \in \mathcal{D}_\alpha} \langle d, w \rangle \leq \sup_{\bar{t} > 0} \sup_{\varpi} \inf_{d \in \mathcal{D}_\alpha} \sum_\mu \varpi_\mu R_{\bar{t}}(\mu, d).$$

The reverse inequality also holds, since

$$v^\star = \inf_{d \in \mathcal{D}_\alpha} \sup_{\mu \in \mathcal{M}} R(\mu, d) \geq \inf_{d \in \mathcal{D}_\alpha} \sup_{\bar{t} > 0} \sup_{\varpi} \sum_\mu \varpi_\mu R_{\bar{t}}(\mu, d) \geq \sup_{\bar{t} > 0} \sup_{\varpi} \inf_{d \in \mathcal{D}_\alpha} \sum_\mu \varpi_\mu R_{\bar{t}}(\mu, d).$$

□

Proof of Theorem C.1. Fix $\bar{t}, \varepsilon > 0$, along with a finitely supported probability measure ϖ on $F \subseteq \mathcal{M}$. For $t \in \mathbb{R}$, $\mu \in \mathbb{R}^{k_\mu}$, and $\overline{S}_n$ the measurable representative of S_n in Definition C.1, write

$$C_n(t, \mu) \equiv P^n_{1/\sqrt{n}, \mu}\big(\theta_0 + n^{-1/2} t \in \overline{S}_n\big)$$

for the coverage probability of the local value t along the path of Assumption C.1 in direction μ.

Ghosh–Pratt inequality. By Part (ii) of Assumption C.1, for each $\mu \in F$ there exists n_μ such that $\mu(P_{1/\sqrt{n},\mu}) = \mu_0 + n^{-1/2}\mu$ for all $n \geq n_\mu$. Since F is finite, $n_F \equiv \max_{\mu \in F} n_\mu$ is as well, and for $n \geq n_F$ the moments are exactly linear along every path in F at once. For any such n and $\mu \in F$, by Lemma C.2,

$$\sqrt{n}\, \mathbb{E}_{n,\mu}\Big[\lambda\Big(\big(S_n \setminus \Theta_n(P_{1/\sqrt{n},\mu})\big) \cap \big[\theta_0 - \bar{t}/\sqrt{n},\ \theta_0 + \bar{t}/\sqrt{n}\big]\Big)\Big] = \int_{-\bar{t}}^{\bar{t}} \mathbb{1}\big\{\theta_0 + n^{-1/2} t \notin \Theta_n(P_{1/\sqrt{n},\mu})\big\} C_n(t, \mu)\, \mathrm{d}t.$$

By Part (a) of Lemma C.1, $\theta_0 + n^{-1/2}t \in \Theta_n(P_{1/\sqrt{n},\mu})$ implies $t \in \Theta(\mu)$, so the indicator in the right-hand side integrand is at least $\mathbb{1}\{t \notin \Theta(\mu)\}$ for every $t \in \mathbb{R}$. Hence, for all $n \geq n_F$ and every $\mu \in F$,

$$\sqrt{n}\,\mathbb{E}_{n,\mu}\big[\lambda(\cdots)\big] \geq \int_{-\bar{t}}^{\bar{t}} \mathbb{1}\{t \notin \Theta(\mu)\} C_n(t,\mu)\,\mathrm{d}t.$$

The maximum over F of the left-hand side is at least its ϖ-weighted average, so by Fatou's lemma,

$$\liminf_{n\to\infty} \max_{\mu\in F} \sqrt{n}\,\mathbb{E}_{n,\mu}\big[\lambda(\cdots)\big] \geq \int_{-\bar{t}}^{\bar{t}} \liminf_{n\to\infty} \sum_{\mu\in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\} C_n(t,\mu)\,\mathrm{d}t. \tag{23}$$

Limit tests for each t. Fix $t \in [-\bar{t},\bar{t}]$, and consider a subsequence along which the integrand in (23) converges to its limit inferior. The non-coverage indicators $\mathbb{1}\{\theta_0 + n^{-1/2}t \notin \overline{S}_n\}$ are $[0,1]$-valued measurable functions of $(W_1,\ldots,W_n,U)$ with expectation $1 - C_n(t,\mu)$ under $P^n_{1/\sqrt{n},\mu}$. By Lemma C.3, then, there is a limit test $\phi_t : \mathbb{R}^{k_\mu} \to [0,1]$ and a further subsequence along which $C_n(t,\mu) \to \mathbb{E}_{P_\mu}[1-\phi_t(X)]$ for every $\mu \in \mathbb{R}^{k_\mu}$. It follows that

$$\liminf_{n\to\infty} \sum_{\mu\in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\} C_n(t,\mu) = \sum_{\mu\in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\}\, \mathbb{E}_{P_\mu}[1-\phi_t(X)].$$

It remains to show that ϕ_t is a level-α test of $H_0 : \mu \in \mathcal{M}(t)$. Let $\mu \in \mathcal{M}(t)$, so $t \in \Theta(\mu)$, and thus $\theta_0 + n^{-1/2}t \in \Theta_n(P_{1/\sqrt{n},\mu})$ for all n sufficiently large by Part (b) of Lemma C.1. Since $s \mapsto P_{s,\mu}$ is differentiable in quadratic mean at P_0, Definition C.1 implies that $\liminf_n C_n(t,\mu) \geq 1-\alpha$, so in particular the subsequential limit $\mathbb{E}_{P_\mu}[1-\phi_t(X)] \geq 1-\alpha$, meaning $\mathbb{E}_{P_\mu}[\phi_t(X)] \leq \alpha$.

Assembly. Let d be the acceptance function of Lemma C.4. Since ϕ_t is a level-α test of $H_0 : \mu \in \mathcal{M}(t)$, the right-hand side of the above display is at least $\sum_{\mu\in F} \varpi_\mu \mathbb{1}\{t \notin \Theta(\mu)\} C(t,\mu,d) - \varepsilon$. This bound holds uniformly over $t \in [-\bar{t},\bar{t}]$, so integrating it as in the right-hand side of (23) yields

$$\liminf_{n\to\infty} \max_{\mu\in F} \sqrt{n}\,\mathbb{E}_{n,\mu}\big[\lambda(\cdots)\big] \geq \sum_{\mu\in F} \varpi_\mu R_{\bar{t}}(\mu,d) - 2\bar{t}\varepsilon \geq \inf_{d\in\mathcal{D}_\alpha} \sum_{\mu\in F} \varpi_\mu R_{\bar{t}}(\mu,d) - 2\bar{t}\varepsilon.$$

Truncation. The left-hand side of the above display does not depend on ε, so letting $\varepsilon \downarrow 0$,

$$\liminf_{n\to\infty} \max_{\mu\in F} \sqrt{n}\,\mathbb{E}_{n,\mu}\big[\lambda(\cdots)\big] \geq \inf_{d\in\mathcal{D}_\alpha} \sum_{\mu\in F} \varpi_\mu R_{\bar{t}}(\mu,d)$$

for every $\bar{t} > 0$, every finite $F \subseteq \mathcal{M}$, and every probability measure ϖ on F. It follows that

$$\sup_F \liminf_{n\to\infty} \max_{\mu\in F} \sqrt{n}\,\mathbb{E}_{n,\mu}\big[\lambda(\cdots)\big] \geq \sup_{\varpi} \inf_{d\in\mathcal{D}_\alpha} \sum_{\mu} \varpi_\mu R_{\bar{t}}(\mu,d) \quad \text{for every } \bar{t} > 0,$$

with the right-hand side supremum taken over finitely supported probability measures on $\mathcal{M}$. Taking the supremum over $\bar{t} > 0$ on both sides, the right-hand side is equal to $v^\star$ by Lemma C.5. □

Lemma C.6 (Feasible Procedure)**.** *Let Assumptions 2.1 to 2.3 hold. Then S_n is a confidence procedure at sample size n, $\overline{S}_n$ in (22) is a measurable representative of it, and for every $(\theta, P) \in \mathbb{R} \times \mathcal{P}$,*

$$P^n(\theta \in \overline{S}_n) = 1 - \mathbb{E}_{P^n}\big[\varphi_{\hat{\Sigma}_n}\big(\sqrt{n}\, Q(\hat{\mu}_n - \theta Kl)\big)\big].$$

Proof. The map $(\theta, w) \mapsto (\hat{\Sigma}_n(w), \sqrt{n}\, Q(\hat{\mu}_n(w) - \theta Kl))$ is measurable, as is $(\Sigma, y) \mapsto \varphi_\Sigma(y)$ by the definition of a family of tests, so

$$(\theta, w, u) \mapsto \mathbb{1}\{\theta \in \overline{S}_n(w, u)\} = \mathbb{1}\big\{u \leq 1 - \varphi_{\hat{\Sigma}_n(w)}\big(\sqrt{n}\, Q(\hat{\mu}_n(w) - \theta Kl)\big)\big\}$$

is measurable. For each $B \in \mathcal{B}$, integrating this indicator over $\theta \in B$ yields $\lambda(\overline{S}_n(w, u) \cap B)$, which is therefore a measurable in (w, u) by Tonelli. Hence S_n is measurable by Part (ii) of Lemma B.1, whose proof applies verbatim with $\mathcal{W}^n$ in place of $\mathbb{R}^{k_\mu}$, and $\overline{S}_n$ is a measurable representative of S_n by Part (v) of Lemma B.1. Finally, since $P^n(\theta \in \overline{S}_n)$ is the expectation of this indicator and U is a uniform random variable independent of the data, by the law of iterated expectations

$$P^n(\theta \in \overline{S}_n) = 1 - \mathbb{E}_{P^n}\big[\varphi_{\hat{\Sigma}_n}\big(\sqrt{n}\, Q(\hat{\mu}_n - \theta Kl)\big)\big]. \qquad \square$$

Proof of Proposition C.1. By Lemma C.6, $\overline{S}_n$ in (22) is a measurable representative of S_n, which I use throughout.

Coverage probabilities. Note that $\tau_0 - \theta_0 l \in \ker(l')$ by Assumption 2.1, and $K \ker(l') \subseteq V = \ker(Q)$ by the proofs of Proposition A.2 and Lemma B.3, so $Q(\mu_0 - \theta_0 Kl) = QK(\tau_0 - \theta_0 l) = 0$. Writing $\hat{X}_n \equiv \sqrt{n}(\hat{\mu}_n - \mu_0)$ and $t = \sqrt{n}(\theta - \theta_0)$, it follows that

$$\sqrt{n}\, Q(\hat{\mu}_n - \theta Kl) = Q\hat{X}_n + \sqrt{n}\, Q(\mu_0 - \theta_0 Kl) - tQKl = Q\hat{X}_n - te_1,$$

since $QKl = e_1$. So, by Lemma C.6, for every $(t, P) \in \mathbb{R} \times \mathcal{P}$,

$$P^n\big(\theta_0 + n^{-1/2}t \in \overline{S}_n\big) = 1 - \mathbb{E}_{P^n}\big[\varphi_{\hat{\Sigma}_n}(Q\hat{X}_n - te_1)\big]. \tag{24}$$

Convergence along local sequences. Let $s \mapsto P_s$ denote a path in $\mathcal{P}$ that is differentiable in quadratic mean at P_0 with score function g. By Part (i) and Lemma 25.23 of van der Vaart (1998), $\hat{\mu}_n$ is regular at P_0, so $\sqrt{n}(\hat{\mu}_n - \mu(P_{1/\sqrt{n}})) \xrightarrow{d} \mathcal{N}(0, \Sigma)$ under $P^n_{1/\sqrt{n}}$, while $\sqrt{n}(\mu(P_{1/\sqrt{n}}) - \mu_0) \to P_0\tilde{\mu}_{P_0} g$ by the assumed differentiability of μ in Part (i) of Assumption C.1. hence $\hat{X}_n \xrightarrow{d} \mathcal{N}(P_0\tilde{\mu}_{P_0} g, \Sigma)$ under $P^n_{1/\sqrt{n}}$. Moreover, $P^n_{1/\sqrt{n}}$ is contiguous to P^n_0 by Lemma 25.14 and Example 6.5 of van der Vaart (1998), so $\hat{\Sigma}_n = \Sigma + o(1)$ in $P^n_{1/\sqrt{n}}$-probability as well, and by Slutsky's lemma $(\hat{X}_n, \hat{\Sigma}_n) \xrightarrow{d} (X, \Sigma)$ under $P^n_{1/\sqrt{n}}$, with $X \sim \mathcal{N}(P_0\tilde{\mu}_{P_0} g, \Sigma)$.

Now, fix $t \in \mathbb{R}$. The map $(\Sigma^\circ, x) \mapsto \varphi_{\Sigma^\circ}(Qx - te_1)$ is measurable and takes values in $[0, 1]$, and it is continuous at (Σ, x) so long as $(\Sigma^\circ, y) \mapsto \varphi_{\Sigma^\circ}(y)$ is continuous at $(\Sigma, Qx - te_1)$, which holds for every $Qx - te_1$ outside a null set by Part (iii). Since $QX - te_1 \sim \mathcal{N}(Q(P_0\tilde{\mu}_{P_0} g - tKl), Q\Sigma Q')$ and $Q\Sigma Q' \succ 0$ by Part (ii) of

Lemma B.3, this null set has probability zero, so by the continuous mapping theorem, boundedness, and (24),

$$P^n_{1/\sqrt{n}}\big(\theta_0 + n^{-1/2}t \in \overline{S}_n\big) \to \mathbb{E}\big[1 - \varphi_\Sigma(QX - te_1)\big] = C(t, P_0\tilde{\mu}_{P_0}g, d_{\varphi_\Sigma}). \tag{25}$$

That is, along every local sequence, the coverage probabilities of the feasible procedure converge to those of the limiting test inversion acceptance function d_{φ_Σ} at the local parameter $P_0\tilde{\mu}_{P_0}g$.

Validity. Suppose that $\theta_0 + n^{-1/2}t \in \Theta_n(P_{1/\sqrt{n}})$ for all sufficiently large n, and let $\mu_n = \sqrt{n}(\mu(P_{1/\sqrt{n}}) - \mu_0)$ so that $\mu(P_{1/\sqrt{n}}) = \mu_0 + n^{-1/2}\mu_n$. By Part (a) of Lemma C.1, $t \in \Theta(\mu_n)$, meaning $\mu_n \in \mathcal{M}(t)$, for all sufficiently large n, and $\mu_n \to P_0\tilde{\mu}_{P_0}g$, so $P_0\tilde{\mu}_{P_0}g$ lies in the closure of $\mathcal{M}(t)$. By Part (ii), $\varphi_\Sigma \in \Phi_\alpha(\Sigma)$, so $\inf_{\mu\in\mathcal{M}(t)} C(t, \mu, d_{\varphi_\Sigma}) \geq 1 - \alpha$ by the proof of Lemma B.9. Since $|C(t, \mu', d_{\varphi_\Sigma}) - C(t, \mu, d_{\varphi_\Sigma})| \leq \|p_{\mu'} - p_\mu\|_1$, which tends to zero as $\mu' \to \mu$ as in the proof of Lemma B.5, the coverage requirement extends to the closure, so $C(t, P_0\tilde{\mu}_{P_0}g, d_{\varphi_\Sigma}) \geq 1 - \alpha$. Hence $\liminf_n P^n_{1/\sqrt{n}}(\theta_0 + n^{-1/2}t \in \overline{S}_n) \geq 1 - \alpha$ by (25), which is Definition C.1.

Risk. Let $\mu \in \mathcal{M}$ and $\bar{t} > 0$, and consider the path $s \mapsto P_{s,\mu}$ of Assumption C.1, whose score function $\mu'\Sigma^{-1}\tilde{\mu}_{P_0}$ sets $P_0\tilde{\mu}_{P_0}g = \mu$ in (25), and along which $\mu(P_{1/\sqrt{n},\mu}) = \mu_0 + n^{-1/2}\mu$ for all sufficiently large n. For such n, writing $C_n(t, \mu)$ as in the proof of Theorem C.1, Lemma C.2 implies that

$$\begin{aligned}
&\sqrt{n}\,\mathbb{E}_{n,\mu}\Big[\lambda\Big(\big(S_n \setminus \Theta_n(P_{1/\sqrt{n},\mu})\big) \cap \big[\theta_0 - \bar{t}/\sqrt{n},\ \theta_0 + \bar{t}/\sqrt{n}\big]\Big)\Big] \\
&= \int_{-\bar{t}}^{\bar{t}} \mathbb{1}\big\{\theta_0 + n^{-1/2}t \notin \Theta_n(P_{1/\sqrt{n},\mu})\big\} C_n(t, \mu)\,\mathrm{d}t.
\end{aligned}$$

By Lemma C.1, the indicator in the above integrand converges to $\mathbb{1}\{t \notin \Theta(\mu)\}$ pointwise in t: by Part (a), it is at least $\mathbb{1}\{t \notin \Theta(\mu)\}$, while by Part (b), it vanishes for sufficiently large n whenever $t \in \Theta(\mu)$. By (25), $C_n(t, \mu) \to C(t, \mu, d_{\varphi_\Sigma})$. By dominated convergence, then,

$$\sqrt{n}\,\mathbb{E}_{n,\mu}\big[\lambda(\cdots)\big] \to \int_{-\bar{t}}^{\bar{t}} \mathbb{1}\{t \notin \Theta(\mu)\}\, C(t, \mu, d_{\varphi_\Sigma})\,\mathrm{d}t = R_{\bar{t}}(\mu, d_{\varphi_\Sigma}) \leq R(\mu, d_{\varphi_\Sigma}) = R(Q_{-1}\mu, \varphi_\Sigma) \leq v^\star$$

The last equality follows from the proof of Theorem 4.1, while the last inequality holds because φ_Σ solves (7) by Part (ii), whose value is $v^\star$ by Theorem 4.1. For finite F, the limit inferior of the maximum of finitely many convergent sequences is the maximum of their limits, so the left-hand side of Theorem C.1 evaluated at $\{S_n\}$ is at most $v^\star$. It is at least $v^\star$ by Theorem C.1, since $\{S_n\}$ is asymptotically valid. □

D Uniform Asymptotic Validity

Theorem C.1 and Proposition C.1 consider local asymptotics around a fixed P_0. This section provides conditions under which the feasible procedure of Section C.3 satisfies the coverage requirement of Section 3.2 for the model at sample size n uniformly over a broad class of DGPs.

Assumption D.1 (Uniformity)**.** For every $P \in \mathcal{P}$, $\Sigma(P)$ is a positive definite matrix, and for every $\varepsilon > 0$,

$$\sup_{P\in\mathcal{P}} d_{\mathrm{BL}}\big(\sqrt{n}(\hat{\mu}_n - \mu(P)),\ \mathcal{N}(0, \Sigma(P))\big) \to 0 \quad \text{and} \quad \sup_{P\in\mathcal{P}} P^n\big(\|\hat{\Sigma}_n - \Sigma(P)\| > \varepsilon\big) \to 0,$$

where d_{BL} denotes the bounded Lipschitz distance between the law of $\sqrt{n}(\hat{\mu}_n - \mu(P))$ under P^n and $\mathcal{N}(0, \Sigma(P))$, and $\|\cdot\|$ is the operator norm.

These are the usual high-level assumptions of uniform convergence in distribution of the moment estimator, and uniform consistency of the covariance matrix estimator. Under Assumption C.1 and Part (i) of Proposition C.1, $\Sigma(P_0) = \Sigma$. Now, write $\Sigma(\mathcal{P}) \equiv \{\Sigma(P) : P \in \mathcal{P}\}$. For $\varepsilon > 0$ and $\Sigma \succ 0$, let

$$\varphi_\Sigma^\varepsilon(y) \equiv \sup\Big\{\varphi_{\Sigma'}(y') - \varepsilon^{-1}\|y - y'\| : y' \in \mathbb{R}^{k_\nu},\ \Sigma' \succ 0,\ \|\Sigma' - \Sigma\| \le \varepsilon\Big\},$$

the smallest function that is ε^{-1}-Lipschitz in y and dominates $\varphi_{\Sigma'}$ for every $\|\Sigma' - \Sigma\| \le \varepsilon$.

Assumption D.2 (Equicontinuity)**.** The family of tests $\{\varphi_\Sigma\}$ satisfies

$$\lim_{\varepsilon \downarrow 0} \sup_{\Sigma \in \Sigma(\mathcal{P})} \sup_{\nu \in Q\mathcal{M}(0)} \mathbb{E}_{\mathcal{N}(\nu, Q\Sigma Q')}\big[\varphi_\Sigma^\varepsilon(Y) - \varphi_\Sigma(Y)\big] = 0.$$

The expectation is the increase in the null rejection probability when φ_Σ is replaced by its envelope: for indicator tests, Assumption D.2 requires the Gaussian mass near the boundary of the rejection region to be small, uniformly over the null parameter space, and the region to vary continuously with Σ. It is the uniform counterpart of the continuity assumption in Part (iii) of Proposition C.1.

Definition D.1 (Uniform Asymptotic Validity)**.** A sequence of confidence procedures $\{S_n\}$ is uniformly asymptotically valid over $\mathcal{P}$ if each S_n admits a measurable representative $\overline{S}_n$ such that

$$\liminf_{n \to \infty} \inf_{P \in \mathcal{P}} \inf_{\theta \in \Theta_n(P)} P^n\big(\theta \in \overline{S}_n\big) \ge 1 - \alpha.$$

Definition D.1 implies Definition C.1 at every $P_0 \in \mathcal{P}$, since if $\theta_0 + n^{-1/2}t \in \Theta_n(P_{1/\sqrt{n}})$ for n sufficiently large, then for such n, $P^n_{1/\sqrt{n}}(\theta_0 + n^{-1/2}t \in \overline{S}_n) \ge \inf_{P \in \mathcal{P}} \inf_{\theta \in \Theta_n(P)} P^n(\theta \in \overline{S}_n)$.

Theorem D.1 (Uniform Asymptotic Validity)**.** *Let Assumptions 2.1 to 2.3 and Assumption D.1 hold, and let $\{S_n\}$ be the feasible procedure of Section C.3 based on a family of tests $\{\varphi_\Sigma\}$ that satisfies Assumption D.2, with $\varphi_\Sigma \in \Phi_\alpha(\Sigma)$ for every $\Sigma \in \Sigma(\mathcal{P})$. Then $\{S_n\}$ is uniformly asymptotically valid over $\mathcal{P}$.*

D.1 Proof of Theorem D.1

Proof. By Lemma C.6, $\overline{S}_n$ in (22) is a measurable representative of S_n, which I use throughout. Fix $\varepsilon > 0$, n, $P \in \mathcal{P}$, and $\theta \in \Theta_n(P)$. The model enters only through the localized reduced moments $\nu \equiv \sqrt{n}\,Q(\mu(P) - \theta Kl)$, which lie in $Q\mathcal{M}(0)$. To see this, write $\mu(P) = K\tau + n^{-1/2}\delta$ with $\theta = l'\tau$ as in (21). Then $\sqrt{n}(\mu(P) - \theta Kl) = \sqrt{n}K(\tau - \theta l) + \delta$, and $\tau - \theta l \in \ker(l')$ by Assumption 2.1, so $\sqrt{n}(\mu(P) - \theta Kl) \in K\ker(l') + \Delta = \mathcal{M}(0)$. Writing $\hat{X}_n \equiv \sqrt{n}(\hat{\mu}_n - \mu(P))$ for the estimator in local units, the argument of the test is $\sqrt{n}\,Q(\hat{\mu}_n - \theta Kl) = \nu + Q\hat{X}_n$, so by Lemma C.6,

$$P^n(\theta \notin \overline{S}_n) = \mathbb{E}_{P^n}\big[\varphi_{\hat{\Sigma}_n}(\nu + Q\hat{X}_n)\big].$$

The first step replaces the test at the estimated covariance matrix by the envelope at $\Sigma(P)$. On the event $\|\hat{\Sigma}_n - \Sigma(P)\| \le \varepsilon$, taking $y' = y$ and $\Sigma' = \hat{\Sigma}_n$ in the definition of the envelope yields $\varphi_{\hat{\Sigma}_n}(y) \le \varphi^{\varepsilon}_{\Sigma(P)}(y)$ for every y; on the complementary event, $\varphi_{\hat{\Sigma}_n} \le 1$. Hence

$$\mathbb{E}_{P^n}\big[\varphi_{\hat{\Sigma}_n}(\nu + Q\hat{X}_n)\big] \le P^n\big(\|\hat{\Sigma}_n - \Sigma(P)\| > \varepsilon\big) + \mathbb{E}_{P^n}\big[\varphi^{\varepsilon}_{\Sigma(P)}(\nu + Q\hat{X}_n)\big].$$

The second step replaces the law of $\hat{X}_n$ by its Gaussian limit, which is the purpose of the envelope. The envelope is is ε^{-1}-Lipschitz and takes values in $[0,1]$, by construction. Consequently, $x \mapsto \varphi^{\varepsilon}_{\Sigma(P)}(\nu + Qx)$ is $\varepsilon^{-1}\|Q\|$-Lipschitz and takes values in $[0,1]$, and by the definition of d_{BL},

$$\mathbb{E}_{P^n}\big[\varphi^{\varepsilon}_{\Sigma(P)}(\nu + Q\hat{X}_n)\big] \le \mathbb{E}_{\mathcal{N}(0,\Sigma(P))}\big[\varphi^{\varepsilon}_{\Sigma(P)}(\nu + QX)\big] + \big(1 + \varepsilon^{-1}\|Q\|\big)\, d_{\mathrm{BL}}\big(\hat{X}_n, \mathcal{N}(0,\Sigma(P))\big).$$

The third step uses the level of the test and the equicontinuity of the family. Write $\Omega(P)$ as shorthand for $Q\Sigma(P)Q'$. Under $X \sim \mathcal{N}(0,\Sigma(P))$, $\nu + QX \sim \mathcal{N}(\nu,\Omega(P))$ with $\nu \in Q\mathcal{M}(0)$ and $\Sigma(P) \in \Sigma(\mathcal{P})$, so the hypothesis $\varphi_{\Sigma(P)} \in \Phi_\alpha(\Sigma(P))$ yields

$$\mathbb{E}_{\mathcal{N}(0,\Sigma(P))}\big[\varphi^{\varepsilon}_{\Sigma(P)}(\nu + QX)\big] = \mathbb{E}_{\mathcal{N}(\nu,\Omega(P))}\big[\varphi_{\Sigma(P)}(Y)\big] + \mathbb{E}_{\mathcal{N}(\nu,\Omega(P))}\big[\varphi^{\varepsilon}_{\Sigma(P)}(Y) - \varphi_{\Sigma(P)}(Y)\big] \le \alpha + \varrho(\varepsilon),$$

where $\varrho(\varepsilon)$ denotes the double supremum in Assumption D.2, which tends to zero as $\varepsilon \downarrow 0$.

Combining the three steps and taking the supremum over $P \in \mathcal{P}$ and $\theta \in \Theta_n(P)$,

$$\begin{aligned}\sup_{P\in\mathcal{P}} \sup_{\theta\in\Theta_n(P)} P^n(\theta \notin \overline{S}_n) \le{}& \sup_{P\in\mathcal{P}} P^n\big(\|\hat{\Sigma}_n - \Sigma(P)\| > \varepsilon\big) \\ &+ \big(1 + \varepsilon^{-1}\|Q\|\big) \sup_{P\in\mathcal{P}} d_{\mathrm{BL}}\big(\sqrt{n}(\hat{\mu}_n - \mu(P)), \mathcal{N}(0,\Sigma(P))\big) + \alpha + \varrho(\varepsilon).\end{aligned}$$

By Assumption D.1, the limit superior in n of the right side is $\alpha + \varrho(\varepsilon)$ for every fixed $\varepsilon > 0$, and letting $\varepsilon \downarrow 0$ completes the proof. □

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